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Consistency Problems for Heath-Jarrow-Morton Interest Rate Models



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Preface

Bond markets differ in one fundamental aspect from standard stock markets. While the latter are built up by a finite number of traded assets, the underlying basis of a bond market is the entire term structure of interest rates: an infinite-dimensional variable which is not directly observable. On the empirical side this necessitates curve-fitting methods for the daily estimation of the term structure. Pricing models on the other hand, are usually built upon stochastic factors representing the term structure in a finite-dimensional state space, making them computationally tractable.

This research monograph brings together curve-fitting methods and factor models for the term structure of interest rates within the Heath–Jarrow–Morton (henceforth HJM) framework [35], which basically unifies all continuous interest rate models. We provide appropriate consistency conditions and explore some important examples.

The HJM framework can be seen as a generic description of the arbitrage-free movements of the forward curve (the term structure of forward rates), driven by a Brownian motion. The no-arbitrage requirement leads to a restriction on the drift of any single forward rate process (HJM drift condition). The (infinite-dimensional) state variable in an HJM model is the entire forward curve. As such, any initial forward curve can be taken as model input. But, by the HJM drift condition, there is a limitation on the shapes of the curves produced by the model. For instance, a flat initial forward curve cannot move only by parallel shifts, unless it remains constant over time. This fact has been known by the experts for a long time. Such consistency considerations for HJM models and finite-dimensional manifolds of curves (such as the set of flat curves) form an essential part of this book.

On the empirical side there exist commonly used methods for fitting the current forward curve. Any curve-fitting method can be represented as a parametrized family of smooth curves $\mathcal{G} = \{G(\cdot, z) \mid z \in \mathcal{Z}\}$ with some finite-dimensional parameter set $\mathcal{Z}$. A lot of cross-sectional data, that is, daily estimations of z , is available, and one may ask for a suitable stochastic model for z that provides accurate prices for interest rate based derivatives. However, this requires the absence of arbitrage. In this book, we characterize all consistent $\mathcal{Z}$ -valued state space Itô processes Z which, by definition, provide an arbitrage-free model when representing the parameter z . That is, the fac-

tor model $G(T - t, Z_t)$ has to satisfy the HJM drift condition. It turns out that selected common curve-fitting methods do not go well with the HJM framework.

We then consider the preceding consistency problem from a geometric point of view, as proposed by Björk and Christensen [8]. First, the HJM methodology is completely recaptured within a functional analytic framework, which also incorporates an infinite-dimensional driving Brownian motion. We then rigorously perform the change of parametrization due to Musiela [46] and arrive at a stochastic equation in a Hilbert space H , describing the arbitrage-free evolution of the forward curve. The family $\mathcal{G}$ can now be seen as a subset of H and the above consistency considerations yield a stochastic invariance problem for the previously derived stochastic equation. Under the assumption that $\mathcal{G}$ is a regular submanifold of H , we derive sufficient and necessary conditions for its invariance. Expressed in local coordinates they turn out to equal the HJM drift condition. In conclusion, we provide a general tool for exploiting the interplay between curve-fitting methods and HJM factor models.

Classical models, such as the Vasicek [56] and Cox–Ingersoll–Ross (henceforth CIR) [18] short rate models, and the popular Brace–Gatarek–Musiela (henceforth BGM) [13] LIBOR rate (short for London Interbank Offered Rate) model, are shown to fit well into our framework. By their very definition, affine HJM models are consistent with finite-dimensional linear submanifolds of H . A straight application of our results yields a complete characterization of all affine HJM models, as obtained by Duffie and Kan [23].

Many phenomena, say in physics or economics, are described by stochastic equations in finite or infinite dimension. Some parts of this book (Chapters 2 and 6) are of general form. The results can (and hopefully will) also be applied to fields other than interest rate theory.

As for the mathematical prerequisites, the reader is expected to be familiar with elementary stochastic calculus and probability theory. This book provides a short introduction both to the basic terminology in the theory of interest rates and to stochastic analysis and equations in infinite dimension.

This monograph is based on my Ph.D. thesis [27], which I wrote under the supervision of F. Delbaen at the Swiss Federal Institute of Technology (ETH) Zurich. It is a great pleasure to thank him here for his guidance throughout my Ph.D. studies. I am deeply grateful to T. Björk for his vital support, which made this publication possible. Further thanks go to B. J. Christensen, G. Da Prato, R. Cont, D. Heath, W. Schachermayer, D. Duffie and in particular to J. Zabczyk for their interest, support and hospitality. I thank Manuela for her love and encouragement over the last years. Financial support from Credit Suisse is gratefully acknowledged.

Stanford, January 2001

Damir Filipović

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1. Introduction

In this chapter we give an informal overview of the problems which will be explored in the present text. First we provide a brief introduction to the basic terminology in the theory of interest rates. A more general and comprehensive treatment of the arbitrage pricing theory is beyond the scope of these notes. For that purpose there are many excellent textbooks, such as [7], [47], [41] and [4] – this list being far from complete.

1.1 Bond Markets

The most basic interest rate contract is a *zero coupon bond*, or simply *bond*, which pays the holder with certainty one unit of cash at a fixed future *maturity* date T . Its price at time $t \leq T$ is denoted by $P(t, T)$. A bond, defined this way, measures the time value of money for default-free borrowing over the time period $[t, T]$. One Euro today is better than one Euro tomorrow and even better than one Euro next year. Thus, the *term structure of bond prices* $\{P(t, T) \mid T \geq t\}$ which we can observe at time t (today) is a (deterministic) non-increasing, positive curve with $P(t, t) = 1$. Whereas for fixed maturity T , $\{P(t, T) \mid t \in [0, T]\}$ is a stochastic process. It is stochastic because the economy, and thus the market beliefs about the future value of money, changes in time and is not known in advance. We remark that in reality the bond prices depend on additional factors, such as the possibility of default by the bond's issuer and the like. These matters are not our concern here and will be excluded throughout.

A more informative measure of the current bond market at time t is the term structure of interest rates or *forward curve* $\{f(t, T) \mid T \geq t\}$ given by

$$P(t, T) = \exp \left(- \int_t^T f(t, s) ds \right). \quad (1.1)$$

The function $f(t, \cdot)$ is a fortiori locally integrable (and $P(t, \cdot)$ in turn absolutely continuous). We call $f(t, T)$ the continuously compounded instantaneous *forward rate* for date T . It should be interpreted as the interest rate over the infinitesimal time interval $[T, T + dT]$ as seen from time t . It follows

from (1.1) that the forward curve contains all the original bond price information, which can be completely recovered. It is therefore a widely accepted (infinite dimensional) state variable of the bond market. Furthermore, it is a vital indicator for central banks in setting monetary policy. However, we remark that in contrast to bond prices, the forward curve is a mathematical idealization rather than a quantity observable in practice.

At the short end $T = t$ of the forward curve we find the *short rate*

$$R(t) := f(t, t).$$

It is the current interest rate for a loan over the infinitesimal interval $[t, t+dt]$ prevailing at time t . If the process $\{R(t) \mid t \in \mathbb{R}_+\}$ is nicely behaved (that is, progressively measurable and integrable with respect to t) one can define the *savings account* process

$$B(t) = \exp \left(\int_0^t R(s) ds \right), \quad t \in \mathbb{R}_+. \quad (1.2)$$

Intuitively, $B(t)$ represents the amount of cash accumulated up to time t starting with one unit at time 0 and continually reinvesting at the short rates $R(s)$, $s \in [0, t]$.

For the definition of an admissible self-financing trading strategy we refer e.g. to [47, Chap. 10]. An arbitrage opportunity is, roughly speaking, such a strategy which yields a risk-less profit. To illustrate things let us consider for the moment a deterministic arbitrage-free world. Here we have necessarily

$$P(t, T) = P(t, S)P(S, T), \quad \forall t \leq S \leq T. \quad (1.3)$$

Indeed, suppose that $P(t, T) > P(t, S)P(S, T)$ for some $t \leq S \leq T$. Then it is easy to see that selling one bond with maturity T and buying $P(S, T)$ bonds with maturity S at time t leads to a risk-less net gain at terminal time T : an arbitrage opportunity. If $P(t, T) < P(t, S)P(S, T)$ the same profit can be realized by changing signs in the strategy, whence (1.3). Notice that in terms of the forward rates (1.3) reads $f(t, T) = f(0, T)$, $\forall t \in [0, T]$.

The exact definition of arbitrage can be formulated in many different ways. Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}_+}, \mathbb{P})$ be a complete filtered probability space satisfying the usual conditions. Delbaen and Schachermayer [20] showed that, under the appropriate assumptions, a particular no-arbitrage condition (referred to as *no free lunch with vanishing risk*) is equivalent to the existence of a probability measure $\mathbb{Q} \sim \mathbb{P}$ under which discounted bond price processes $P(t, T)/B(t)$, $t \in [0, T]$, follow local martingales. In particular, they show that traded assets (in the present framework the bond price processes) have to be semimartingales. The measure $\mathbb{Q}$ is called an *equivalent local martingale measure (ELMM)* or *risk neutral measure*. By the absence of arbitrage we shall mean from now on the existence of such a measure $\mathbb{Q}$.

1.2 Forward Curve Fitting Methods and Factor Models

Since the forward curve cannot directly be observed on the market, it has to be estimated. Several algorithms for constructing the current term structure of interest rates from market data are applied in practice. Any common method can be represented as a parametrized family

$$\mathcal{G} = \{G(\cdot, z) \mid z \in \mathcal{Z}\}$$

of smooth curves where $\mathcal{Z} \subset \mathbb{R}^m$ is a finite dimensional parameter set, for some $m \in \mathbb{N}$. By appropriate choice of the parameter $z \in \mathcal{Z}$, an optimal fit of the forward curve $x \mapsto G(x, z)$ to the observed data is achieved. Here $x \geq 0$ denotes *time to maturity*. In that sense z represents the current state of the economy, taking values in the *state space* $\mathcal{Z}$.

Prominent examples are the Nelson–Siegel [48] family with curve shape

$$G_{NS}(x, z) = z_1 + (z_2 + z_3 x)e^{-z_4 x} \quad (1.4)$$

and the Svensson [53] family, an extension of Nelson–Siegel,

$$G_S(x, z) = z_1 + (z_2 + z_3 x)e^{-z_5 x} + z_4 x e^{-z_6 x}. \quad (1.5)$$

A recent document from the Bank for International Settlements (BIS [6]) provides an overview of all estimation methods used by some of the most important central banks, see Table 1.1.

Table 1.1. Forward rate curve fitting procedures

Central bank	Curve fitting procedure
Belgium	Nelson–Siegel, Svensson
Canada	Svensson
Finland	Nelson–Siegel
France	Nelson–Siegel, Svensson
Germany	Svensson
Italy	Nelson–Siegel
Japan	Smoothing splines
Norway	Svensson
Spain	Nelson–Siegel (before 1995), Svensson
Sweden	Svensson
UK	Svensson
USA	Smoothing splines

According to [6] a lot of cross-sectional data, that is, daily estimations of z , is available for selected estimation methods $\mathcal{G}$, such as the Nelson–Siegel and Svensson families. In addition there exists an extensive literature on statistical inference for diffusion processes based on discrete time observations,

see e.g. [5]. It therefore seems promising to explore the stochastic evolution of the parameter z continuously over time. Let $Z = (Z_t)_{t \in \mathbb{R}_+}$ be a diffusion model for the above parameter z . One would think that now

$$f(t, T) = G(T - t, Z_t), \quad t \leq T, \quad (1.6)$$

provides an accurate *factor model* for the interest rates. However, this is where the first consistency problems arise:

(P1) What are the conditions on Z (and G) such that (1.6) provides an arbitrage-free interest rate model?

Necessary and sufficient conditions for the absence of arbitrage can be formulated in terms of the generator of Z applied to G . These conditions turn out to be very restrictive. For fixed G we arrive at a type of *inverse problem* for the generator of Z . State space processes Z which comply with these constraints, are called *consistent* with $\mathcal{G}$. It can be shown for example that the Nelson–Siegel family admits no non-trivial consistent diffusion Z . We achieve these negative results even for generic state space Itô processes Z (thereby the inverse problem applies to the characteristics of Z instead of the generator as for diffusions).

Factor models (1.6) based on a diffusion process Z are very popular and have been exploited by many authors. New is our systematic study of the consistency problem (P1) for Z , which we investigate in a general Itô process framework.

1.3 The HJM Methodology

An application of Itô’s formula shows that the preceding factor models (1.6) are part of the framework introduced by Heath, Jarrow and Morton (HJM) [35]. For arbitrary but fixed $T \in \mathbb{R}_+$, HJM let $f(\cdot, T)$ evolve as an Itô process

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) + \int_0^t \sigma(s, T) dW_s, \quad t \in [0, T]. \quad (1.7)$$

Originally in [35], W is a finite dimensional Brownian motion. We will extend this approach at once by considering an *infinite dimensional* Brownian motion W . This generalization is straightforward but one has to take care of the definition of the stochastic integral. It is important to note that the HJM methodology basically unifies *all* continuous interest rate models (that is, models which are driven by a Brownian motion).

The no-arbitrage requirement cumulates in the well-known *HJM drift condition*: existence of an equivalent local martingale measure $\mathbb{Q}$ yields a representation for the $\mathbb{Q}$ -drift of $f(\cdot, T)$ in terms of σ (HJM let the filtration be

generated by W). Thus, pricing formula for interest rate sensitive contingent claims do only depend on σ . The above mentioned inverse problem for the characteristics of Z is simply this drift condition.

Note that (1.7) provides a generic description of the arbitrage-free evolution rather than a model for the interest rates. By HJM's result it is convenient and standard to describe the evolution of $f(\cdot, T)$ under the risk neutral measure $\mathbb{Q}$, involving only σ and the Girsanov transform $\tilde{W}$ of W .

Later, Musiela [46] proposed the reparametrization

$$r_t(x) = f(t, x + t), \quad (1.8)$$

taking better into account the nature of the forward curve $x \mapsto r_t(x)$ as an (infinite dimensional) state variable, where $x \geq 0$ denotes time to maturity. There are indeed several reasons for doing so.

Non-varying state space. The basic underlying state variable is the time t forward curve $\{f(t, T) \mid T \geq t\}$. In the HJM time framework, however, this means that the state space is a space of functions on an interval varying with t . The above modification eliminates this difficulty.

Non-local state dependence. Models within the classical HJM framework have state dependent volatility coefficients of the form $\sigma(t, T, \omega, f(t, \omega, T))$, see [35], [45] and [44]. Hence the dependence on the state variable is *local*. It would be natural if the volatility depended on the whole forward curve. Hence if also non-local (like LIBOR rate) dependence were allowed, incorporating all LIBOR or simple rate models in one framework. This aim is provided by considering r_t as a state variable in a function space, evolving according to functional sensitive dynamics. It allows in particular to enclose into our framework also existence results for short rate models, where the dependence is on $R(t) = f(t, t) = r_t(0)$, a linear functional evaluated at r_t .

Naive inversion is impossible. We have mentioned above that, by Itô's formula, any factor interest rate model of the form (1.6) where Z denotes an Itô process is part of the HJM framework. Now suppose conversely that we are given a family of HJM forward rate processes (1.7), a parametrized family $\mathcal{G}$ and *any* $\mathcal{Z}$ -valued process Z such that (1.6) holds. We encounter the following consistency problems:

(P2) Can one solve equation (1.6) for Z ?

(P3) If so, what type of process is Z ? Is it necessarily an Itô process?

To illustrate the problem let us consider the Nelson–Siegel family (1.4). A naive approach is to fix a tenor $0 < T_1 < \dots < T_5$, define the mapping

$$\tilde{G}(t, z) := (G_{NS}(T_1 - t, z), \dots, G_{NS}(T_5 - t, z)) : [0, T_1] \times \mathcal{Z} \rightarrow \mathbb{R}^5$$

and try to (locally) invert $\tilde{G}$, which yields

$$(t, Z_t) = \tilde{G}^{-1}(f(t, T_1), \dots, f(t, T_5)), \quad t \in (0, T_1).$$

The inverse mapping theorem requires non-singularity of the 5×5 -matrix

$$\begin{bmatrix} -\partial_x G_{NS}(T_1 - t, z) & \partial_{z_1} G_{NS}(T_1 - t, z) & \dots & \partial_{z_4} G_{NS}(T_1 - t, z) \\ \vdots & \vdots & & \vdots \\ -\partial_x G_{NS}(T_5 - t, z) & \partial_{z_1} G_{NS}(T_5 - t, z) & \dots & \partial_{z_4} G_{NS}(T_5 - t, z) \end{bmatrix}.$$

But this is impossible since

$$\begin{aligned} \partial_x G_{NS}(x, z) &= (-z_2 z_4 + z_3 - z_3 z_4 x) e^{-z_4 x} \\ \partial_{z_1} G_{NS}(x, z) &= 1 \\ \partial_{z_2} G_{NS}(x, z) &= e^{-z_4 x} \\ \partial_{z_3} G_{NS}(x, z) &= x e^{-z_4 x} \\ \partial_{z_4} G_{NS}(x, z) &= (-z_2 x - z_3 x^2) e^{-z_4 x} \end{aligned}$$

are linearly dependent functions in x . The problem originates from the appearance of $\partial_x G_{NS}(x, z)$ in the above matrix, which is due to the explicit t -dependence of $G_{NS}(T - t, Z_t)$. A way out is given by the new parametrization (1.8) since then (1.6) reads

$$G(x, Z_t) = r_t(x), \quad t, x \in \mathbb{R}_+. \quad (1.9)$$

This however causes a new difficulty: $(r_t(x))_{t \in \mathbb{R}_+}$ is not an Itô process in general. Problem **(P3)** remains, even if (1.9) can be inverted for any $x \in \mathbb{R}_+$. We encounter in addition:

(P4) Can Z be chosen independently of x ?

One of our principle objects will be the inversion of equation (1.6), or equivalently (1.9). We are thus interested in the implied stochastic dynamics for $(r_t(x))_{t \in \mathbb{R}_+}$. The semigroup theory for stochastic equations in a Hilbert space provided by Da Prato and Zabczyk [19] is tailor-made for that approach. We give an axiomatic scheme for the choice of an admissible Hilbert space H of functions $h : \mathbb{R}_+ \rightarrow \mathbb{R}$ in which $(r_t)_{t \in \mathbb{R}_+}$ can be realized as a solution to the stochastic equation

$$\begin{cases} dr_t = (Ar_t + \alpha_{HJM}(t, r_t)) dt + \sigma(t, r_t) d\tilde{W}_t \\ r_0 = h_0. \end{cases} \quad (1.10)$$

Here A is the generator of the semigroup of right-shifts $S(t)h(x) = h(x + t)$; that is, $Ah(x) = \partial_x h(x)$ for any function in the domain of A . The coefficients¹ $\sigma = \sigma(t, \omega, h)$ and $\alpha_{HJM} = \alpha_{HJM}(t, \omega, h)$ are random mappings. The latter is fully specified by σ according to the HJM drift condition under the measure $\mathbb{Q}$ (under $\mathbb{P}$ the market price of risk is also involved). Equation (1.10) with $h_0 = f(0, \cdot)$ represents the system of infinitely many Itô processes (1.7) as

¹ By a slight abuse of notation we use the same letter σ in (1.10) as well as (1.7), although they represent different objects.

one dynamical system in infinite dimensions, now allowing for arbitrary initial curves $r_0 \in H$.

The axiomatic exposure of an admissible Hilbert space of forward curves is new. We propose a particular class of admissible Hilbert spaces H_w which are both economically reasonable and appropriate for providing existence and uniqueness results for (1.10) in full generality. Each H_w is a type of weighted Sobolev space. Any Lipschitz continuous and bounded volatility coefficient σ provides an HJM model therein. Classical models are shown to fit well into that framework. These include the short rate models of Vasicek [56] and Cox, Ingersoll and Ross (CIR) [18], and also the popular LIBOR rate model by Brace, Gątarek and Musiela (BGM) [13].

Equation (1.10) was introduced and analyzed first in [46] and [14], but only for deterministic volatility structure. In that Gaussian framework Musiela also characterized the set of attainable forward curves by the model. This is an issue which is strongly related to the present discussion. Further aspects of equation (1.10) have been exploited by several authors. Among them are Björk et al. [8], [9], [11], Zabczyk [59], Vargiolu [55], [54], Kusuoka [40], Douady [22], Guiotto and Roncoroni [33], [34] and Milian [43]. Research is ongoing, see e.g. [31]. A rigorous exposure of the no-arbitrage property for equation (1.10) as provided by this monograph therefore seems to be needed.

1.4 Invariant Manifolds

In view of the previously, in Section 1.2, mentioned negative consistency results for common curve fitting methods, one can proceed in two possible ways. Either one considers more general non-continuous (Markovian) state space processes Z , or one stays within the continuous (that is HJM) regime and looks for appropriate forward curve families $\mathcal{G}$. The first approach is not a topic of this book and will be exploited elsewhere, see e.g. [25]. The second approach leads naturally to the complex of *consistency problems for HJM models* introduced by Björk and Christensen [8]. They propose the following geometric point of view which will eventually allow us to solve the problems (P2)–(P4).

Considering $\mathcal{G}$ as a subset of H , we can study the *stochastic invariance problem* related to (1.10) for $\mathcal{G}$. The set $\mathcal{G}$ is called *(locally) invariant* for (1.10) if for any space-time initial point $(t_0, h_0) \in \mathbb{R}_+ \times \mathcal{G}$, the solution $r = r^{(t_0, h_0)}$ of (1.10) stays (locally) in $\mathcal{G}$ almost surely. Accordingly, we call the HJM model provided by (1.10) *consistent* with $\mathcal{G}$.

By a solution to (1.10) we mean a *mild* or a *weak* solution rather than a *strong* solution. These three concepts will be defined below, see Definition 2.4.1. It is essential, however, to notice that a mild or a weak solution is *not* an H -valued semimartingale in general.

The existence of a state space diffusion process Z which is (locally) consistent with $\mathcal{G}$, for any initial point $(t_0, z_0) \in \mathbb{R}_+ \times \mathcal{Z}$, obviously implies

consistency of the induced HJM model $G(\cdot, Z)$ with $\mathcal{G}$. The converse is not true in general: if the model (1.10) is consistent with $\mathcal{G}$ it is not clear what the Z -valued *coordinate process* Z determined by (1.9) looks like. We cannot expect it to be an Itô process.

We provide a general regularity result for this problem. Suppose $\mathcal{G}$ is a finite dimensional regular submanifold of H and locally invariant for a stochastic equation like (1.10). Then the coordinate process Z in (1.9) is a Z -valued diffusion. Furthermore, we find Nagumo-type *consistency conditions* in terms of σ and α_{HJM} and the tangent bundle of $\mathcal{G}$. Expressing these consistency conditions in local coordinates we arrive at exactly the above inverse problem for the characteristics of the coordinate (i.e. state space) process Z . That way the consistency problem for $\mathcal{G}$ is completely solved.

These results are then applied to the estimation methods provided by [6]. Under some technical restrictions, we show that *exponential-polynomial families*, such as those of Nelson–Siegel or Svensson, are regular submanifolds of H_w . Thus, the previously derived negative consistency results can be restated within the present setup.

A further application yields the characterization of all HJM models which are consistent with linear submanifolds. These are just the *affine* HJM models. We derive the well-known results of Duffie and Kan [23] from our general point of view.

The present consistency results have been basically established by Björk and Christensen [8]. However, they proceeded under much stronger assumptions which are not necessarily convenient for the HJM framework, see Remark 4.5.1 below. Moreover, they did not address the difference between a regular and an immersed submanifold, see Figure 7.1 on page 115. By exploiting the strong structure of the regular submanifold $\mathcal{G}$ we derive *global* invariance results in contrast to [8] where only local results are provided. New is also the representation of the consistency conditions in local coordinates, making them feasible for applications. These work mainly in one direction: given a curve fitting method $\mathcal{G}$, we can determine the consistent HJM models. The converse problem of the existence of an invariant finite dimensional submanifold, given a particular HJM model, is treated in Björk and Svensson [11], by using the Frobenius theorem, see also [31]. The non-consistency of exponential-polynomial families has been proved in [8] for the particular case of a deterministic volatility structure.

Similar stochastic invariance problems have been studied by Zabczyk [59], Jachimiak [37] and, for finite dimensional systems, by Milian [42] (see also the references therein). Björk et al. [9], [11] discuss further the existence of (minimal) finite dimensional realizations for (1.10). In [59] the invariance question for finite dimensional linear subspaces with respect to Ornstein–Uhlenbeck processes is resolved. Applications to HJM models, the BGM model and to a second order term structure model by Cont [17] are exploited. But the support theorem methods in [59] are not yet established for the general equation

(1.10). Jachimiak [37] exploits invariance for closed sets $K \subset H$. However, his methods cannot be applied directly to (1.10).

1.5 Outline

In Chapter 2 we repeat the relevant material from [19] about stochastic analysis in infinite dimensions. We define an infinite dimensional Brownian motion W and show how it can be realized as a Hilbert space valued Wiener process. The construction of the stochastic integral with respect to W is then sketched, and we provide the main tools from stochastic analysis: Itô's formula, the stochastic Fubini theorem and Girsanov's theorem. We introduce stochastic equations in a Hilbert space, define the concepts for a solution and state an existence and uniqueness result.

In Chapter 3 we give the rigorous setup for the factor model (1.6). We prove a general result for the characterization of consistent state space Itô processes, which replies to problem (P1). This is then applied to the large class of exponential-polynomial families. We obtain remarkably restrictive results. In particular, we identify the only non-trivial factor model which is consistent with the Svensson family: an extended Vasicek short rate model. Chapter 3 was motivated by the examples given in [8] (see Remarks 5.3 and 7.1 therein).

Chapter 4 recaptures the HJM [35] methodology and extends it to incorporate an infinite dimensional driving Brownian motion W . We perform the change of parametrization (1.8) and arrive at the stochastic equation (1.10) in a Hilbert space H which is axiomatically scheduled. The no-arbitrage (or HJM drift) condition is rigorously exposed. We give a sketch for contingent claim valuation in the preceding bond market and discuss in particular the forward measure, forward LIBOR rates and caplets. This is introductory for the BGM model to be recaptured in the subsequent chapter. Finally, we define what we mean by an HJM model.

In Chapter 5 we introduce a class of Hilbert spaces H_w which are both economically reasonable and convenient for analyzing the previously derived stochastic equation (1.10). We provide a general existence and uniqueness result for Lipschitz continuous and bounded volatility coefficients σ . Classical models are shown to fit well into that framework. So does the classical HJM model, which corresponds to locally state dependent volatility coefficients σ . We also recapture the popular BGM model and provide further promising examples.

Chapter 6 is intermediary and not directly related to interest rate models. It starts with a comprehensive discussion on finite dimensional submanifolds in Banach spaces. Their crucial properties are deduced. Then we provide some general results on stochastic invariance for finite dimensional submanifolds in a Hilbert space. The main results of [8] are considerably extended.

Chapter 7 unifies the results obtained in the earlier chapters. We prove the main consistency result for regular families $\mathcal{G}$ and thus solve completely the problems (P2)–(P4). We discuss some pitfalls which arise from the subtle but important difference between a regular and an immersed submanifold. A simple criterion for differentiability in H_w is presented and applied to the class of exponential-polynomial families. We restate the (negative) consistency results for the common Nelson–Siegel and Svensson families. Finally, we apply our methods for identifying the class of affine term structure models which are characterized by possessing linear invariant submanifolds. We provide some of the well-known results from [23] and recapture the Vasicek and CIR short rate models.

For the convenience of the reader we summarize in Chapter 8 all the assumptions made in the text.

1.6 Remark on Notation

The reader is assumed to be familiar with the usual notation of stochastic analysis and probability theory, such as introduced in [49] and [19].

Let E and H be Banach spaces. The norm of H is denoted by $\|\cdot\|_H$ and we write $\mathcal{B}(H)$ for the σ -algebra generated by its topology. If H is a Hilbert space we use the notation $\langle \cdot, \cdot \rangle_H$, or simply $\langle \cdot, \cdot \rangle$ if there is no ambiguity, for its scalar product. The closure of a subset $U \subset H$ is denoted by $\overline{U}$, and we write $\mathcal{B}(U)$ for the σ -algebra $\mathcal{B}(H)$ restricted to U . We define the open ball of radius $R > 0$

$$B_R(H) := \{h \in H \mid \|h\|_H < R\}.$$

The space of continuous linear operators from E into H is denoted $L(E; H)$ and $L(H) = L(H; H)$ for short. We write $D(A)$ for the domain of an (unbounded) linear operator $A : E \rightarrow H$ and A^* for its adjoint.

We write $C(E; H)$ and $C^k(E; H)$ for the space of continuous and k times continuously differentiable mappings $\phi : E \rightarrow H$, respectively. The derivatives are denoted by $D\phi$ and $D^k\phi$. We write $C_b^k(E; H)$ for the space of all uniformly bounded mappings $\phi \in C^k(E; H)$ whose derivatives up to order k are uniformly bounded and $C_c^k(E; H)$ for those $\phi \in C^k(E; H)$ which have compact support in E . In all definitions we can replace E by any reasonable subset $U \subset E$. If $H = \mathbb{R}$ we write shortly $C(U)$ for $C(U; \mathbb{R})$, etc.

If $(\Lambda, \mathcal{H}, \mu)$ is a measure space and $p \geq 1$, we write $L^p(\Lambda, \mathcal{H}, \mu)$ for the Banach space of integrable functions² $X : \Lambda \rightarrow \mathbb{R}$ equipped with the norm

$$\|X\|_{L^p(\Lambda, \mathcal{H}, \mu)}^p = \int_{\Lambda} |X|^p d\mu.$$

² In fact, equivalence classes of functions where $X \sim Y$ if $X = Y$ μ -a.s.

If Λ is a Borel subset of $\mathbb{R}$ and μ the Lebesgue measure, we just write $L^p(\Lambda)$. The space of locally integrable functions³ $h : \Lambda \rightarrow \mathbb{R}$ is denoted by $L^1_{\text{loc}}(\Lambda)$.

Letters such as $C, \tilde{C}, K, \dots$ represent positive real constants which, if not otherwise stated, can vary from line to line.

All remaining notation is either standard and self-explanatory or it is introduced and explained in the text.

³ That is, equivalence classes ($h \sim g$ if $h = g$ a.s.) of measurable functions h satisfying

$$\int_{B_R(\mathbb{R}) \cap \Lambda} |h(x)| dx < \infty \quad \forall R \in \mathbb{R}_+.$$

2. Stochastic Equations in Infinite Dimensions

In this chapter we provide a brief introduction to infinite dimensional stochastic analysis and sketch the relevant material from Da Prato and Zabczyk [19] without giving detailed proofs.

There are substantial difficulties to define the stochastic integral of a Banach space valued process, even with respect to a scalar Brownian motion, see [58]. It is therefore convenient to work within a Hilbert space: throughout, H stands for a separable Hilbert space.

Henceforth we let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}_+}, \mathbb{P})$ denote a given complete filtered probability space satisfying the usual conditions; that is, $\mathcal{F}$ is $\mathbb{P}$ -complete, $(\mathcal{F}_t)$ is increasing and right continuous, $\mathcal{F}_0$ contains all $\mathbb{P}$ -nullsets of $\mathcal{F}$. For ease of notation we shall make the convention that $\mathcal{F} = \mathcal{F}_\infty := \bigvee_{t \in \mathbb{R}_+} \mathcal{F}_t$ (this is crucial when we apply Girsanov's theorem). The predictable σ -field is denoted by $\mathcal{P}$. We will suppress the notational dependence of stochastic variables on ω when no confusion can arise.

Let $I \subset \mathbb{R}_+$. A stochastic process $(t, \omega) \mapsto X_t(\omega) : I \times \Omega \rightarrow H$ will be written indifferently $(X_t)_{t \in I}$, X or (X_t) if there is no ambiguity about the index set I . Differing notations, such as $(B(t))_{t \in \mathbb{R}_+}$ or $(P(t, T))_{t \in [0, T]}$ are self-explanatory. Note that $X_t : \Omega \rightarrow H$ is a random variable for each $t \in I$. A process $(X_t)_{t \in I}$ is a *version* of $(Y_t)_{t \in I}$ if $X_t = Y_t$, $\mathbb{P}$ -a.s. for any $t \in I$. The two processes are *indistinguishable* if

$$\mathbb{P}[X_t = Y_t, \forall t \in I] = 1.$$

In the latter case we do not distinguish between them and write $X = Y$. If both X and Y have right-continuous paths, then $X = Y$ if and only if X is a version of Y .

2.1 Infinite Dimensional Brownian Motion

What is an H -valued “standard Brownian motion” W ? A obvious requirement would be that $\langle W, h \rangle_H$ were a real Brownian motion with

$$\mathbb{E}[\langle W_t, h \rangle_H \langle W_t, h' \rangle_H] = t \langle h, h' \rangle_H, \quad \forall h, h' \in H.$$

If $\dim H < \infty$ this certainly applies. But in general

$$\mathbb{E} [\|W_t\|_H^2] = \sum_{j \in \mathbb{N}} \mathbb{E} [|\langle W_t, h_j \rangle_H|^2] = \sum_{j \in \mathbb{N}} t = \infty,$$

where $\{h_j \mid j \in \mathbb{N}\}$ is an orthonormal basis in H . Whence W_t does not exist in H , see [57]. Usually one relaxes the condition on W_t to belonging to H . Yor [57] gives a comprehensive overview of the different concepts for defining W .

We will use a more direct approach here. For us, an *infinite dimensional Brownian motion* is a sequence

$$W = (\beta^j)_{j \in \mathbb{N}} \quad (2.1)$$

of independent real-valued standard $(\mathcal{F}_t)$ -Brownian motions. For being consistent with [19] we first show how W can be realized as a Hilbert space valued Wiener process.

For that purpose we introduce the usual Hilbert sequence space

$$\ell^2 := \left\{ v = (v_j)_{j \in \mathbb{N}} \in \mathbb{R}^{\mathbb{N}} \mid \|v\|_{\ell^2}^2 := \sum_{j \in \mathbb{N}} |v_j|^2 < \infty \right\}$$

and denote by $\{g_j \mid j \in \mathbb{N}\}$ the standard orthonormal basis¹ in ℓ^2 . As mentioned above, $W = (\beta^j)_{j \in \mathbb{N}} \equiv \sum_{j \in \mathbb{N}} \beta^j g_j$ does not exist in ℓ^2 . In the notation of [19, Section 4.3.1], W is a *cylindrical Wiener process* in ℓ^2 . It can be realized in a larger space as follows.

Let $\lambda = (\lambda_j)_{j \in \mathbb{N}}$ be a sequence of strictly positive numbers with

$$\sum_{j \in \mathbb{N}} \lambda_j < \infty.$$

We define the weighted sequence Hilbert space

$$\ell_\lambda^2 := \left\{ v = (v_j)_{j \in \mathbb{N}} \in \mathbb{R}^{\mathbb{N}} \mid \|v\|_{\ell_\lambda^2}^2 := \sum_{j \in \mathbb{N}} \lambda_j |v_j|^2 < \infty \right\}.$$

Clearly, $\ell^2 \subset \ell_\lambda^2$ with Hilbert-Schmidt embedding, and $e_j := (\lambda_j)^{-\frac{1}{2}} g_j$ form an orthonormal basis in ℓ_λ^2 . Define $Q \in L(\ell_\lambda^2)$ by $Qe_j := \lambda_j e_j$. Obviously, the operator Q is strictly positive, self-adjoint and nuclear. Moreover we have $Q^{\frac{1}{2}}(\ell_\lambda^2) = \ell^2$ and $Q^{-\frac{1}{2}} : \ell^2 \rightarrow \ell_\lambda^2$ is an isometry since for $u, v \in \ell^2$

$$\langle u, v \rangle_{\ell^2} = \sum_{j \in \mathbb{N}} u_j v_j = \sum_{j \in \mathbb{N}} \lambda_j \frac{u_j}{\sqrt{\lambda_j}} \frac{v_j}{\sqrt{\lambda_j}} = \langle Q^{-\frac{1}{2}} u, Q^{-\frac{1}{2}} v \rangle_{\ell_\lambda^2}. \quad (2.2)$$

Fix $T \in \mathbb{R}_+$.

Definition 2.1.1. We denote by $\mathcal{H}_T^2(H)$ the space of H -valued continuous martingales² M on $[0, T]$ such that

$$\|M\|_{\mathcal{H}_T^2(H)}^2 := \mathbb{E} [\|M_T\|_H^2] < \infty. \quad (2.3)$$

¹ That is, $g_1 = (1, 0, \dots)$, $g_2 = (0, 1, 0, \dots)$, ...

² see [19, Section 3.4]

Write $M_T^* := \sup_{t \in [0, T]} \|M_t\|_H$. Doob's inequality [19, Theorem 3.8] says

$$\mathbb{E} [|M_T^*|^2] \leq 4 \|M\|_{\mathcal{H}_T^2(H)}^2, \quad \text{for } M \in \mathcal{H}_T^2. \quad (2.4)$$

Using (2.4) it can be shown as in [19, Proposition 3.9] that $\mathcal{H}_T^2(H)$ is a Hilbert space for the norm (2.3). Denote by $\mathcal{H}_T^{0,2}(H)$ the closed subspace consisting of $M \in \mathcal{H}_T^2(H)$ with $M_0 = 0$.

We adopt the terminology of [19, Section 4.1].

Definition 2.1.2. An ℓ_λ^2 -valued continuous martingale X is called a Q -Wiener process if

- i) $X_0 = 0$
- ii) X has independent increments
- iii) $X_t - X_s$ is centered Gaussian³ with covariance operator $(t - s)Q$, for all $0 \leq s \leq t$.

Proposition 2.1.1. The series

$$W = \sum_{j \in \mathbb{N}} \beta^j g_j$$

converges in $\mathcal{H}_T^{0,2}(\ell_\lambda^2)$ for all $T \in \mathbb{R}_+$ and defines a Q -Wiener process in ℓ_λ^2 .

Observe that this realization of W as a Hilbert space valued Wiener process depends on the choice of λ . Indeed, the same procedure applies for any given sequence λ sharing the above properties. Yet the class of integrable processes does not depend on λ as we shall see.

Proof. Fix $T \in \mathbb{R}_+$. We define $W^n \in \mathcal{H}_T^{0,2}(\ell_\lambda^2)$ by

$$W^n = \sum_{j=1}^n \beta^j g_j, \quad n \in \mathbb{N}. \quad (2.5)$$

Obviously, $\mathbb{E} [\|W_T^n - W_T^m\|_{\ell_\lambda^2}^2] = T \sum_{j=m+1}^n \lambda_j \rightarrow 0$ for $m, n \rightarrow \infty$. Hence $(W^n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathcal{H}_T^{0,2}(\ell_\lambda^2)$, converging to W . Since this holds for any $T \in \mathbb{R}_+$, we get that W is an ℓ_λ^2 -valued continuous martingale.

For any $v \in \ell_\lambda^2$ the $\mathbb{R}$ -valued random variables $\langle W_t^n - W_s^n, v \rangle_{\ell_\lambda^2}$ are centered Gaussian, converging in $L^2(\Omega, \mathcal{F}, \mathbb{P})$ to $\langle W_t - W_s, v \rangle_{\ell_\lambda^2}$ which is therefore centered Gaussian too. By the same reasoning

$$\mathbb{E} [\langle W_t - W_s, e_i \rangle_{\ell_\lambda^2} \langle W_t - W_s, e_j \rangle_{\ell_\lambda^2}] = (t - s) \langle Q e_i, e_j \rangle_{\ell_\lambda^2}.$$

Whence Condition iii) of Definition 2.1.2 holds. Similarly one shows ii) and clearly $W_0 = 0$. $\square$

Remark 2.1.1. We notice that any Hilbert space valued (cylindrical) Wiener process in the terminology of Da Prato and Zabczyk [19] can be considered in the preceding $\mathbb{R}^N$ -framework (2.1), see [19, Propositions 4.1 and 4.11].

³ see [19, Section 2.3.2]

2.2 The Stochastic Integral

For the definition of the stochastic integral with respect to W , Da Prato and Zabczyk [19] introduce the space $L_2^0(H)$ of Hilbert Schmidt operators from ℓ^2 into H .⁴ It consists of linear mappings $\Phi: \ell^2 \rightarrow H$ with

$$\|\Phi\|_{L_2^0(H)}^2 := \sum_{j \in \mathbb{N}} \|\Phi^j\|_H^2 < \infty, \quad (2.6)$$

where $\Phi^j := \Phi(g_j)$. In the sequel we shall identify Φ with $(\Phi^j)_{j \in \mathbb{N}}$. On the other hand, observe that any sequence $(\Phi^j)_{j \in \mathbb{N}}$ of H -valued predictable processes satisfying (2.6) provides an $L_2^0(H)$ -valued predictable process Φ by $\Phi(g_j) := \Phi^j$. Here we have used that $\mathcal{B}(L_2^0(H)) = \bigotimes_{\mathbb{N}} \mathcal{B}(H)$, which follows by the separability of H .

Notice the particular case $L_2^0(\mathbb{R}) = \ell^2$.

We now summarize the construction of the stochastic integral, based on the theory from [19, Sections 4.1-4.4]. Fix $T \in \mathbb{R}_+$.

Definition 2.2.1. *We call $\mathcal{L}_T^2(H)$ the Hilbert space of equivalence classes of $L_2^0(H)$ -valued predictable processes Φ with norm*

$$\|\Phi\|_{\mathcal{L}_T^2(H)}^2 := \mathbb{E} \left[\int_0^T \|\Phi_t\|_{L_2^0(H)}^2 dt \right]. \quad (2.7)$$

Then there exists an isometry $\Phi \mapsto \Phi \cdot W$ between $\mathcal{L}_T^2(H)$ and $\mathcal{H}_T^{0,2}(H)$, see [19, Proposition 4.5], defined on the dense⁵ subset of elementary $L(\ell_\lambda^2; H)$ -valued⁶ processes

$$\Phi = \sum_{m=0}^{k-1} \Phi_m 1_{(t_m, t_{m+1}]}, \quad \Phi_m \text{ is } \mathcal{F}_{t_m} \text{-measurable}, \quad (2.8)$$

by

$$(\Phi \cdot W)_t := \sum_{m=0}^{k-1} \Phi_m (W_{t_{m+1} \wedge t} - W_{t_m \wedge t}), \quad t \in [0, T].$$

Definition 2.2.2. *The H -valued continuous martingale $(\Phi \cdot W)_{t \in [0, T]}$ is called the stochastic integral of Φ with respect to W and is also denoted by*

$$(\Phi \cdot W)_t = \int_0^t \Phi_s dW_s.$$

⁴ Remember that $\ell^2 = Q^{\frac{1}{2}}(\ell_\lambda^2)$ endowed with the implied scalar product, see (2.2) and compare with [19, (4.7)].

⁵ see [19, Proposition 4.7]

⁶ for an arbitrary choice of $\lambda = (\lambda_j)_{j \in \mathbb{N}}$ as in Section 2.1

It is clear, how stochastic integration with respect to the d -dimensional Brownian motion $W^d = (\beta^1, \dots, \beta^d)$, $d \in \mathbb{N}$, is included in this concept: any H^d -valued predictable process $\Phi = (\Phi^1, \dots, \Phi^d)$ can be considered as $L_2^0(H)$ -valued predictable process $\tilde{\Phi}$ by setting $\tilde{\Phi}^j = \Phi^j$ if $j \leq d$ and $\tilde{\Phi}^j \equiv 0$ for $j > d$. This way one gets easily

$$(\Phi \cdot W^d)_t = (\tilde{\Phi} \cdot W)_t = \sum_{j=1}^d \int_0^t \Phi_s^j d\beta_s^j. \quad (2.9)$$

Here are some elementary properties of the stochastic integral. Denote by E a separable Hilbert space.

Proposition 2.2.1. *Let $\Phi \in \mathcal{L}_T^2(H)$.*

i) *For any stopping time $\tau \leq T$*

$$(\Phi \cdot W)_{t \wedge \tau} = ((\Phi 1_{[0, \tau]}) \cdot W)_t. \quad (2.10)$$

ii) *The following series converges uniformly on $[0, T]$ in probability*

$$(\Phi \cdot W)_t = \sum_{j \in \mathbb{N}} \int_0^t \Phi_s^j d\beta_s^j. \quad (2.11)$$

iii) *Let $A \in L(H; E)$. Then $A \circ \Phi \in \mathcal{L}_T^2(E)$ and*

$$A(\Phi \cdot W) = (A \circ \Phi) \cdot W. \quad (2.12)$$

Proof. Property i) is [19, Lemma 4.9]. Property ii) follows by considering (2.9) and $(\Phi^1, \dots, \Phi^d, 0, \dots)$ which converges in $\mathcal{L}_T^2(H)$ towards Φ for $d \rightarrow \infty$. Now apply Doob's inequality (2.4). Finally, iii) can be shown in the same way as [19, Proposition 4.15].

Notice that all equalities hold in $\mathcal{H}_T^{0,2}(H)$, hence up to indistinguishability. $\square$

By the approximation (2.11) of the stochastic integral many definitions and theorems from the finite dimensional stochastic analysis can be carried over to our infinite dimensional setting. An example is the following

Proposition 2.2.2. *Let $\Phi, \Psi \in \mathcal{L}_T^2(\mathbb{R})$ and define*

$$Z_t := \sum_{j \in \mathbb{N}} \int_0^t \Phi_s^j \Psi_s^j ds.$$

Then $(\Phi \cdot W)(\Psi \cdot W) - Z$ is a continuous local martingale. Moreover, Z is the unique real-valued continuous finite variation process with this property. We call Z the bracket process of $\Phi \cdot W$ and $\Psi \cdot W$, and we write

$$Z_t =: \langle (\Phi \cdot W), (\Psi \cdot W) \rangle_t.$$

Proof. First observe that Z is a well-defined $\mathbb{R}$ -valued continuous predictable process of finite variation. Write $X := (\Phi \cdot W)(\Psi \cdot W) - Z$, which is an $\mathbb{R}$ -valued continuous predictable process. We approximate the stochastic integrals by $M^d := \Phi \cdot W^d$ and $N^d := \Psi \cdot W^d$, see (2.9), which converge in $\mathcal{H}_T^{0,2}(\mathbb{R})$. We have from the finite dimensional theory that $X^d := M^d N^d - \sum_{j=1}^d \int_0^t \Phi_s^j \Psi_s^j ds$ is a martingale. But $X_t^d \rightarrow X_t$ in $L^1(\Omega, \mathcal{F}, \mathbb{P})$ for any fixed $t \in [0, T]$. Hence X is a martingale. The uniqueness of Z follows by [49, Proposition (1.2), Chap. IV]. $\square$

There is a larger class of integrable processes than $\mathcal{L}_T^2(H)$.

Definition 2.2.3. We call $\mathcal{L}_T^{\text{loc}}(H)$ the space of all $L_2^0(H)$ -valued predictable processes Φ such that

$$\mathbb{P} \left[\int_0^T \|\Phi_t\|_{L_2^0(H)}^2 dt < \infty \right] = 1.$$

Observe that for any $\Phi \in \mathcal{L}_T^{\text{loc}}(H)$ there exists an increasing sequence of stopping times $\tau_n \uparrow T$ such that $\Phi 1_{[0, \tau_n]}$ is in $\mathcal{L}_T^2(H)$ for all $n \in \mathbb{N}$. This allows us to extend the notion of the stochastic integral, see [19, p. 94–96].

Proposition 2.2.3. Let $\Phi \in \mathcal{L}_T^{\text{loc}}(H)$. Then there exists a unique H -valued continuous local martingale $(M_t)_{t \in [0, T]}$ characterized by

$$M_{t \wedge \tau} = ((\Phi 1_{[0, \tau]}) \cdot W)_t$$

whenever $\Phi 1_{[0, \tau]} \in \mathcal{L}_T^2(H)$. Again, M is called the stochastic integral of Φ with respect to W and it is written

$$M_t =: (\Phi \cdot W)_t = \int_0^t \Phi_s dW_s.$$

Many properties of the stochastic integral for $\mathcal{L}_T^2(H)$ -integrands carry over for $\mathcal{L}_T^{\text{loc}}(H)$ -integrands by *localization*. Here is one feature of this method.

Lemma 2.2.1. Let $(M^n)_{n \in \mathbb{N}}$ be a sequence of H -valued continuous local martingales. Assume that for all $\delta > 0$ there exists a stopping time τ with $\mathbb{P}[\tau < T] \leq \delta$ such that $(M_{t \wedge \tau}^n) \rightarrow 0$ in $\mathcal{H}_T^2(H)$. Then $\sup_{t \in [0, T]} \|M_t^n\|_H \rightarrow 0$ in probability.

Proof. Let $\delta, \varepsilon > 0$ and τ be as in the lemma. Then

$$\mathbb{P} \left[\sup_{t \in [0, T]} \|M_t^n\|_H > \varepsilon \right] \leq \delta + \mathbb{P} \left[\sup_{t \in [0, T]} \|M_{t \wedge \tau}^n\|_H > \varepsilon \right]$$

and the last term goes to zero as $n \rightarrow \infty$ by Doob's inequality (2.4). $\square$

Using Lemma 2.2.1 the following can be shown.

Proposition 2.2.4. *Equality (2.9) and Propositions 2.2.1–2.2.2 remain valid if $\mathcal{L}_T^2(H)$ is replaced by $\mathcal{L}_T^{\text{loc}}(H)$.*

Up to now, the stochastic integral was defined only on the finite time interval $[0, T]$. But we can consider integrands in $\mathcal{L}^2(H) := \mathcal{L}_\infty^2(H)$ (with T in (2.7) replaced by ∞) and $\mathcal{L}^{\text{loc}}(H) := \bigcap_{T \in \mathbb{R}_+} \mathcal{L}_T^{\text{loc}}(H)$, respectively.

Let $\Phi \in \mathcal{L}^2(H)$. Write $M^n := \Phi 1_{[0, n]} \cdot W$ for the stochastic integral on $[0, n]$. By (2.10), M^{n+1} coincides with M^n on $[0, n]$ for all $n \in \mathbb{N}$. Therefore we can define unambiguously a process $\Phi \cdot W$ – the *stochastic integral* of Φ with respect to W – by stipulating that it is equal to M^n on $[0, n]$. This process is obviously an H -valued continuous martingale.

By localization the same procedure applies for $\Phi \in \mathcal{L}^{\text{loc}}(H)$, and $\Phi \cdot W$ is an H -valued continuous local martingale.

There is now a straightforward extension of the notion of an $\mathbb{R}^m$ -valued Itô process, see e.g. [49].

Definition 2.2.4. *An H -valued continuous adapted process X is called an Itô process if it is of the form*

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \Phi_s dW_s, \quad (2.13)$$

where $\Phi \in \mathcal{L}^{\text{loc}}(H)$, b is an H -valued predictable process such that

$$\mathbb{P} \left[\int_0^t \|b_s\|_H ds < \infty \right] = 1, \quad \forall t \in \mathbb{R}_+, \quad (2.14)$$

and X_0 is $\mathcal{F}_0$ -measurable.

We notice that an Itô process, being the sum of an $\mathcal{F}_0$ -measurable random variable, a finite variation process and a local martingale, belongs to the important class of *semimartingales*. Moreover, that decomposition is unique.

2.3 Fundamental Tools

Three fundamental theorems of stochastic analysis are *Itô's formula*, the *stochastic Fubini theorem* and *Girsanov's theorem*. This section provides them.

2.3.1 Itô's Formula

Let E denote a separable Hilbert space. A slight extension of [19, Theorem 4.17] leads to the following version of Itô's formula.

Theorem 2.3.1. *Suppose that X is an H -valued Itô process given by (2.13) and let $\Psi \in C_b^2(H; E)$. Then $(D\Psi(X)\Phi^j)_{j \in \mathbb{N}} \in \mathcal{L}^{loc}(E)$ and*

$$J_t := \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\Psi(X_t)(\Phi_t^j, \Phi_t^j)$$

forms an E -valued predictable process such that

$$\mathbb{P} \left[\int_0^t (\|D\Psi(X_s)b_s\|_H + \|J_s\|_H) ds < \infty \right] = 1, \quad \forall t \in \mathbb{R}_+. \quad (2.15)$$

Moreover, $\Psi \circ X$ is an E -valued Itô process given by

$$(\Psi \circ X)_t = (\Psi \circ X)_0 + \int_0^t (D\Psi(X_s)b_s + J_s) ds + \sum_{j \in \mathbb{N}} \int_0^t D\Psi(X_s)\Phi_s^j d\beta_s^j. \quad (2.16)$$

Proof. The statement on $(D\Psi(X)\Phi^j)_{j \in \mathbb{N}}$ holds by inspection. Pointwise convergence of the series defining J_t follows from (2.6). Hence (J_t) is predictable. The integrability property (2.15) follows by (2.14) together with the fact that $\Phi \in \mathcal{L}^{loc}(H)$. Denote the right hand side of (2.16) with I_t . We have shown that (I_t) is an E -valued Itô process. Choose an orthonormal basis $\{e_i \mid i \in \mathbb{N}\}$ in E . It is enough to show equality (2.16) componentwise with respect to each e_i . Define $\Psi_i(h) := \langle e_i, \Psi(h) \rangle$. Then $\Psi_i \in C_b^2(H; \mathbb{R})$ and [19, Theorem 4.17] applies. It follows that

$$\begin{aligned} \langle e_i, (\Psi \circ X)_t \rangle &= (\Psi_i \circ X)_0 + \int_0^t \left(D\Psi_i(X_s)b_s + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\Psi_i(X_s)(\Phi_s^j, \Phi_s^j) \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t D\Psi_i(X_s)\Phi_s^j d\beta_s^j \\ &= \langle e_i, I_t \rangle, \end{aligned}$$

which yields the assertion. □

2.3.2 The Stochastic Fubini Theorem

Let $(E, \mathcal{E})$ be a measurable space and let

$$\Phi : (t, \omega, x) \mapsto \Phi_t(\omega, x)$$

be a measurable mapping from $(\mathbb{R}_+ \times \Omega \times E, \mathcal{P} \otimes \mathcal{E})$ into $(L_2^0(H), \mathcal{B}(L_2^0(H)))$. In addition let μ denote a finite positive measure on $(E, \mathcal{E})$.

Fix $T > 0$. By localization we get the following version of the stochastic Fubini theorem [19, Theorem 4.18].

Theorem 2.3.2. *Assume that*

$$\int_0^T \int_E \|\Phi_t(x)\|_{L_2^0(H)}^2 \mu(dx) dt < \infty, \quad \mathbb{P}\text{-a.s.}$$

Then there exists an $\mathcal{F}_T \otimes \mathcal{E}$ -measurable version $\xi(\omega, x)$ of the stochastic integral $\int_0^T \Phi_t(x) dW_t$ which is μ -integrable $\mathbb{P}$ -a.s. such that

$$\int_E \xi(x) \mu(dx) = \int_0^T \left(\int_E \Phi_t(x) \mu(dx) \right) dW_t, \quad \mathbb{P}\text{-a.s.}$$

2.3.3 Girsanov's Theorem

Let $\gamma = (\gamma^j)_{j \in \mathbb{N}} \in \mathcal{L}^{\text{loc}}(\mathbb{R})$ and recall that $L_2^0(\mathbb{R}) = \ell^2$. We define the *stochastic exponential* of $\gamma \cdot W$ by

$$\mathcal{E}(\gamma \cdot W)_t := \exp \left((\gamma \cdot W)_t - \frac{1}{2} \int_0^t \|\gamma_s\|_{\ell^2}^2 ds \right). \quad (2.17)$$

By [19, Lemma 10.15] there exists a real-valued standard $(\mathcal{F}_t)$ -Brownian motion β^0 such that $(\gamma \cdot W)_t = \int_0^t \|\gamma_s\|_{\ell^2} d\beta_s^0$. This way we are lead to the real-valued theory.

Lemma 2.3.1. *The stochastic exponential $\mathcal{E}(\gamma \cdot W)$ is a non-negative local martingale. It is a martingale if and only if $\mathbb{E}[\mathcal{E}(\gamma \cdot W)_t] = 1$ for all $t \in \mathbb{R}_+$.*

Proof. By Itô's formula

$$\mathcal{E}(\gamma \cdot W)_t = 1 + \int_0^t (\mathcal{E}(\gamma \cdot W)_s \|\gamma_s\|_{\ell^2}^2) d\beta_s^0.$$

Whence the first assertion. For the second statement we refer the reader to [49, Remark 3, p. 141]. $\square$

We recall [49, Proposition (1.15), Chap. VIII].

Lemma 2.3.2 (Novikov's criterion). *If*

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_{\mathbb{R}_+} \|\gamma_t\|_{\ell^2}^2 dt \right) \right] < \infty, \quad (2.18)$$

then $\mathcal{E}(\gamma \cdot W)$ is a uniformly integrable martingale. Accordingly, $\mathcal{E}(\gamma \cdot W)_t \rightarrow \mathcal{E}(\gamma \cdot W)_\infty$ in $L^1(\Omega, \mathcal{F}, \mathbb{P})$ for $t \rightarrow \infty$ and $\mathcal{E}(\gamma \cdot W)_\infty > 0$, $\mathbb{P}$ -a.s.

The last statement is a direct consequence of the fact that $\mathcal{E}(\gamma \cdot W)_\infty > 0$, $\mathbb{P}$ -a.s. on the set $\{\int_{\mathbb{R}_+} \|\gamma_t\|_{\ell^2}^2 dt < \infty\}$, see [49, Proposition (1.26), Chap. IV].

Let $T \in \mathbb{R}_+$. The next theorem will assure the existence of an equivalent local martingale measure.

Theorem 2.3.3 (Girsanov's Theorem). *Let $\gamma = (\gamma^j)_{j \in \mathbb{N}} \in \mathcal{L}_T^{\text{loc}}(\mathbb{R})$ satisfy*

$$\mathbb{E}[\mathcal{E}(\gamma \cdot W)_T] = 1.$$

Then

$$\tilde{\beta}_t^j := \beta_t^j - \int_0^t \gamma_s^j ds, \quad t \in [0, T], \quad j \in \mathbb{N}$$

are independent real-valued standard $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motions with respect to the measure $\tilde{\mathbb{P}}_T \sim \mathbb{P}$ on $\mathcal{F}_T$ given by

$$\frac{d\tilde{\mathbb{P}}_T}{d\mathbb{P}} = \mathcal{E}(\gamma \cdot W)_T.$$

Proof. See [19, Theorem 10.14]. □

We can now define, as above, the spaces $\tilde{\mathcal{H}}_T^2(H)$, $\tilde{\mathcal{H}}_T^{0,2}(H)$, $\tilde{\mathcal{L}}_T^2(H)$ and $\tilde{\mathcal{L}}_T^{\text{loc}}(H)$ on the filtered probability space $(\Omega, \mathcal{F}_T, (\mathcal{F}_t)_{t \in [0, T]}, \tilde{\mathbb{P}}_T)$. Obviously we have $\tilde{\mathcal{L}}_T^{\text{loc}}(H) = \mathcal{L}_T^{\text{loc}}(H)$. Notice that a sequence of $\mathcal{F}_T$ -measurable random variables converging in probability $\mathbb{P}$ also converges in probability $\tilde{\mathbb{P}}_T$ and vice versa.

Proposition 2.3.1. *The series*

$$\tilde{W} = \sum_{j \in \mathbb{N}} \tilde{\beta}^j g_j$$

converges in $\tilde{\mathcal{H}}_T^{0,2}(\ell_\lambda^2)$ and defines a Q -Wiener process in ℓ_λ^2 with respect to $\tilde{\mathbb{P}}_T$. Moreover, for any $\Phi \in \tilde{\mathcal{L}}_T^{\text{loc}}(H)$ we have

$$(\Phi \cdot \tilde{W})_t = (\Phi \cdot W)_t - \int_0^t \Phi_s(\gamma_s) ds, \quad \forall t \in [0, T], \quad \mathbb{P}\text{-a.s.} \quad (2.19)$$

Proof. For $n \in \mathbb{N}$ we define $\tilde{W}^n := \sum_{j=1}^n \tilde{\beta}^j g_j$. By Proposition 2.1.1, the sequence $(\tilde{W}^n)_{n \in \mathbb{N}}$ converges to $\tilde{W}$ in $\tilde{\mathcal{H}}_T^{0,2}(H)$ and $\tilde{W}$ is a Q -Wiener process in ℓ_λ^2 for the measure $\tilde{\mathbb{P}}_T$. In view of (2.5) and Theorem 2.3.3

$$\tilde{W}_t^n = W_t^n - \sum_{j=1}^n \int_0^t \gamma_s^j g_j ds, \quad \forall n \in \mathbb{N}.$$

By Doob's inequality (2.4) and Proposition 2.1.1, the right hand side converges to $W_t - \sum_{j \in \mathbb{N}} \int_0^t \gamma_s^j g_j ds$ uniformly on $[0, T]$ in probability, which is therefore indistinguishable from $\tilde{W}$ on $[0, T]$.

Equality (2.19) is certainly true for elementary $L(\ell_\lambda^2; H)$ -valued processes, recall (2.8). Now let $\Phi \in \tilde{\mathcal{L}}_T^{\text{loc}}(H) = \mathcal{L}_T^{\text{loc}}(H)$. We define the stopping times

$$\tau_n := T \wedge \inf \left\{ t \in \mathbb{R}_+ \mid \int_0^t \|\Phi_s\|_{L^2_0(H)}^2 ds \geq n \right\}, \quad n \in \mathbb{N}.$$

Then $\tau_n \uparrow T$ and $\Phi^n := \Phi 1_{[0, \tau_n]} \in \tilde{\mathcal{L}}_T^2(H) \cap \mathcal{L}_T^2(H)$. By Lemma 2.3.3 below, (2.19) holds for each Φ^n . By (2.10) therefore also for Φ . $\square$

Lemma 2.3.3. *Let $\Phi \in \tilde{\mathcal{L}}_T^2(H) \cap \mathcal{L}_T^2(H)$. Then there exists a sequence $(\Phi^n)_{n \in \mathbb{N}}$ of elementary $L(\ell_\gamma^2; H)$ -valued processes converging to Φ in $\tilde{\mathcal{L}}_T^2(H)$ and in $\mathcal{L}_T^2(H)$ simultaneously.*

Proof. Standard measure theory. $\square$

2.4 Stochastic Equations

This section provides the basic concepts and results for stochastic equations in infinite dimensions. Let $W = (\beta^j)_{j \in \mathbb{N}}$ be an infinite dimensional Brownian motion as introduced in Section 2.1. Our standing set of ingredients is the following:

- A strongly continuous semigroup $\{S(t) \mid t \in \mathbb{R}_+\}$ on H with infinitesimal generator A .
- Two measurable mappings $F(t, \omega, h)$ and $B(t, \omega, h) = (B^j(t, \omega, h))_{j \in \mathbb{N}}$ from $(\mathbb{R}_+ \times \Omega \times H, \mathcal{P} \otimes \mathcal{B}(H))$ into $(H, \mathcal{B}(H))$ and $(L_2^0(H), \mathcal{B}(L_2^0(H)))$, respectively.
- A non-random initial value $h_0 \in H$.

2.4.1 Mild, Weak and Strong Solutions

We shall assign a meaning to the *stochastic equation* in H

$$\begin{cases} dX_t = (AX_t + F(t, X_t)) dt + B(t, X_t) dW_t \\ X_0 = h_0. \end{cases} \quad (2.20)$$

In view of (2.11) this can be rewritten equivalently

$$\begin{cases} dX_t = (AX_t + F(t, X_t)) dt + \sum_{j \in \mathbb{N}} B^j(t, X_t) d\beta_t^j \\ X_0 = h_0. \end{cases} \quad (2.20')$$

Remark 2.4.1. Notice that F and B are random mappings and may depend explicitly on ω . In accordance with the notation of [19], however, we abbreviate $F(t, \omega, X_t(\omega))$ and $B(t, \omega, X_t(\omega))$ to $F(t, X_t)$ and $B(t, X_t)$, respectively.

Before we define the concept of a solution let us recall the following facts.

Lemma 2.4.1. *Let $\Phi \in \mathcal{L}^{loc}(H)$, then the stochastic convolution*

$$\sum_{j \in \mathbb{N}} \int_0^t S(t-s) \Phi_s^j d\beta_s^j,$$

which is defined for any $t \in \mathbb{R}_+$, has a predictable version.

Proof. See [19, Proposition 6.2]. □

Lemma 2.4.2. *Let X be an H -valued predictable process. Then the random set $\{X \in D(A)\}$ and the process*

$$Y_t(\omega) := \begin{cases} AX_t(\omega), & \text{if } X_t(\omega) \in D(A) \\ 0, & \text{else,} \end{cases} \quad (2.21)$$

are predictable.

Proof. We only give a sketch. Denote by A_n the Yosida approximations of A , see [19, Section A.2]. We have that $X \in D(A)$ if and only if $(A_n X)_{n \in \mathbb{N}}$ has a limit in H , which then equals AX . But since H is separable and complete we know that the set where $\lim A_n X$ exists in H , and thus the process Y , are predictable. □

Definition 2.4.1. *Suppose that X is an H -valued predictable process and $\tau > 0$ a stopping time satisfying*

$$\mathbb{P} \left[\int_0^{t \wedge \tau} \left(\|X_s\|_H + \|F(s, X_s)\|_H + \|B(s, X_s)\|_{L_2^0(H)}^2 \right) ds < \infty \right] = 1,$$

for all $t \in \mathbb{R}_+$. We call X

i) a local mild solution to (2.20), if the stopped variation of constants formula holds

$$\begin{aligned} X_t &= S(t \wedge \tau) h_0 + \int_0^{t \wedge \tau} S((t \wedge \tau) - s) F(s, X_s) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^{t \wedge \tau} S((t \wedge \tau) - s) B^j(s, X_s) d\beta_s^j, \quad \mathbb{P}\text{-a.s.} \quad \forall t \in \mathbb{R}_+. \end{aligned}$$

ii) a local weak solution to (2.20), if for arbitrary $\zeta \in D(A^)$*

$$\begin{aligned} \langle \zeta, X_t \rangle_H &= \langle \zeta, h_0 \rangle_H + \int_0^{t \wedge \tau} \left(\langle A^* \zeta, X_s \rangle_H + \langle \zeta, F(s, X_s) \rangle_H \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^{t \wedge \tau} \langle \zeta, B^j(s, X_s) \rangle_H d\beta_s^j, \quad \mathbb{P}\text{-a.s.} \quad \forall t \in \mathbb{R}_+. \end{aligned} \quad (2.22)$$

iii) a local strong solution to (2.20), if $X \in D(A)$, $dt \otimes d\mathbb{P}$ -a.s.

$$\mathbb{P} \left[\int_0^{t \wedge \tau} \|AX_s\|_H ds < \infty \right] = 1, \quad \forall t \in \mathbb{R}_+ \quad (2.23)$$

and the integral version

$$X_t = h_0 + \int_0^{t \wedge \tau} (AX_s + F(s, X_s)) ds + \int_0^{t \wedge \tau} B(s, X_s) dW_s \quad (2.24)$$

of (2.20) holds $\mathbb{P}$ -a.s. $\forall t \in \mathbb{R}_+$. By a slight abuse of notation we have written AX for the predictable process given by (2.21).

We call τ the lifetime of X . If $\tau = \infty$ we just refer to X as mild, weak or strong solution, respectively.

Definition 2.4.1 may be straightly extended for random $\mathcal{F}_0$ -measurable initial values h_0 . Notice that the lifetime τ is by no means maximal.

A (local) strong solution X perforce is a version of an H -valued Itô process. But a (local) weak solution has not even a continuous version in general.

Remark 2.4.2. For stochastic equations in finite dimensions one usually distinguishes between strong and weak solutions, see e.g. [39, Chap. 5]. There, however, *weak* has a different meaning than in the present context. Here we follow the terminology of [19].

The concepts in Definition 2.4.1 are related by

$$iii) \Rightarrow ii) \Rightarrow i),$$

and $i) \Rightarrow ii)$ if $B(\cdot, X) \in \mathcal{L}^2(H)$, see [19, Theorem 6.5].

Since the coefficients F and B are not homogeneous in time, we will frequently consider the time shifted version of (2.20). Let τ_0 denote a bounded stopping time. Then $(\mathcal{F}_t^{(\tau_0)}) := (\mathcal{F}_{\tau_0+t})$ is a filtration satisfying the usual conditions. The following result is classic and can be proved by Lévy's characterization theorem, see [49, Theorem (3.6), Chap. IV].

Lemma 2.4.3. *The processes*

$$\beta_t^{(\tau_0),j} := \beta_{\tau_0+t}^j - \beta_{\tau_0}^j, \quad t \in \mathbb{R}_+, \quad j \in \mathbb{N}$$

are independent standard $(\mathcal{F}_t^{(\tau_0)})$ -Brownian motions.

We write $W^{(\tau_0)} = (\beta_t^{(\tau_0),j})_{j \in \mathbb{N}}$. It is clear how $W^{(\tau_0)}$ can be realized as Hilbert space valued $(\mathcal{F}_t^{(\tau_0)})$ -Wiener process and how one has to define $\mathcal{L}^{\text{loc}}(H, (\mathcal{F}_t^{(\tau_0)}))$ etc. In fact, we have $(\Phi_t) \in \mathcal{L}^{\text{loc}}(H)$ if and only if $(\Phi_{\tau_0+t}) \in \mathcal{L}^{\text{loc}}(H, (\mathcal{F}_t^{(\tau_0)}))$.

The *time τ_0 -shifted version* of (2.20) now reads

$$\begin{cases} dX_t = (AX_t + F(\tau_0 + t, X_t)) dt + B(\tau_0 + t, X_t) dW_t^{(\tau_0)} \\ X_0 = h_0 \end{cases} \quad (2.25)$$

and similarly the expanded form (2.20'). A local mild (weak, strong) solution to (2.25) will be denoted by $X^{(\tau_0, h_0)}$.

Lemma 2.4.4. *Suppose that $X = X^{(0, h_0)}$ is a local mild (weak, strong) solution to (2.20) with lifetime $\tau > \tau_0$. Then (X_{τ_0+t}) is a local mild (weak, strong) solution to (2.25) where h_0 is replaced by X_{τ_0} , with lifetime $\tau - \tau_0$.*

Proof. First notice the identity, for $\Phi \in \mathcal{L}^{\text{loc}}(H)$,

$$\int_{\tau_0}^{\tau_0+t} \Phi_s dW_s = \int_0^t \Phi_{\tau_0+s} dW_s^{(\tau_0)}. \quad (2.26)$$

This is clear if Φ is elementary and if τ_0 takes only a finite number of values. The general case follows by a well-known localization procedure, see [19, Lemma 4.9]. The rest of the claim holds by inspection. $\square$

In the sequel we shall mostly consider deterministic times $\tau_0 \equiv t_0 \in \mathbb{R}_+$.

2.4.2 Existence and Uniqueness

Lipschitz continuity and linear growth conditions play an important role in the theory of stochastic equations. Assume that D and E are Banach spaces and let $\Phi = \Phi(t, \omega, h) : \mathbb{R}_+ \times \Omega \times D \rightarrow E$ be a given mapping.

Definition 2.4.2. *The mapping $\Phi(t, \omega, h)$ is called locally Lipschitz continuous in h if for all $R \in \mathbb{R}_+$ there exists a number $C = C(R)$, the Lipschitz constant for Φ , such that for all $(t, \omega) \in \mathbb{R}_+ \times \Omega$*

$$\|\Phi(t, \omega, h_1) - \Phi(t, \omega, h_2)\|_E \leq C \|h_1 - h_2\|_D, \quad \forall h_1, h_2 \in B_R(D). \quad (2.27)$$

If C does not depend on R we call Φ simply Lipschitz continuous in h .

Remark 2.4.3. The Lipschitz constant C may actually depend on T and then (2.27) holds for all $(t, \omega) \in [0, T] \times \Omega$. The subsequent existence results remain valid. For ease of notation we abandon this generalization.

Definition 2.4.3. *The mapping $\Phi(t, \omega, h)$ is called locally bounded in h if for all $R \in \mathbb{R}_+$ there exists a number $K = K(R)$ such that*

$$\|\Phi(t, \omega, h)\|_E \leq K, \quad \forall (t, \omega, h) \in \mathbb{R}_+ \times \Omega \times B_R(D). \quad (2.28)$$

Suppose that $\Phi(t, \omega, h)$ is locally Lipschitz continuous in h . Then it is easy to see that Φ is locally bounded in h if and only if

$$\|\Phi(t, \omega, h_0)\|_E \leq \tilde{C}, \quad \forall (t, \omega) \in \mathbb{R}_+ \times \Omega,$$

for some $h_0 \in D$ and a constant $\tilde{C}$.

We can now formulate the main result on stochastic equations, which is a combination of Theorems 6.5 and 7.4 in [19]. It is based on a classical fixed point argument.

Theorem 2.4.1. *Suppose $F(t, \omega, h)$ and $B(t, \omega, h)$ are Lipschitz continuous in h and satisfy the linear growth condition*

$$\|F(t, \omega, h)\|_H^2 + \|B(t, \omega, h)\|_{L^2_0(H)}^2 \leq C(1 + \|h\|_H^2),$$

for all $(t, \omega, h) \in \mathbb{R}_+ \times \Omega \times H$, where C does not depend on (t, ω, h) .

Then for any space time initial point $(t_0, h_0) \in \mathbb{R}_+ \times H$ there exists a unique continuous weak solution $X = X^{(t_0, h_0)}$ to the time t_0 -shifted equation (2.25). Moreover, for any $T \in \mathbb{R}_+$ there exists a constant $C = C(T)$ such that

$$\mathbb{E} \left[\sup_{t \in [0, T]} \|X_t\|_H^2 \right] \leq C(1 + \|h_0\|_H^2).$$

Here is the local analogue. A local weak solution X to (2.25) with lifetime τ_X is called *unique* if $(X_{t \wedge \tau_X \wedge \tau_Y}) = (Y_{t \wedge \tau_X \wedge \tau_Y})$ for any local weak solution Y with lifetime τ_Y .

Corollary 2.4.1. *Suppose that $F(t, \omega, h)$ and $B(t, \omega, h)$ are locally bounded and locally Lipschitz continuous in h .*

Then for any space time initial point $(t_0, h_0) \in \mathbb{R}_+ \times H$ there exists a unique continuous local weak solution $X = X^{(t_0, h_0)}$ to the time t_0 -shifted equation (2.25). Moreover, any continuous weak solution to (2.25) is unique.

Proof. We may assume that $t_0 = 0$. Let $h_0 \in H$. Set $l := 2\|h_0\|_H$ and define

$$\tilde{F}(t, \omega, h) := \begin{cases} F(t, \omega, h), & \text{if } \|h\|_H \leq l \\ F(t, \omega, \frac{l}{\|h\|_H} h), & \text{if } \|h\|_H > l \end{cases}$$

and similarly $\tilde{B}(t, \omega, h)$. Then $\tilde{F}$ and $\tilde{B}$ satisfy the assumptions of Theorem 2.4.1. Hence there exists a unique continuous weak solution $\tilde{X}$ to (2.25) where F and B are replaced by $\tilde{F}$ and $\tilde{B}$, respectively. Define the stopping time $\tau := \inf\{t \geq 0 \mid \|\tilde{X}_t\|_H \geq l\}$. Then $\tau > 0$ and $X_t := \tilde{X}_{t \wedge \tau}$ is a continuous local weak solution to (2.25) with lifetime τ .

If X is a continuous weak solution to (2.25) then by the above arguments it is unique on $[0, \tau_n]$ for $n \geq 2$, where $\tau_n := \inf\{t \geq 0 \mid \|X\|_H \geq n\|h_0\|_H\}$. Now use that $\tau_n \uparrow \infty$. $\square$

Remark 2.4.4. Theorem 2.4.1 and its corollary indicate that a (local) weak solution is the proper concept for equation (2.20). Indeed, (local) strong solutions exist very rarely in general. Even if $F, B^j \in D(A)$ and $D(A)$ is invariant for $S(t)$, this does not imply (2.23) yet. See e.g. [16, Proposition 3.26] for the case $F \equiv 0$.

3. Consistent State Space Processes

In this chapter we discuss the consistency problems for factor models which were introduced in Section 1.2. We first characterize the consistent Itô processes which have the property to provide an arbitrage-free interest rate model when representing the parameters of a given forward curve family $\mathcal{G}$. This includes diffusion processes in particular and replies to problem (P1). We then illustrate the consequences of these general results by applying them to the class of exponential-polynomial families. Important examples are the Nelson–Siegel (1.4) and the Svensson (1.5) families, which are widely used to estimate the current forward rate curve (see Table 1.1 in Section 1.2). It turns out that there are strong limitations on the choice of their stochastic factors. Exponential-polynomial families are to be used with care for modeling the term structure of interest rates. This chapter is a summary of the articles [26] and [28].

3.1 Itô Process Factor Models

Motivated by Section 1.2 we shall subsequently formulate the rigorous setup for a factor model. Fix $m \in \mathbb{N}$ and let Z be a Borel set in $\mathbb{R}^m$. Suppose we are given a parametrized family $\mathcal{G} = \{G(\cdot, z) \mid z \in Z\}$ where $G \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^m)$. That is, G and the partial derivatives $\partial G/\partial x$, $\partial G/\partial z_i$, $\partial^2 G/\partial z_i \partial z_j$, which exist for all $1 \leq i, j \leq m$, are continuous functions on $\mathbb{R}_+ \times \mathbb{R}^m$.

We shall consider an $\mathbb{R}^m$ -valued Itô process $Z = (Z^1, \dots, Z^m)$ of the form

$$Z_t^i = Z_0^i + \int_0^t b_s^i ds + \sum_{j \in \mathbb{N}} \int_0^t \rho_s^{ij} d\beta_s^j, \quad i = 1, \dots, m. \quad (3.1)$$

Notice that $\rho_t = (\rho_t^j)_{j \in \mathbb{N}} \in L_2^0(\mathbb{R}^m)$ and we have made (and will from now on use) the notational convention

$$\rho_t^j = (\rho_t^{1j}, \dots, \rho_t^{mj}) \in \mathbb{R}^m.$$

Write $a := \rho \rho^*$ for the diffusion coefficient of Z . That is, $a^{ij} = \sum_{k \in \mathbb{N}} \rho^{ik} \rho^{jk}$, for $1 \leq i, j \leq m$. Then a is a predictable process with values in the symmetric nonnegative definite $m \times m$ -matrices.

We now let the forward rates $f(t, T)$ be given by (1.6). Observe that $\Pi \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^m)$ where

$$\Pi(x, z) := \exp \left(- \int_0^x G(\eta, z) d\eta \right).$$

By Itô's formula, the bond price processes

$$P(t, T) = \Pi(T - t, Z_t), \quad t \in [0, T], \quad T \in \mathbb{R}_+, \quad (3.2)$$

are Itô processes. The same applies for the process of the savings account

$$B(t) = \exp \left(\int_0^t G(0, Z_s) ds \right), \quad t \in \mathbb{R}_+,$$

see (1.2). Recall our definition of no-arbitrage at the end of Section 1.1.

Definition 3.1.1. *A probability measure $\mathbb{Q} \sim \mathbb{P}$ is called an equivalent local martingale measure (ELMM) if the discounted bond price process*

$$\left(\frac{P(t, T)}{B(t)} \right)_{t \in [0, T]} \quad (3.3)$$

is a $\mathbb{Q}$ -local martingale for any $T \in \mathbb{R}_+$. If $\mathbb{P}$ is an ELMM, we call it simply a local martingale measure (LMM).

Whether there exists an ELMM or not depends on Z .

Definition 3.1.2. *The process Z is called consistent with $\mathcal{G}$ (e-consistent with $\mathcal{G}$) if $Z_t \in \mathcal{Z}$ for all $t \in \mathbb{R}_+$ and $\mathbb{P}$ is a LMM (there exists an ELMM $\mathbb{Q}$).*

Consistency trivially implies e-consistency. Using Itô's formula, the dynamics of (3.3) can be decomposed into finite variation and local martingale part. If consistency of Z is required, the former has to vanish. This is the well-known HJM drift condition and is stated explicitly in the next proposition.

Proposition 3.1.1. *A $\mathcal{Z}$ -valued Itô process Z given by (3.1) is consistent with $\mathcal{G}$ if and only if*

$$\begin{aligned} \frac{\partial G(x, Z)}{\partial x} &= \sum_{i=1}^m b^i \frac{\partial G(x, Z)}{\partial z_i} \\ &+ \sum_{i,j=1}^m a^{ij} \left(\frac{1}{2} \frac{\partial^2 G(x, Z)}{\partial z_i \partial z_j} - \frac{\partial G(x, Z)}{\partial z_i} \int_0^x \frac{\partial G(\eta, Z)}{\partial z_j} d\eta \right) \end{aligned} \quad (3.4)$$

for all $x \in \mathbb{R}_+$, $dt \otimes d\mathbb{P}$ -a.s.

Remark 3.1.1. Proposition 3.1.1 is folklore if Z is a diffusion process; that is, if $b_t(\omega) = b(t, Z_t(\omega))$ and $\rho_t(\omega) = \rho(t, Z_t(\omega))$ for nice Borel mappings b and ρ from $\mathbb{R}_+ \times \mathcal{Z}$ into the corresponding spaces. In that case the generator of Z induces a PDE for the function Π on $\mathbb{R}_+ \times \mathcal{Z}$ with initial condition $\Pi(0, z) = 1$. Then Z is a fortiori consistent with the forward curve family implied by Π . Here we ask the other way round: given $\mathcal{G}$ (or equivalently Π), we intend to derive conditions on b and ρ being *necessary* for the consistency of Z with $\mathcal{G}$. Moreover, b and ρ are predictable processes which may depend on the whole history of Z . Hence Z is not a Markov process in general. Therefore we do not get a PDE for G but merely the condition (3.4), which holds $dt \otimes d\mathbb{P}$ -a.s.

We also notice that consistency of a diffusion process Z with $\mathcal{G}$ does not imply the validity of the PDE condition for Π on $\mathbb{R}_+ \times \mathcal{Z}$ in general (the support of Z_t might be a strict subset of $\mathcal{Z}$ for some t).

Proof. For $f \in C^2(\mathbb{R}^m)$ we set

$$\mathcal{A}_t(\omega)f(z) := \sum_{i=1}^m b_t^i(\omega) \frac{\partial f(z)}{\partial z_i} + \frac{1}{2} \sum_{i,j=1}^m a_t^{ij}(\omega) \frac{\partial^2 f(z)}{\partial z_i \partial z_j}, \quad t \in \mathbb{R}_+, \quad z \in \mathbb{R}^m.$$

Using Itô's formula we get, for any $T \in \mathbb{R}_+$,

$$\begin{aligned} P(t, T) &= P(0, T) + \int_0^t \left(\mathcal{A}_s \Pi(T-s, Z_s) - \frac{\partial}{\partial x} \Pi(T-s, Z_s) \right) ds \\ &\quad + \int_0^t D_z \Pi(T-s, Z_s) \rho_s dW_s, \quad 0 \leq t \leq T, \end{aligned}$$

where D_z denotes the derivative with respect to z , and

$$\frac{1}{B(t)} = 1 + \int_0^t \frac{1}{B(s)} \frac{\partial}{\partial x} \Pi(0, Z_s) ds, \quad t \in \mathbb{R}_+.$$

For $0 \leq t \leq T$ define

$$H(t, T) := \frac{\mathcal{A}_t \Pi(T-t, Z_t) - \frac{\partial \Pi(T-t, Z_t)}{\partial x} + \frac{\partial \Pi(0, Z_t)}{\partial x} \Pi(T-t, Z_t)}{B(t)}$$

and the local martingale

$$M(t, T) := \int_0^t \frac{1}{B(s)} D_z \Pi(T-s, Z_s) \rho_s dW_s.$$

Integration by parts then yields

$$\frac{1}{B(t)} P(t, T) = P(0, T) + \int_0^t H(s, T) ds + M(t, T), \quad 0 \leq t \leq T.$$

By the uniqueness of this decomposition we see that Z is consistent with $\mathcal{G}$ if and only if

$$\int_0^t H(s, T) ds = 0, \quad \forall t \in [0, T], \quad \mathbb{P}\text{-a.s.} \quad (3.5)$$

Since b and ρ are predictable and Π is smooth enough, $H(\cdot, T)$ is predictable on $[0, T] \times \Omega$. We claim that (3.5) is equivalent to

$$H(\cdot, T) = 0, \quad \text{on } [0, T] \times \Omega, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (3.6)$$

Indeed, sufficiency of (3.6) is clear. Now suppose that (3.5) holds and define the predictable set $N := \{(t, \omega) \in [0, T] \times \Omega \mid H(t, T)(\omega) > 0\}$. Since $H(\cdot, T)$ is positive on N we can use Tonelli's theorem

$$\int_N H(t, T)(\omega) dt \otimes d\mathbb{P} = \int_\Omega \left(\int_{N_\omega} H(t, T)(\omega) dt \right) d\mathbb{P}(\omega) = 0,$$

where $N_\omega := \{t \mid (t, \omega) \in N\} \in \mathcal{B}(\mathbb{R}_+)$ and we have used (3.5) and the inner regularity of the measure dt . We therefore conclude that N has $dt \otimes d\mathbb{P}$ -measure zero. By using a similar argument for $-H(\cdot, T)$ we have proved (3.6).

Note that (3.6) holds for all $T \in \mathbb{R}_+$, where the $dt \otimes d\mathbb{P}$ -nullset depends on T . But since $H(t, T)$ is continuous in T , a standard argument yields

$$H(t, t+x)(\omega) = 0, \quad \forall x \in \mathbb{R}_+, \quad \text{for } dt \otimes d\mathbb{P}\text{-a.e. } (t, \omega).$$

Multiplying this equation with $B(t)$ and using again the full form this reads

$$\mathcal{A} \Pi(x, Z) - \frac{\partial \Pi(x, Z)}{\partial x} + \frac{\partial \Pi(0, Z)}{\partial x} \Pi(x, Z) = 0, \quad \forall x \in \mathbb{R}_+, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (3.7)$$

Differentiation gives

$$\begin{aligned} \int_0^x \mathcal{A} G(\eta, Z) d\eta - \frac{1}{2} \sum_{i,j=1}^m a^{ij} \left(\int_0^x \frac{\partial}{\partial z_i} G(\eta, Z) d\eta \right) \left(\int_0^x \frac{\partial}{\partial z_j} G(\eta, Z) d\eta \right) \\ - G(x, Z) + G(0, Z) = 0, \quad \forall x \in \mathbb{R}_+, \quad dt \otimes d\mathbb{P}\text{-a.s.} \end{aligned}$$

where we have divided by $\Pi(x, Z)$, since $\Pi > 0$ on $\mathbb{R}_+ \times \mathcal{Z}$. Differentiating with respect to x finally yields (3.4). $\square$

Remark 3.1.2. Definition 3.1.2 can be extended in a natural way to a wider class of state space processes Z . We mention here just two possible directions:

- a) Z a *time homogeneous Markov process with infinitesimal generator* $\mathcal{L}$. The corresponding version of Proposition 3.1.1 can be formulated in terms of equation (3.7), where $\mathcal{A}$ has to be replaced by $\mathcal{L}$. The difficulty here consists of checking whether Π goes well together with Z , i.e. $x \mapsto \Pi(x, \cdot)$ has to be a nice mapping from $\mathbb{R}_+$ into the domain of $\mathcal{L}$.

b) Z an Itô process with jumps. Again one could reformulate Proposition 3.1.1 on the basis of equation (3.7) by adding the corresponding stochastic integral with respect to the compensator of the jump measure, which we assume to be absolutely continuous with respect to dt . The jump measure could be implied for example by a homogeneous Poisson process.

Recall the fact that there is a one-to-one correspondence between the Itô processes Z starting in Z_0 (up to indistinguishability) and the equivalence classes of b and ρ with respect to the $dt \otimes d\mathbb{P}$ -nullsets in $\mathcal{P}$, see Definition 2.2.1 and (3.6) (in Lemma 3.3.1 below we will actually formulate a stronger version of this fact). In view of Proposition 3.1.1 we now may restate the consistency problem (P1) from Section 1.2 as the following *inverse problem* to equation (3.4):

(P1') Given a family of forward curves $\mathcal{G}$, for which choices of (equivalence classes of) predictable coefficients b and ρ do we obtain a consistent Itô process Z starting at Z_0 ?

In the rest of the chapter, we shall exemplarily investigate this problem for the large class of exponential-polynomial families. This includes the Nelson–Siegel (1.4) and Svensson (1.5) families G_{NS} and G_S in particular. Of course, the procedures which will be introduced subsequently can be applied to any other given family $\mathcal{G}$.

We remark at this point that the present Itô process framework is too general for applications, where one usually considers diffusion factor models. But we shall have interesting insights into the interplay between the local behavior of the predictable coefficients b and ρ and of the Itô process Z . The price we have to pay for this mathematical generality is the somewhat complicated way of obtaining our first results. The reader who is mainly interested in applications may, after having read the main definitions below, continue with Section 3.6 at first.

3.2 Exponential-Polynomial Families

Let K denote a positive integer and let $n = (n_1, \dots, n_K)$ be a vector with components $n_i \in \mathbb{N}_0$. We write $|n| := n_1 + \dots + n_K$. For a point

$$z = (z_{1,0}, \dots, z_{1,n_1+1}, z_{2,0}, \dots, z_{2,n_2+1}, \dots, z_{K,0}, \dots, z_{K,n_K+1}) \in \mathbb{R}^{|n|+2K}, \quad (3.8)$$

we define the K polynomials

$$p_i(z) = p_i(x, z) := \sum_{\mu=0}^{n_i} z_{i,\mu} x^\mu, \quad 1 \leq i \leq K.$$

They induce the function

$$G(x, z) := \sum_{i=1}^K p_i(x, z) e^{-z_{i, n_i+1} x}. \quad (3.9)$$

Obviously, we have $G \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R}^{|n|+2K})$. Hence the previous section applies with $m = |n| + 2K$.

From an economic point of view it is reasonable to introduce the parameter set $\mathcal{Z} := \{z \in \mathbb{R}^m \mid \sup_{x \in \mathbb{R}_+} |G(x, z)| < \infty\}$.

Definition 3.2.1. *The exponential-polynomial family $EP(K, n)$ is defined as the set of forward curves $\{G(\cdot, z) \mid z \in \mathbb{R}^m\}$.*

The bounded exponential-polynomial family $BEP(K, n) \subset EP(K, n)$ is defined as the set of forward curves $\{G(\cdot, z) \mid z \in \mathcal{Z}\}$.

Clearly, $G_{NS}(\cdot, z) \in BEP(2, (0, 1))$ and $G_S(\cdot, z) \in BEP(3, (0, 1, 1))$ if in each case the parameter z is chosen such that the curve is bounded. From now on the Nelson–Siegel and Svensson families are considered as subsets of $BEP(2, (0, 1))$ and $BEP(3, (0, 1, 1))$, respectively.

If two exponents z_{i, n_i+1} and z_{j, n_j+1} coincide, the sum (3.9) defining G reduces to a linear combination of $K - 1$ exponential functions. Thus for any $z \in \mathbb{R}^m$ we introduce the equivalence relation

$$i \sim_z j : \Longleftrightarrow z_{i, n_i+1} = z_{j, n_j+1} \quad (3.10)$$

on the set $\{1, \dots, K\}$ and denote by $[i] = [i]_z$ the equivalence class of i . We will use the notation

$$\begin{aligned} n_{[i]} &= n_{[i]}(z) := \max\{n_j \mid j \in [i]_z\} \\ \mathcal{I}_{[i], \mu} &= \mathcal{I}_{[i], \mu}(z) := \{j \in [i]_z \mid n_j \geq \mu\}, \quad 0 \leq \mu \leq n_{[i]}(z) \\ z_{[i], \mu} &= z_{[i], \mu}(z) := \sum_{j \in \mathcal{I}_{[i], \mu}(z)} z_{j, \mu}, \quad 0 \leq \mu \leq n_{[i]}(z) \\ p_{[i]}(z) &:= \sum_{j \in [i]_z} p_j(z). \end{aligned} \quad (3.11)$$

In particular $p_{[i]}(z) = \sum_{\mu=0}^{n_{[i]}} z_{[i], \mu} x^\mu$, and (3.9) now reads

$$G(x, z) = \sum_{[i] \in \{1, \dots, K\} / \sim} p_{[i]}(z) e^{-z_{i, n_i+1} x}.$$

Observe that for $z \in \mathcal{Z}$ we have

$$z_{i, n_i+1} \begin{cases} = 0, & \text{only if } p_{[i]}(z) = z_{[i], 0} \\ < 0, & \text{only if } p_{[i]}(z) = 0. \end{cases} \quad (3.12)$$

We shall write the $\mathbb{R}^m$ -valued Itô process Z with the same indices as we use for a point $z \in \mathbb{R}^m$ (see (3.8))

$$Z_t^{i,\mu} = Z_0^{i,\mu} + \int_0^t b_s^{i,\mu} ds + \sum_{\lambda \in \mathbb{N}} \int_0^t \rho_s^{i,\mu;\lambda} d\beta_s^\lambda, \quad 0 \leq \mu \leq n_i + 1, \quad 1 \leq i \leq K. \quad (3.13)$$

Its diffusion matrix a consists of the components

$$a^{i,\mu;j,\nu} = \sum_{\lambda \in \mathbb{N}} \rho^{i,\mu;\lambda} \rho^{j,\nu;\lambda}, \quad 0 \leq \mu \leq n_i + 1, \quad 0 \leq \nu \leq n_j + 1, \quad 1 \leq i, j \leq K.$$

Notice that, for any $1 \leq i \leq K$, the set

$$\begin{aligned} \{z \mid p_{[i]}(z) = 0\} = & \bigcup_{J \subset \{1, \dots, K\} \atop J \ni i} \left(\{z \mid z_{j,n_j+1} = z_{i,n_i+1} \text{ for all } j \in J\} \right. \\ & \cap \bigcap_{\mu=0}^{\max\{n_j \mid j \in J\}} \{z \mid \sum_{\substack{j \in J \\ n_j \geq \mu}} z_{j,\mu} = 0\} \\ & \left. \setminus \bigcup_{l \in J^c} \{z \mid z_{l,n_l+1} = z_{i,n_i+1}\} \right) \end{aligned} \quad (3.14)$$

is not closed in general but nevertheless a Borel set in $\mathbb{R}^m$. We introduce the predictable sets of singular points (t, ω)

$$\begin{aligned} \mathcal{A}_i &:= \{p_i(Z) = 0 \text{ or } p_{[i]}(Z) = 0\}, \quad 1 \leq i \leq K \\ \mathcal{B} &:= \bigcup_{\substack{i,j=1 \\ i \neq j}}^K \{Z^{i,n_i+1} = Z^{j,n_j+1}\} \\ \mathcal{C} &:= \bigcup_{\substack{i,j=1 \\ i \neq j}}^K \{2Z^{i,n_i+1} = Z^{j,n_j+1}\} \end{aligned}$$

and the predictable sets of regular points (t, ω)

$$\begin{aligned} \mathcal{D} &:= (\mathbb{R}_+ \times \Omega) \setminus \left(\bigcup_{i=1}^K \mathcal{A}_i \cup \mathcal{B} \cup \mathcal{C} \right) \\ \mathcal{D}' &:= (\mathbb{R}_+ \times \Omega) \setminus (\mathcal{B} \cup \mathcal{C}). \end{aligned}$$

Let us recall that for stopping times S and T , a stochastic interval like $[S, T]$ is a subset of $\mathbb{R}_+ \times \Omega$. Hence $[S] = [S, S]$ is the restriction of the graph of the mapping $S : \Omega \rightarrow [0, \infty]$ to the set $\mathbb{R}_+ \times \Omega$.

For any stopping time τ with $[\tau] \in (\mathbb{R}_+ \times \Omega) \setminus \mathcal{A}_i$, we define

$$\tau'(\omega) := \inf\{t > \tau(\omega) \mid (t, \omega) \in \mathcal{A}_i\},$$

the *début* of the predictable set $] \tau, \infty[\cap \mathcal{A}_i$, which is again a stopping time (see e.g. [38]). Observe that in general we do not have $\tau' > \tau$ on $\{\tau < \infty\}$. This can be seen from the following example. Consider

$$G(x, z) = z_{1,0}e^{-z_{1,1}x} + z_{2,0}e^{-z_{2,1}x} + z_{3,0}e^{-z_{3,1}x} \in BEP(3, (0, 0, 0))$$

and let

$$\begin{aligned} Z_t^{1,0} &= Z_t^{1,1} = Z_t^{2,1} = Z_t^{3,0} = 1 \\ Z_t^{2,0} &= -1 \\ Z_t^{3,1} &= 1 + t \end{aligned}$$

for $t \in [0, 1]$. Then $p_1(Z_0) = p_{[1]}(Z_0) = 1$ and $p_{[1]}(Z_t) = 0$ for all $t \in (0, 1]$. Hence $[0] \in (\mathbb{R}_+ \times \mathcal{O}) \setminus \mathcal{A}_1$, but $\tau' = 0$. However, by continuity of Z we always have

$$\tau < \tau' \quad \mathbb{P}\text{-a.s. on } \{\omega \mid (\tau(\omega), \omega) \in \mathcal{D}'\}. \quad (3.15)$$

The following characterization of all consistent Itô processes is remarkably restrictive. The proof of the theorem will be given in Sections 3.4–3.5.

Theorem 3.2.1. *Let $K \in \mathbb{N}$, $n = (n_1, \dots, n_K) \in \mathbb{N}_0^K$ and Z be as above. If Z is consistent with $BEP(K, n)$ then necessarily, for $1 \leq i \leq K$,*

$$a^{i, n_i+1; i, n_i+1} = 0 \quad \text{on } \{p_i(Z) \neq 0\}, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (3.16)$$

$$b^{i, n_i+1} = 0 \quad \text{on } \{p_i(Z) \neq 0\} \cap \{p_{[i]}(Z) \neq 0\}, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (3.17)$$

As a consequence, Z^{i, n_i+1} is constant on intervals where $p_i(Z) \neq 0$ and $p_{[i]}(Z) \neq 0$. That is, for $\mathbb{P}$ -almost every ω ,

$$Z_t^{i, n_i+1}(\omega) = Z_u^{i, n_i+1}(\omega) \quad \forall t \in [u, v],$$

if $p_i(Z_t(\omega)) \neq 0$ and $p_{[i]}(Z_t(\omega)) \neq 0$ for all $t \in (u, v)$.

For a stopping time τ with $[\tau] \subset \mathcal{D}'$, let

$$\tau'(\omega) := \inf\{t > \tau(\omega) \mid (t, \omega) \notin \mathcal{D}'\}$$

denote the *début* of the predictable set $(\mathcal{B} \cup \mathcal{C}) \cap] \tau, \infty[$. Then we have furthermore that $\tau < \tau'$ on $\{\tau < \infty\}$ and

$$\begin{aligned} Z_{\tau+t}^{i, \mu} &= Z_{\tau}^{i, \mu} e^{-Z_{\tau}^{i, n_i+1} t} + Z_{\tau}^{i, \mu+1} t e^{-Z_{\tau}^{i, n_i+1} t} \\ Z_{\tau+t}^{i, n_i} &= Z_{\tau}^{i, n_i} e^{-Z_{\tau}^{i, n_i+1} t} \end{aligned}$$

on $[0, \tau' - \tau]$, for $0 \leq \mu \leq n_i - 1$ and $1 \leq i \leq K$, up to evanescence.

If $\mathcal{D}'$ above is replaced by $\mathcal{D}$ and τ' is the *début* of $(\cup_{i=1}^K \mathcal{A}_i \cup \mathcal{B} \cup \mathcal{C}) \cap] \tau, \infty[$, then $\tau' = \infty$ and in addition

$$Z_{\tau+t}^{i, n_i+1} = Z_{\tau}^{i, n_i+1}$$

for $1 \leq i \leq K$, $\mathbb{P}$ -a.s. on $\{\tau < \infty\}$.

Remark 3.2.1. It will be made clear in the proof of the theorem that it is actually sufficient to assume Z to be consistent with $EP(K, n)$ for (3.16) to hold.

As an immediate consequence we may state the following corollaries. The notation is the same as in Theorem 3.2.1.

Corollary 3.2.1. *If Z is consistent with $BEP(K, n)$ and if the predictable sets $\{p_i(Z) = 0\}$ and $\{p_{[i]}(Z) = 0\}$ have $dt \otimes d\mathbb{P}$ -measure zero, then the exponent Z^{i, n_i+1} is indistinguishable from Z_0^{i, n_i+1} , $1 \leq i \leq K$.*

Proof. By assumption, $\{p_i(Z) \neq 0\} \cap \{p_{[i]}(Z) \neq 0\} = \mathbb{R}_+ \times \Omega$ up to a $dt \otimes d\mathbb{P}$ -nullset. The claim follows using (3.16) and (3.17). $\square$

Corollary 3.2.2. *If Z is consistent with $BEP(K, n)$ and if the following three points are $\mathbb{P}$ -a.s. satisfied*

- i) $p_i(Z_0) \neq 0$, for all $1 \leq i \leq K$,
- ii) there exists no pair of indices $i \neq j$ with $Z_0^{i, n_i+1} = Z_0^{j, n_j+1}$,
- iii) there exists no pair of indices $i \neq j$ with $2Z_0^{i, n_i+1} = Z_0^{j, n_j+1}$,

then Z and hence the interest rate model $G(x, Z)$ is quasi-deterministic; that is, all randomness remains $\mathcal{F}_0$ -measurable. In particular, the exponents Z^{i, n_i+1} are indistinguishable from Z_0^{i, n_i+1} , for $1 \leq i \leq K$.

Proof. By assumptions i)–iii), we have $[0] \subset \mathcal{D}$. The claim follows from the second part of Theorem 3.2.1 setting $\tau = 0$. $\square$

One may argue that it is enough to consider e-consistent Itô processes. However, if we pose some more conditions on the filtration $(\mathcal{F}_t)$ we can even give the following stronger result. We denote by $(\mathcal{F}_t^W)$ the filtration generated by the Brownian motion W .

Theorem 3.2.2. *Suppose that $(\mathcal{F}_t) = (\mathcal{F}_t^W)$. Then Theorem 3.2.1 and Corollaries 3.2.1–3.2.2 remain valid if consistent is replaced by e-consistent.*

Proof. Let $\mathbb{Q}$ be an ELMM. By hypothesis, W has the predictable representation property (see [49, Chap. V]). By Girsanov's theorem it follows that Z remains an Itô process under $\mathbb{Q}$, which is consistent with $BEP(K, n)$ with respect to $\mathbb{Q}$. The drift coefficients $b^{i, \mu}$ change under $\mathbb{Q}$ into $\tilde{b}^{i, \mu}$. Whereas $b^{i, \mu} = \tilde{b}^{i, \mu}$ on $\{a^{i, \mu; i, \mu} = 0\}$, $dt \otimes d\mathbb{P}$ -a.s. The diffusion matrix a remains the same. Therefore and since the measures $dt \otimes d\mathbb{Q}$ and $dt \otimes d\mathbb{P}$ are equivalent on $\mathcal{B}(\mathbb{R}_+) \otimes \mathcal{F}$, the Itô process Z is of the form as stated in Theorem 3.2.1. $\square$

Notice that under the assumptions of Theorem 3.2.2, the expression *quasi-deterministic* in Corollary 3.2.2 can be replaced by *deterministic*, since $\mathcal{F}_0^W$ is trivial.

3.3 Auxiliary Results

For the proof of Theorem 3.2.1 we need three auxiliary lemmas, presented in this section. First, there is a result on the identification of the coefficients of Itô processes.

Lemma 3.3.1. *Let*

$$\begin{aligned} X_t &= X_0 + \int_0^t \beta_s^X ds + \sum_{j \in \mathbb{N}} \int_0^t \gamma_s^{X,j} d\beta_s^j, \\ Y_t &= Y_0 + \int_0^t \beta_s^Y ds + \sum_{j \in \mathbb{N}} \int_0^t \gamma_s^{Y,j} d\beta_s^j \end{aligned}$$

be two real-valued Itô processes. Then $dt \otimes d\mathbb{P}$ -a.s.

$$\begin{aligned} 1_{\{X=Y\}} \sum_{j \in \mathbb{N}} (\gamma^{X,j})^2 &= 1_{\{X=Y\}} \sum_{j \in \mathbb{N}} \gamma^{X,j} \gamma^{Y,j} = 1_{\{X=Y\}} \sum_{j \in \mathbb{N}} (\gamma^{Y,j})^2 \\ 1_{\{X=Y\}} \beta^X &= 1_{\{X=Y\}} \beta^Y. \end{aligned}$$

Proof. We write $\langle \cdot, \cdot \rangle$ for the scalar product in ℓ^2 . Then

$$\begin{aligned} |\langle \gamma^X, \gamma^X \rangle - \langle \gamma^X, \gamma^Y \rangle| &= |\langle \gamma^X, \gamma^X - \gamma^Y \rangle| \\ &\leq \sqrt{\langle \gamma^X, \gamma^X \rangle} \sqrt{\langle \gamma^X - \gamma^Y, \gamma^X - \gamma^Y \rangle}. \end{aligned}$$

By the occupation times formula, see [49, Corollary (1.6), Chap. VI],

$$\int_0^t 1_{\{X_s=Y_s\}} \langle \gamma_s^X - \gamma_s^Y, \gamma_s^X - \gamma_s^Y \rangle ds = 0, \quad \text{for all } t \in \mathbb{R}_+, \mathbb{P}\text{-a.s.}$$

Hence, by the Hölder inequality,

$$\begin{aligned} &\int_0^t 1_{\{X_s=Y_s\}} |\langle \gamma_s^X, \gamma_s^X \rangle - \langle \gamma_s^X, \gamma_s^Y \rangle| ds \\ &\leq \int_0^t 1_{\{X_s=Y_s\}} \sqrt{\langle \gamma_s^X, \gamma_s^X \rangle} \sqrt{\langle \gamma_s^X - \gamma_s^Y, \gamma_s^X - \gamma_s^Y \rangle} ds \\ &\leq \left(\int_0^t 1_{\{X_s=Y_s\}} \langle \gamma_s^X, \gamma_s^X \rangle ds \right)^{\frac{1}{2}} \left(\int_0^t 1_{\{X_s=Y_s\}} \langle \gamma_s^X - \gamma_s^Y, \gamma_s^X - \gamma_s^Y \rangle ds \right)^{\frac{1}{2}} \\ &= 0, \quad \text{for all } t \in \mathbb{R}_+, \mathbb{P}\text{-a.s.} \end{aligned}$$

Thus, by symmetry,

$$1_{\{X=Y\}} \langle \gamma^X, \gamma^X \rangle = 1_{\{X=Y\}} \langle \gamma^X, \gamma^Y \rangle = 1_{\{X=Y\}} \langle \gamma^Y, \gamma^Y \rangle, \quad dt \otimes d\mathbb{P}\text{-a.s.}$$

By continuity of the processes X and Y , there are sequences of stopping times (S_n) and (T_n) , $S_n \leq T_n$, with $[S_m, T_m] \cap [S_n, T_n] = \emptyset$ for all $m \neq n$ and

$$\{X = Y\} = \bigcup_{n \in \mathbb{N}} [S_n, T_n], \quad \text{up to evanescence.}$$

To see this, let $n \in \mathbb{N}$ and let $S(n, 1) := \inf\{t > 0 \mid |X_t - Y_t| = 0\}$. Define $T(n, p) := \inf\{t > S(n, p) \mid |X_t - Y_t| > 0\}$ and inductively

$$S(n, p+1) := \inf\{t > S(n, p) \mid |X_t - Y_t| = 0 \\ \text{and } \sup_{S(n, p) \leq s \leq t} |X_s - Y_s| > 2^{-n}\}.$$

Then by continuity we have $\lim_{p \rightarrow \infty} S(n, p) = \infty$ for all $n \in \mathbb{N}$ and it follows that $\{X = Y\} = \bigcup_{n, p \in \mathbb{N}} [S(n, p), T(n, p)]$. Now proceed as in [38, Lemma I.1.31] to find the sequences (S_n) and (T_n) with the desired properties.

From above we have $1_{\{X=Y\}}(\gamma^X - \gamma^Y)^2 = 0$, $dt \otimes d\mathbb{P}$ -a.s. For any $t \in \mathbb{R}_+$ therefore

$$\int_{S_n \wedge t}^{T_n \wedge t} (\gamma_s^X - \gamma_s^Y) dW_s = 0, \quad \mathbb{P}\text{-a.s.}$$

Hence

$$0 = (X - Y)_{T_n \wedge t} - (X - Y)_{S_n \wedge t} = \int_{S_n \wedge t}^{T_n \wedge t} (\beta_s^X - \beta_s^Y) ds, \quad \mathbb{P}\text{-a.s.}$$

and we conclude that

$$\int_0^t 1_{\{X_s=Y_s\}} (\beta_s^X - \beta_s^Y) ds = \sum_{n \in \mathbb{N}} \int_{S_n \wedge t}^{T_n \wedge t} (\beta_s^X - \beta_s^Y) ds = 0,$$

for all $t \in \mathbb{R}_+$, $\mathbb{P}$ -a.s. Using the same arguments as for the proof of (3.6) we derive the desired result. $\square$

Next, we list two results in matrix algebra.

Lemma 3.3.2. *Let $\gamma = (\gamma_\lambda)_{\lambda \in \mathbb{N}} \in L_2^0(\mathbb{R}^m)$ and define the symmetric non-negative definite $m \times m$ -matrix $\alpha := \gamma\gamma^*$, i.e. $\alpha_{i,j} = \alpha_{j,i} = \sum_{\lambda \in \mathbb{N}} \gamma_{i,\lambda} \gamma_{j,\lambda}$. Let I and J denote two arbitrary subsets of $\{1, \dots, m\}$. Define*

$$\alpha_{I,J} = \alpha_{J,I} := \sum_{j \in J} \sum_{i \in I} \alpha_{i,j}.$$

Then we have $\alpha_{I,I} \geq 0$ and $|\alpha_{I,J}| \leq \sqrt{\alpha_{I,I}} \sqrt{\alpha_{J,J}}$.

Proof. For $\lambda \in \mathbb{N}$, define $\gamma_{I,\lambda} := \sum_{i \in I} \gamma_{i,\lambda}$. Then by definition

$$\alpha_{I,J} = \sum_{j \in J} \sum_{i \in I} \sum_{\lambda \in \mathbb{N}} \gamma_{i,\lambda} \gamma_{j,\lambda} = \sum_{\lambda \in \mathbb{N}} \left(\sum_{i \in I} \gamma_{i,\lambda} \right) \left(\sum_{j \in J} \gamma_{j,\lambda} \right) = \sum_{\lambda \in \mathbb{N}} \gamma_{I,\lambda} \gamma_{J,\lambda}.$$

Hence

$$\alpha_{I,I} = \sum_{\lambda \in \mathbb{N}} (\gamma_{I,\lambda})^2 \geq 0$$

and by the Cauchy–Schwarz inequality

$$|\alpha_{I,J}| \leq \sqrt{\sum_{\lambda \in \mathbb{N}} (\gamma_{I,\lambda})^2} \sqrt{\sum_{\lambda \in \mathbb{N}} (\gamma_{J,\lambda})^2} = \sqrt{\alpha_{I,I}} \sqrt{\alpha_{J,J}}.$$

□

Lemma 3.3.3. *Let $\alpha = (\alpha_{i,j})$ be an $n \times n$ -matrix, $n \in \mathbb{N}$, which is diagonally dominant from the right; that is,*

$$|\alpha_{i,i}| \geq \sum_{\substack{j=1 \\ j \neq i}}^n |\alpha_{i,j}|$$

$$|\alpha_{i,i}| > \sum_{j=i+1}^n |\alpha_{i,j}|, \quad \left(\text{set } \sum_{j=n+1}^n \cdots := 0 \right),$$

for all $1 \leq i \leq n$. Then α is regular.

Proof. The proof, involving Gaussian elimination, is a slight modification of an argument given in [51, Theorem 1.5].

By assumption $|\alpha_{1,1}| > \sum_{j=2}^n |\alpha_{1,j}| \geq 0$, in particular $\alpha_{1,1} \neq 0$. If $n = 1$ we are done. If $n > 1$, the elimination step

$$\alpha_{i,j}^{(1)} := \alpha_{i,j} - \frac{\alpha_{i,1}}{\alpha_{1,1}} \alpha_{1,j}, \quad 2 \leq i, j \leq n,$$

leads to the $(n-1) \times (n-1)$ -matrix $\alpha^{(1)} = (\alpha_{i,j}^{(1)})_{2 \leq i,j \leq n}$. We show that $\alpha^{(1)}$ is diagonally dominant from the right. If $\alpha_{i,1} = 0$, there is nothing to prove for the i th row. Let $\alpha_{i,1} \neq 0$, for some $2 \leq i \leq n$. We have

$$|\alpha_{i,j}^{(1)}| \geq |\alpha_{i,j}| - \left| \frac{\alpha_{i,1}}{\alpha_{1,1}} \right| |\alpha_{1,j}|, \quad 2 \leq j \leq n.$$

Therefore

$$\begin{aligned} \sum_{j=i+1}^n |\alpha_{i,j}^{(1)}| &\leq \sum_{\substack{j=2 \\ j \neq i}}^n |\alpha_{i,j}^{(1)}| = \sum_{\substack{j=2 \\ j \neq i}}^n \left| \alpha_{i,j} - \frac{\alpha_{i,1}}{\alpha_{1,1}} \alpha_{1,j} \right| \leq \sum_{\substack{j=2 \\ j \neq i}}^n |\alpha_{i,j}| + \left| \frac{\alpha_{i,1}}{\alpha_{1,1}} \right| \sum_{\substack{j=2 \\ j \neq i}}^n |\alpha_{1,j}| \\ &= \sum_{\substack{j=1 \\ j \neq i}}^n |\alpha_{i,j}| - |\alpha_{i,1}| + \left| \frac{\alpha_{i,1}}{\alpha_{1,1}} \right| \left(\sum_{j=2}^n |\alpha_{1,j}| - |\alpha_{1,i}| \right) \\ &< |\alpha_{i,i}| - |\alpha_{i,1}| + \left| \frac{\alpha_{i,1}}{\alpha_{1,1}} \right| (|\alpha_{1,1}| - |\alpha_{1,i}|) \\ &= |\alpha_{i,i}| - \left| \frac{\alpha_{i,1}}{\alpha_{1,1}} \right| |\alpha_{1,i}| \leq |\alpha_{i,i}^{(1)}|. \end{aligned}$$

Proceed inductively to $\alpha^{(2)}, \dots, \alpha^{(n-1)}$. $\square$

3.4 The Case $BEP(1, n)$

We will treat the case $K = 1$ separately, since it represents a key step in the proof of the general $BEP(K, n)$ case. For simplicity we shall skip the index $i = 1$ and write $n = n_1 \in \mathbb{N}_0$, $p = p_1$, $b^j = b^{1,j}$, $a^{i,j} = a^{1,i;1,j}$, etc. In particular, we use the notation of Section 3.1 with $m = n + 2$.

Lemma 3.4.1. *Let $n \in \mathbb{N}_0$ and Z be as above. If Z is consistent with $BEP(1, n)$, then necessarily*

$$\begin{aligned} Z_t^i &= Z_0^i e^{-Z_0^{n+1}t} + Z_0^{i+1} t e^{-Z_0^{n+1}t} \\ Z_t^n &= Z_0^n e^{-Z_0^{n+1}t} \\ Z_t^{n+1} &= Z_0^{n+1} + \left(\int_0^t b_s^{n+1} ds + \sum_{j \in \mathbb{N}} \int_0^t \rho_s^{n+1,j} d\beta_s^j \right) 1_{\Omega_0}, \end{aligned}$$

for $0 \leq i \leq n - 1$ and $t \in \mathbb{R}_+$, $\mathbb{P}$ -a.s., where $\Omega_0 := \{p(Z_0) = 0\}$.

Consequently, if Z is consistent with $BEP(1, n)$ then $\{p(Z) = 0\} = \mathbb{R}_+ \times \Omega_0$. Hence $\{Z^{n+1} \neq Z_0^{n+1}\} \subset \{p(Z) = 0\}$. Therefore we may state

Corollary 3.4.1. *If Z is consistent with $BEP(1, n)$ then*

$$G(x, Z) = p(x, Z) e^{-Z_0^{n+1}x}.$$

Hence the corresponding interest rate model is quasi-deterministic; that is, all randomness remains $\mathcal{F}_0$ -measurable.

Remark 3.4.1. Suppose that $(\mathcal{F}_t) = (\mathcal{F}_t^W)$. Then Lemma 3.4.1 and Corollary 3.4.1 remain valid if *consistent* is replaced by *e-consistent*. Moreover, *quasi-deterministic* is the same as *deterministic* in this case. This follows immediately from the proof of Theorem 3.2.2.

Proof (of Lemma 3.4.1). Let Z be an Itô process which is consistent with $BEP(1, n)$. Fix a point (t, ω) in $\mathbb{R}_+ \times \Omega$. For simplicity we write z_i for $Z_t^i(\omega)$, $a_{i,j}$ for $a_t^{i,j}(\omega)$ and b_i for $b_t^i(\omega)$. The proof relies on expanding equation (3.4) at the point $z = (z_0, \dots, z_{n+1})$. The terms involved are

$$\frac{\partial}{\partial x} G(x, z) = \left(\frac{\partial}{\partial x} p(x, z) - z_{n+1} p(x, z) \right) e^{-z_{n+1}x} \quad (3.18)$$

$$\frac{\partial}{\partial z_i} G(x, z) = \begin{cases} x^i e^{-z_{n+1}x}, & 0 \leq i \leq n \\ -x p(x, z) e^{-z_{n+1}x}, & i = n + 1 \end{cases} \quad (3.19)$$

$$\frac{\partial^2 G(x, z)}{\partial z_i \partial z_j} = \frac{\partial^2 G(x, z)}{\partial z_j \partial z_i} = \begin{cases} 0, & 0 \leq i, j \leq n \\ -x^{i+1} e^{-z_{n+1}x}, & 0 \leq i \leq n, j = n + 1 \\ x^2 p(x, z) e^{-z_{n+1}x}, & i = j = n + 1. \end{cases} \quad (3.20)$$

Finally, the following relation for $k \in \mathbb{N}_0$ is useful:

$$\int_0^x \eta^k e^{-z_{n+1}\eta} d\eta = \begin{cases} -r_k(x)e^{-z_{n+1}x} + \frac{k!}{z_{n+1}^{k+1}}, & z_{n+1} \neq 0 \\ \frac{x^{k+1}}{k+1}, & z_{n+1} = 0, \end{cases} \quad (3.21)$$

where

$$r_k(x) = \sum_{l=0}^k \frac{k!}{(k-l)!} \frac{x^{k-l}}{z_{n+1}^{l+1}}$$

is a polynomial in x of order k .

Let us first suppose that $z_{n+1} \neq 0$. Thus, subtracting $\frac{\partial}{\partial x} G(x, Z)$ from both sides of (3.4) we obtain a null equation of the form

$$q_1(x)e^{-z_{n+1}x} + q_2(x)e^{-2z_{n+1}x} = 0, \quad (3.22)$$

which has to hold simultaneously for all $x \in \mathbb{R}_+$. The polynomials q_1 and q_2 depend on the z_i , b_i and $a_{i,j}$. Equality (3.22) implies $q_1 = q_2 = 0$. This again yields that all coefficients of the q_i have to be zero.

To proceed we have to distinguish the two cases $p(z) \neq 0$ and $p(z) = 0$. Let us first suppose the former is true. Then there exists an index $i \in \{0, \dots, n\}$ such that $z_i \neq 0$. Set $\mu_m := \max\{i \leq n \mid z_i \neq 0\}$. With regard to (3.19)–(3.21) it follows that $\deg q_2 = 2\mu_m + 2$. In particular,

$$q_2(x) = a_{n+1, n+1} \frac{z_{\mu_m}^2}{z_{n+1}} x^{2\mu_m+2} + \dots,$$

where $\dots$ denotes terms of lower order in x . Hence $a_{n+1, n+1} = 0$. But the matrix a has to be nonnegative definite, so necessarily

$$a_{n+1, j} = a_{j, n+1} = 0, \quad \text{for all } 1 \leq j \leq n+1.$$

In view of Lemma 3.3.1 (setting $Y = 0$), since we are characterizing a and b up to $dt \otimes d\mathbb{P}$ -nullsets, we may assume $a_{i,j} = a_{j,i} = 0$, for $0 \leq j \leq n+1$, for all $i \geq \mu_m + 1$. Thus the degree of q_2 reduces to $2\mu_m$. Explicitly

$$q_2(x) = \frac{a_{\mu_m, \mu_m}}{z_{n+1}} x^{2\mu_m} + \dots$$

Hence $a_{\mu_m, \mu_m} = 0$ and so $a_{\mu_m, j} = a_{j, \mu_m} = 0$, for $0 \leq j \leq n+1$. Proceeding inductively for $i = \mu_m - 1, \mu_m - 2, \dots, 0$, we finally obtain that the diffusion matrix a is equal to zero and hence $q_2 = 0$ is fulfilled.

Now we determine the drift b . By Lemma 3.3.1, we may assume $b_i = 0$ for $\mu_m + 1 \leq i \leq n$. With regard to (3.18) and (3.19), q_1 reduces therefore to

$$q_1(x) = -b_{n+1} z_{\mu_m} x^{\mu_m+1} + \dots$$

It follows that $b_{n+1} = 0$ and we are left with

$$\begin{aligned}
q_1(x) &= (b_{\mu_m} + z_{n+1}z_{\mu_m})x^{\mu_m} + \sum_{i=0}^{\mu_m-1} (b_i - z_{i+1} + z_{n+1}z_i)x^i \\
&= (b_n + z_{n+1}z_n)x^n + \sum_{i=0}^{n-1} (b_i - z_{i+1} + z_{n+1}z_i)x^i.
\end{aligned}$$

We now turn to the singular cases. If $p(z) = 0$, that is $z_0 = \dots = z_n = 0$, we may assume $a_{i,j} = a_{j,i} = b_i = 0$, $0 \leq j \leq n+1$, for all $i \leq n$. But this means that $q_1 = q_2 = 0$, independently of the choice of b_{n+1} and $a_{n+1,n+1}$.

For the case where $z_{n+1} = 0$ we need the boundedness assumption $z \in \mathcal{Z}$. By (3.12) it follows that $z_1 = \dots = z_n = 0$. So by Lemma 3.3.1 again $a_{i,j} = a_{j,i} = b_i = 0$, for $0 \leq j \leq n+1$, for all $i \geq 1$. Thus in this case equation (3.4) reduces to

$$0 = b_0 - a_{0,0}$$

and therefore $b_0 = a_{0,0} = 0$.

Summarizing all cases, we conclude that necessarily

$$\begin{aligned}
b_i &= -z_{n+1}z_i + z_{i+1}, \quad 0 \leq i \leq n-1, \\
b_n &= -z_{n+1}z_n, \\
a_{i,j} &= 0, \quad \text{for } (i, j) \neq (n+1, n+1);
\end{aligned}$$

while b_{n+1} and $a_{n+1,n+1}$ are arbitrary real and nonnegative real numbers, respectively, whenever $p(z) = 0$. Otherwise $b_{n+1} = a_{n+1,n+1} = 0$.

This has to hold for $dt \otimes d\mathbb{P}$ -a.e. (t, ω) . So Z is uniquely (up to indistinguishability) determined and satisfies

$$\begin{aligned}
Z_t^i &= Z_0^i + \int_0^t (Z_s^{i+1} - Z_s^{n+1}Z_s^i) ds, \quad 0 \leq i \leq n-1, \\
Z_t^n &= Z_0^n - \int_0^t Z_s^{n+1}Z_s^n ds, \\
Z_t^{n+1} &= Z_0^{n+1} + \int_0^t b_s^{n+1} ds + \sum_{j \in \mathbb{N}} \int_0^t \rho_s^{n+1,j} d\beta_s^j
\end{aligned}$$

for some predictable processes b^{n+1} and $(\rho^{n+1,j})_{j \in \mathbb{N}} \in \mathcal{L}^{\text{loc}}(\mathbb{R})$ which vanish outside the predictable set $\{p(Z) = 0\}$.

Note that $Z^1, \dots, Z^n$ satisfy (path-wise) a system of linear ODEs with continuous coefficients. Hence they are indistinguishable from zero on Ω_0 . So the statement of the lemma is proved on Ω_0 . It remains to prove it on $\Omega_1 := \Omega \setminus \Omega_0$.

Introduce the stopping time $\tau := \inf\{s > 0 \mid p(Z_s) = 0\}$. We have just argued that $\Omega_0 \subset \{\tau = 0\}$. By continuity of Z also the reverse inclusion holds, hence $\Omega_0 = \{\tau = 0\}$. The stopped process $Z_{t \wedge \tau} =: Y_t$ satisfies (path-wise) the following system of linear stochastic integral equations

$$\begin{aligned}
Y_t^i &= Z_0^i + \int_0^{t \wedge \tau} (Y_s^{i+1} - Y_s^{n+1} Y_s^i) ds, \quad 0 \leq i \leq n-1, \\
Y_t^n &= Z_0^n - \int_0^{t \wedge \tau} Y_s^{n+1} Y_s^n ds, \\
Y_t^{n+1} &= Z_0^{n+1}.
\end{aligned} \tag{3.23}$$

We have used (2.10) and the fact that we can replace b^{n+1} and ρ^{n+1} by $b^{n+1}1_{] \tau, \infty]}$ and $\rho^{n+1}1_{] \tau, \infty]}$, respectively. The system (3.23) has the unique global solution

$$\begin{aligned}
Y_t^i &= Z_0^i e^{-Z_0^{n+1}(t \wedge \tau)} + Z_0^{i+1}(t \wedge \tau) e^{-Z_0^{n+1}(t \wedge \tau)}, \quad 0 \leq i \leq n-1, \\
Y_t^n &= Z_0^n e^{-Z_0^{n+1}(t \wedge \tau)}, \\
Y_t^{n+1} &= Z_0^{n+1}.
\end{aligned}$$

Since $Z = Y$ on the stochastic interval $[0, \tau]$ and since $Y_t \neq 0$, $\forall t \in \mathbb{R}_+$, $\mathbb{P}$ -a.s. on Ω_1 , it follows by the continuity of Z that $\Omega_1 = \{\tau > 0\} = \{\tau = \infty\}$. Inserting this in the above solution, the lemma is proved also on Ω_1 . $\square$

3.5 The General Case $BEP(K, n)$

Using again the notation of Section 3.2 we now give the proof of Theorem 3.2.1 for the case $K \geq 2$. The exposition is somewhat complicated, which is due to the multidimensionality of the problem. The idea, however, is simple. For a fixed point $(t, \omega) \in \mathbb{R}_+ \times \Omega$ we expand equation (3.4), which turns out to be a linear combination of linearly independent exponential functions over the ring of polynomials, equaling zero. Consequently, many of the coefficients have to vanish, which leads to our assertion.

The difficulty is that some exponents may coincide. This causes a considerable number of singular cases which require a separate discussion.

Let $K \geq 2$, $n = (n_1, \dots, n_K) \in \mathbb{N}_0^K$, and let Z be consistent with $BEP(K, n)$. As in the proof of Lemma 3.4.1, we fix a point $(t, \omega) \in \mathbb{R}_+ \times \Omega$ and use the shorthand notation $z_{i,\mu}$ for $Z_t^{i,\mu}(\omega)$, $a_{i,\mu;j,\nu}$ for $a_t^{i,\mu;j,\nu}(\omega)$ and $b_{i,\mu}$ for $b_t^{i,\mu}(\omega)$, etc. Since we are characterizing a and b up to a $dt \otimes d\mathbb{P}$ -nullset, we assume that (t, ω) is chosen outside an exceptional $dt \otimes d\mathbb{P}$ -nullset. In particular, the lemmas from Section 3.3 shall apply each time this choice is made.

The strategy is the same as for the case $K = 1$. Thus we expand equation (3.4) at the point $z = (z_{1,0}, \dots, z_{K,n_K+1})$ to get a linear combination of (ideally) linearly independent exponential functions over the ring of polynomials

$$\sum_{i=1}^K q_i(x) e^{-z_{i,n_i+1}x} + \sum_{1 \leq i \leq j \leq K} q_{i,j}(x) e^{-(z_{i,n_i+1} + z_{j,n_j+1})x} = 0. \tag{3.24}$$

Consequently, all polynomials q_i and $q_{i,j}$ have to be zero. The main difference between this case and the case $K = 1$ is that representation (3.24) may not be unique due to the possibly multiple occurrence of the following singular cases

- i) $z_{i,n_i+1} = z_{j,n_j+1}$, for $i \neq j$,
- ii) $2z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1}$,
- iii) $2z_{i,n_i+1} = z_{j,n_j+1}$,
- iv) $z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1}$,

for some indices $1 \leq i, j, k \leq K$. However, the lemmas in Section 3.3 and the boundedness assumption $z \in \mathcal{Z}$ are good enough to settle these four cases.

Let us suppose first that $p_i(z) \neq 0$, for all $i \in \{1, \dots, K\}$. To settle case i), let $\sim$ denote the equivalence relation defined in (3.10). After reparametrization if necessary, we may assume that

$$\{1, \dots, K\} / \sim = \{[1], \dots, [\tilde{K}]\}$$

and $z_{1,n_1+1} < \dots < z_{\tilde{K},n_{\tilde{K}}+1}$ for some integer $\tilde{K} \leq K$. Write $I := \{1, \dots, \tilde{K}\}$. In view of Lemma 3.3.1 we may assume

$$a_{j,n_j+1;j,n_j+1} = a_{i,n_i+1;i,n_i+1} \quad \text{and} \quad b_{j,n_j+1} = b_{i,n_i+1}, \quad \forall j \in [i], i \in I. \quad (3.25)$$

The proof of (3.16) and (3.17) is divided into four lemmas.

Lemma 3.5.1. $a_{i,n_i+1;i,n_i+1} = 0$, for all $i \in I$.

Proof. Expression (3.24) takes the form

$$\sum_{i \in I} \tilde{q}_i(x) e^{-z_{i,n_i+1}x} + \sum_{\substack{i,j \in I \\ i \leq j}} \tilde{q}_{i,j}(x) e^{-(z_{i,n_i+1} + z_{j,n_j+1})x} = 0, \quad (3.26)$$

for some polynomials $\tilde{q}_i$ and $\tilde{q}_{i,j}$. Taking into account cases ii)–iv), this representation may still not be unique. However, if for an index $i \in I$ there exist no $j, k \in I$ such that $2z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1}$ or $2z_{i,n_i+1} = z_{j,n_j+1}$ (in particular, $z_{i,n_i+1} \neq 0$) then we have

$$\tilde{q}_{i,i}(x) = a_{i,n_i+1;i,n_i+1} \frac{\sum_{j \in \mathcal{I}_{[i], \mu_m}} z_{j, \mu_m}^2}{z_{i,n_i+1}} x^{2\mu_m+2} + \dots,$$

where $\mu_m := \max\{\nu \mid \nu \leq n_j \text{ and } z_{j,\nu} \neq 0 \text{ for some } j \in [i]\} \in \mathbb{N}_0$. Hence $a_{i,n_i+1;i,n_i+1} = 0$ and the lemma is proved for the regular case.

For the singular cases observe first that $z_{i,n_i+1} = 0$ implies $a_{i,n_i+1;i,n_i+1} = 0$, which follows from Lemma 3.3.1. Now we split I into two disjoint subsets I_1 and I_2 , where

$$I_1 := \{i \in I \mid z_{i,n_i+1} \neq 0, \text{ and there exist } j, k \in I \text{ such that} \\ 2z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1} \text{ or } 2z_{i,n_i+1} = z_{j,n_j+1}\}, \\ I_2 := I \setminus I_1.$$

Observe that $z_{\tilde{K}, n_{\tilde{K}}+1} > 0$ implies $\tilde{K} \in I_2$ and $z_{1,n_1+1} < 0$ implies $1 \in I_2$. Since at least one of these events has to happen, the set I_2 is not empty. We have shown above that $a_{i,n_i+1,i,n_i+1} = 0$, for $i \in I_2$. If I_1 is not empty, we will show that for each $i \in I_1$, the parameter z_{i,n_i+1} can be written as a linear combination of z_{j,n_j+1} s with $j \in I_2$. From this it follows by Lemmas 3.3.1 and 3.3.2 that $a_{i,n_i+1,i,n_i+1} = 0$ for all $i \in I_1$ and the lemma is completely proved. We proceed as follows. Write $I_1 = \{i_1, \dots, i_r\}$ with $z_{i_1,n_{i_1}+1} < \dots < z_{i_r,n_{i_r}+1}$. For each $i_k \in I_1$ there exists one linear equation of the form

$$(*, \dots, *, 2, *, \dots, *) \begin{pmatrix} z_{i_1,n_{i_1}+1} \\ \vdots \\ z_{i_k,n_{i_k}+1} \\ \vdots \\ z_{i_r,n_{i_r}+1} \end{pmatrix} = \alpha_k,$$

where $*$ stands for 0 or -1 , but with at most one -1 on each side of 2. The α_k on the right hand side is 0 or z_{i,n_i+1} or $z_{i,n_i+1} + z_{j,n_j+1}$ for some indices $i, j \in I_2$. Hence we obtain the system of linear equations

$$\begin{pmatrix} 2 & * & \dots & * \\ * & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & * \\ * & \dots & * & 2 \end{pmatrix} \begin{pmatrix} z_{i_1,n_{i_1}+1} \\ \vdots \\ \vdots \\ z_{i_r,n_{i_r}+1} \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \vdots \\ \alpha_r \end{pmatrix}.$$

By Lemma 3.3.3, the matrix on the left hand side is invertible, from which our assertion follows. $\square$

Lemma 3.5.2. $a_{j,n_j+1;k,\nu} = a_{k,\nu;j,n_j+1} = 0$, for $0 \leq \nu \leq n_k$, for all $1 \leq j, k \leq K$.

Proof. In view of (3.25), the lemma follows immediately from Lemmas 3.5.1 and 3.3.2. $\square$

Analogous to the notation introduced in (3.11), we set

$$b_{[i],\mu} := \sum_{j \in \mathcal{I}_{[i],\mu}} b_{j,\mu} \\ \rho_{[i],\mu;\lambda} := \sum_{j \in \mathcal{I}_{[i],\mu}} \rho_{j,\mu;\lambda} \\ a_{[i],\mu;k,\nu} := \sum_{j \in \mathcal{I}_{[i],\mu}} a_{j,\mu;k,\nu},$$

for $0 \leq \mu \leq n_{[i]}$, $0 \leq \nu \leq n_k$, $1 \leq k \leq K$, $1 \leq \lambda \leq d$, $i \in I$, and

$$a_{[i], \mu; [k], \nu} := \sum_{l \in \mathcal{I}_{[k], \nu}} \sum_{j \in \mathcal{I}_{[i], \mu}} a_{j, \mu; l, \nu},$$

for $0 \leq \mu \leq n_{[i]}$, $0 \leq \nu \leq n_{[k]}$, $i, k \in I$.

Lemma 3.5.3. *If $z_{[i], \mu} = 0$, for $i \in I$ and $\mu \in \{0, \dots, n_{[i]}\}$, then*

$$b_{[i], \mu} = a_{[i], \mu; [i], \mu} = a_{[i], \mu; k, \nu} = a_{k, \nu; [i], \mu} = 0,$$

for all $0 \leq \nu \leq n_k$, $1 \leq k \leq K$.

Proof. Notice that $a_{[i], \mu; [i], \mu} = \sum_{\lambda \in \mathbb{N}} \rho_{[i], \mu; \lambda}^2$. Hence Lemma 3.5.3 follows by Lemmas 3.3.1 and 3.3.2. $\square$

Lemma 3.5.4. *$b_{i, n_i+1} = 0$, for all $i \in I$ such that $p_{[i]}(z) \neq 0$.*

Proof. Suppose first that $z_{i, n_i+1} \neq 0$, for all $i \in I$. Let $i \in I$ such that $p_{[i]}(z) \neq 0$, and let us assume there exist no $j, k \in I$ with $z_{i, n_i+1} = z_{j, n_j+1} + z_{k, n_k+1}$. What does the polynomial $\tilde{q}_i$ in (3.26) look like? With regard to (3.25), Lemmas 3.5.2 and 3.3.1, and equalities (3.18)–(3.21), the contributing terms are

$$\begin{aligned} \frac{\partial p_j(x, z)}{\partial x} e^{-z_{j, n_j+1}x} &= \left(\left(\sum_{\mu=1}^{\mu_m \wedge n_j} z_{j, \mu} x^{\mu-1} \right) - z_{i, n_i+1} \left(\sum_{\mu=0}^{\mu_m \wedge n_j} z_{j, \mu} x^{\mu} \right) \right) \\ &\quad \times e^{-z_{i, n_i+1}x}, \end{aligned} \quad (3.27)$$

$$\begin{aligned} \sum_{\mu=0}^{n_j+1} b_{j, \mu} \frac{\partial G(x, z)}{\partial z_{j, \mu}} &= \left(\left(\sum_{\mu=0}^{\mu_m \wedge n_j} b_{j, \mu} x^{\mu} \right) - b_{i, n_i+1} \left(\sum_{\mu=0}^{\mu_m \wedge n_j} z_{j, \mu} x^{\mu+1} \right) \right) \\ &\quad \times e^{-z_{i, n_i+1}x} \end{aligned} \quad (3.28)$$

and

$$\begin{aligned} \sum_{\mu=0}^{n_j} a_{j, \mu; k, \nu} \frac{\partial G(x, z)}{\partial z_{j, \mu}} \int_0^x \frac{\partial G(\eta, z)}{\partial z_{k, \nu}} d\eta &= \left(\sum_{\mu=0}^{\mu_m \wedge n_j} a_{j, \mu; k, \nu} \frac{n_k!}{z_{k, n_k+1}^{n_k+1}} x^{\mu} \right) e^{-z_{i, n_i+1}x} \\ &\quad + (\text{polynomial in } x) e^{-(z_{i, n_i+1} + z_{k, n_k+1})x}, \end{aligned} \quad (3.29)$$

for $0 \leq \nu \leq n_k$, for all $1 \leq k \leq K$ and $j \in [i]$. We have used the integer

$$\mu_m := \max\{\lambda \mid \lambda \leq n_l \text{ and } z_{l, \lambda} \neq 0 \text{ for some } l \in [i]\}.$$

Define $\tilde{\mu}_m := \max\{\lambda \mid \lambda \leq n_{[i]} \text{ and } z_{[i], \lambda} \neq 0\} \in \mathbb{N}_0$. Obviously $\tilde{\mu}_m \leq \mu_m$. By Lemma 3.5.3 we have $a_{[i], \mu; k, \nu} = 0$, for all $\tilde{\mu}_m < \mu \leq n_{[i]}$. Thus summing the above expressions over $j \in [i]$, we obtain

$$\tilde{q}_i(x) = -b_{i,n_i+1}z_{[i],\bar{\mu}_m}x^{\bar{\mu}_m+1} + \dots \quad (3.30)$$

Consequently, $b_{i,n_i+1} = 0$ in the regular case.

For the singular cases the boundedness assumption $z \in \mathcal{Z}$ is essential. We split I into two disjoint subsets J_1 and J_2 , where

$$\begin{aligned} J_1 &:= \{i \in I \mid \text{there exist } j, k \in I, \text{ such that } z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1} \\ &\quad \text{and } z_{j,n_j+1} > 0 \text{ and } z_{k,n_k+1} > 0\} \\ J_2 &:= I \setminus J_1. \end{aligned}$$

Notice that in any case $1 \in J_2$. We have shown above that for each $i \in J_2$ such that z_{i,n_i+1} is not the sum of two other z_{j,n_j+1} s it follows that $b_{i,n_i+1} = 0$. We will now show that $b_{i,n_i+1} = 0$ for all $i \in J_2$. Let $i \in J_2$ and assume there exist $j, k \in I$ with $z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1}$. Then necessarily one of the summands is strictly less than zero. Without loss of generality $z_{j,n_j+1} < 0$. Since $z \in \mathcal{Z}$, we have $p_{[j]}(z) = 0$ (see (3.12)). Thus $a_{[j],\mu;[j],\mu} = 0$ by Lemma 3.5.3 and therefore $a_{[j],\mu;k,\nu} = 0$, for all $0 \leq \mu \leq n_{[j]}$, $0 \leq \nu \leq n_k$, $1 \leq k \leq K$. The terms contributing to the polynomial in front of $e^{-z_{i,n_i+1}x}$, that is, $\tilde{q}_i + \tilde{q}_{j,k} + \dots$, are those in (3.27)–(3.29) and also

$$a_{l,\mu;r,\nu} \frac{\partial G(x,z)}{\partial z_{l,\mu}} \int_0^x \frac{\partial G(\eta,z)}{\partial z_{r,\nu}} d\eta = a_{l,\mu;r,\nu} x^\mu e^{-z_{j,n_j+1}x} \int_0^x \eta^\nu e^{-z_{k,n_k+1}\eta} d\eta, \quad (3.31)$$

for $0 \leq \mu \leq n_l$, $0 \leq \nu \leq n_r$, $l \in [j]$, $r \in [k]$. However, summing – for fixed μ, r and ν – the right hand side of (3.31) over $l \in \mathcal{I}_{[j],\mu}$ gives zero. Hence the terms in (3.31) do not actually contribute to the polynomial. The same conclusion can be drawn for all $j, k \in I$ with the property that $z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1}$. It finally follows, as in the regular case, that $b_{i,n_i+1} = 0$ for all $i \in J_2$.

If J_1 is not empty, we show that for each $i \in J_1$, the parameter z_{i,n_i+1} can be written as a linear combination of z_{j,n_j+1} s with $j \in J_2$. From this it follows by Lemma 3.3.1 that $b_{i,n_i+1} = 0$ for all $i \in J_1$. We proceed as follows. Write $J_1 = \{i_1, \dots, i_{r'}\}$ with $z_{i_1,n_{i_1}+1} < \dots < z_{i_{r'},n_{i_{r'}}+1}$. For each $i_k \in J_1$ there exists one linear equation of the form

$$(*, \dots, *, 1, 0, \dots, 0) \begin{pmatrix} z_{i_1,n_{i_1}+1} \\ \vdots \\ z_{i_k,n_{i_k}+1} \\ \vdots \\ z_{i_{r'},n_{i_{r'}}+1} \end{pmatrix} = \alpha'_k,$$

where the $*$ stand for 0 or -1 , but at most two of them are -1 . The α'_k on the right hand side is 0 or z_{i,n_i+1} or $z_{i,n_i+1} + z_{j,n_j+1}$ for some indices $i, j \in J_2$ with $z_{i,n_i+1} > 0$ and $z_{j,n_j+1} > 0$. Obviously α'_1 is of the latter form. Hence we obtain the system of linear equations

$$\begin{pmatrix} 1 & 0 & \dots & 0 \\ * & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ * & \dots & * & 1 \end{pmatrix} \begin{pmatrix} z_{i_1, n_{i_1}+1} \\ \vdots \\ \vdots \\ z_{i_{r'}, n_{i_{r'}}+1} \end{pmatrix} = \begin{pmatrix} z_{i, n_i+1} + z_{j, n_j+1} \\ \alpha'_2 \\ \vdots \\ \alpha'_{r'} \end{pmatrix},$$

for some $i, j \in J_2$. On the left hand side is a lower-triangular matrix, which is therefore invertible. Hence the lemma is proved in the case where $z_{i, n_i+1} \neq 0$ for all $i \in I$.

Assume now that there exists $i \in I$ with $z_{i, n_i+1} = 0$. Then $i \in J_2$. We have to make sure that also in this case $b_{j, n_j+1} = 0$, for all $j \in J_2$. Clearly b_{i, n_i+1} is zero by Lemma 3.3.1. The problem is that $z_{j, n_j+1} = z_{i, n_i+1} + z_{j, n_j+1}$ for all $j \in J_2$. But following the lines above, it is enough to show $a_{[i], \mu; [i], \mu} = 0$, for all $0 \leq \mu \leq n_{[i]}$. From the boundedness assumption $z \in \mathcal{Z}$ we know that $p_{[i]}(z) = z_{[i], 0}$ (see (3.12)). Hence $a_{[i], \mu; [i], \mu} = 0$, for $1 \leq \mu \leq n_{[i]}$. Suppose there is no pair of indices $j, k \in I \setminus \{i\}$ with $z_{j, n_j+1} + z_{k, n_k+1} = 0$. Summing the contributing terms in (3.27)–(3.29) over $j \in [i]$, we obtain the polynomial in front of e^0 , that is,

$$\tilde{q}_i(x) + \tilde{q}_{i, i}(x) = -a_{[i], 0; [i], 0}x + \dots, \quad (3.32)$$

hence $a_{[i], 0; [i], 0} = 0$. If there exist a pair of indices $j, k \in I \setminus \{i\}$ with $z_{j, n_j+1} + z_{k, n_k+1} = 0$, then one of these summands is strictly less than zero. Arguing as before, the polynomial in front of e^0 remains of the form (3.32) and again $a_{[i], 0; [i], 0} = 0$. Thus the lemma is completely proved. $\square$

So far we have established (3.16) and (3.17) under the hypothesis that $p_i(z) \neq 0$, for all $i \in \{1, \dots, K\}$. Suppose now that there exists an index $i \in \{1, \dots, K\}$ with $p_i(z) = 0$. By Lemma 3.3.1, we may assume $a_{i, \mu; i, \mu} = b_{i, \mu} = 0$, for all $0 \leq \mu \leq n_i$. But then Lemma 3.3.2 tells us that none of the terms including the index i appears in (3.24). In particular, $a_{i, n_i+1; i, n_i+1}$ and b_{i, n_i+1} can be chosen arbitrarily without affecting equation (3.24). This means that we may skip i and proceed, after a reparametrization if necessary, with the remaining index set $\{1, \dots, K-1\}$ to establish Lemmas 3.5.1–3.5.4 as above.

This all has to hold for $dt \otimes d\mathbb{P}$ -a.e. (t, ω) . Hence (3.16) and (3.17) are fully proved. A closer look to the proof of (3.16), that is, Lemma 3.5.1, shows that the boundedness assumption $z \in \mathcal{Z}$ was not explicitly used there – whence Remark 3.2.1.

Next we prove that the exponents Z^{i, n_i+1} are locally constant on intervals where $p_i(Z)$ and $p_{[i]}(Z)$ do not vanish. Let $v \geq 0$ be a rational number and let $T_v := \inf\{t > v \mid p_i(Z_t) = 0 \text{ or } p_{[i]}(Z_t) = 0\}$ denote the début of the predictable set $[v, \infty[\cap \mathcal{A}_i$. By (3.16) and (3.17) and the continuity of Z we have that Z^{i, n_i+1} is $\mathbb{P}$ -a.s. constant on $[v, T_v]$, hence $\mathbb{P}$ -a.s. constant on every such interval $[v, T_v]$. Since every open interval where $p_i(Z_t) \neq 0$ or $p_{[i]}(Z_t) \neq 0$

is covered by a countable union of intervals $[v, T_v]$ and by continuity of Z , the assertion follows and the first part of the theorem is proved.

To establish the second part of the theorem, let τ be a stopping time with $[\tau] \in \mathcal{D}'$ and $\mathbb{P}[\tau < \infty] > 0$. Define the stopping time $\tau'(\omega) := \inf\{t > \tau(\omega) \mid (t, \omega) \notin \mathcal{D}'\}$. By continuity of Z , we conclude that $\tau < \tau'$ on $\{\tau < \infty\}$. Choose a point (t, ω) in $[\tau, \tau']$. We use shorthand notation as above.

By definition of $\mathcal{D}'$ we can exclude the singular cases $z_{i,n_i+1} = z_{j,n_j+1}$ or $2z_{i,n_i+1} = z_{j,n_j+1}$, for $i \neq j$. In particular, $\tilde{K} = K$, hence $I = \{1, \dots, K\}$. First, we show that the diffusion matrix for the coefficients of the polynomials $p_i(z)$ vanishes.

Lemma 3.5.5. $a_{i,\mu;j,\nu} = a_{j,\nu;i,\mu} = 0$, for $0 \leq \mu \leq n_i$ and $0 \leq \nu \leq n_j$, for all $i, j \in I$.

Proof. By Lemma 3.3.1 it is enough to prove that the diagonal $a_{i,\mu;i,\mu}$ vanishes for $0 \leq \mu \leq n_i$ and $i \in I$. If there is an index $i \in I$ with $p_i(z) = 0$ then, arguing as above, $a_{i,\mu;i,\mu} = b_{i,\mu} = 0$, for all $0 \leq \mu \leq n_i$, and we may skip the index i . Hence we assume now that there is a $K' \leq K$ such that $p_i(z) \neq 0$ (and thus $z_{i,n_i+1} \geq 0$, since $z \in \mathcal{Z}$) for all $1 \leq i \leq K'$. Let $I' := \{1, \dots, K'\}$. To handle the singular cases, we split I' into two disjoint subsets I'_1 and I'_2 , where

$$\begin{aligned} I'_1 &:= \{i \in I' \mid z_{i,n_i+1} > 0, \text{ and there exist } j, k \in I' \\ &\quad \text{such that } 2z_{i,n_i+1} = z_{j,n_j+1} + z_{k,n_k+1}\}, \\ I'_2 &:= I' \setminus I'_1. \end{aligned}$$

Hence $z_{i,n_i+1} = 0$ for $i \in I'$ implies $i \in I'_2$. We have already shown in the proof of Lemma 3.5.4 that in this case $a_{i,\mu;i,\mu} = 0$, for all $0 \leq \mu \leq n_i$. The same follows for $i \in I'_2$ with $z_{i,n_i+1} > 0$, as was demonstrated for the case $K = 1$.

Now let $i \in I'_1$ and let $l, r \in I'$, such that $l \leq r$ and $2z_{i,n_i+1} = z_{l,n_l+1} + z_{r,n_r+1}$. Thus the polynomial in front of $e^{-2z_{i,n_i+1}x}$ is $q_{i,i} + q_{l,r} + \dots$, and among the contributing terms are also those in (3.31). If l or r is in I'_2 , those are all zero. Write $I'_1 = \{i_1, \dots, i_{r''}\}$ with $z_{i_1,n_{i_1}+1} < \dots < z_{i_{r''},n_{i_{r''}}+1}$. Then necessarily $l \in I'_2$ in the above representation for $z_{i_1,n_{i_1}+1}$. Thus the polynomial in front of $e^{-2z_{i_1,n_{i_1}+1}x}$ is q_{i_1,i_1} . It follows that $a_{i_1,\mu;i_1,\mu} = 0$, for all $0 \leq \mu \leq n_{i_1}$, as was demonstrated for the case $K = 1$. Proceeding inductively for $i_2, \dots, i_{r''}$, we eventually derive that $a_{i,\mu;i,\mu} = 0$, for all $0 \leq \mu \leq n_i$ and $i \in I'$. This establishes the lemma. $\square$

We are left with the task of determining the drift of the coefficients in $p_i(z)$. By (3.17), we have $b_{i,n_i+1} = 0$ for all $i \in I'$. Straightforward calculations show that (3.24) reduces to

$$\sum_{i=1}^{K'} q_i(x) e^{-z_{i,n_i+1}x} = 0,$$

with

$$q_i(x) = (b_{i,n_i} + z_{i,n_i+1}z_{i,n_i})x^{n_i} + \sum_{\mu=0}^{n_i-1} (b_{i,\mu} - z_{i,\mu+1} + z_{i,n_i+1}z_{i,\mu})x^\mu.$$

We conclude that for all $1 \leq i \leq K$ (in particular, if $p_i(z) = 0$)

$$\begin{aligned} b_{i,\mu} &= -z_{i,n_i+1}z_{i,\mu} + z_{i,\mu+1}, \quad 0 \leq \mu \leq n_i - 1 \\ b_{i,n_i} &= -z_{i,n_i+1}z_{i,n_i}. \end{aligned} \quad (3.33)$$

Lemma 3.5.5 and (3.33) hold $dt \otimes d\mathbb{P}$ -a.s. on the stochastic interval $[\tau, \tau']$. Arguing as in the proof of Lemma 3.4.1, we obtain the claimed form of $Z_{\tau+}$ on $[0, \tau' - \tau]$.

Now replace $\mathcal{D}'$ by $\mathcal{D}$ and proceed as above. By (3.15) we have $\tau < \tau'$ on $\{\tau < \infty\}$, and, since $\mathcal{D} \subset \mathcal{D}'$, all the above results remain valid. In addition, $p_i(z) = p_{[i]}(z) \neq 0$ and thus $a_{i,n_i+1;i,n_i+1} = b_{i,n_i+1} = 0$, for all $1 \leq i \leq K$, by (3.16) and (3.17). Hence $Z_{\tau+}^{i,n_i+1} = Z_\tau^{i,n_i+1}$ on $[0, \tau' - \tau]$, for all $1 \leq i \leq K$, up to evanescence. But this again implies $\tau' = \infty$ by the continuity of Z , and Theorem 3.2.1 is completely proved.

3.6 The Diffusion Case

Theorem 3.2.1 is much clearer for diffusion processes. In all applications the $\mathbb{R}^m$ -valued Itô process Z given by (3.13) is rather the strong solution of a stochastic differential equation

$$Z_t^{i,\mu} = Z_0^{i,\mu} + \int_0^t b^{i,\mu}(s, Z_s) ds + \sum_{\lambda \in \mathbb{N}} \int_0^t \rho^{i,\mu;\lambda}(s, Z_s) d\beta_s^\lambda, \quad (3.34)$$

for $0 \leq \mu \leq n_i + 1$ and $1 \leq i \leq K$, where b and ρ are some Borel measurable mappings from $\mathbb{R}_+ \times \mathbb{R}^m$ to $\mathbb{R}^m$ and $L_2^0(\mathbb{R}^m)$, respectively.

The coefficients b and ρ could be derived by statistical inference methods from the daily observations of the diffusion Z made by some central bank. These observations are of course made under the objective probability measure. Hence $\mathbb{P}$ is not a LMM in applications of this kind.

On the other hand, we want an arbitrage-free model for pricing interest rate sensitive securities. Thus the diffusion has to be e-consistent. For the rest of the chapter we assume that $(\mathcal{F}_t) = (\mathcal{F}_t^W)$. Then Theorem 3.2.2 applies. To stress the fact that $\mathcal{F}_0^W$ -measurable functions are deterministic, we denote the initial values of the diffusion in (3.34) with small letters $z_0^{i,\mu}$.

Since all reasonable theory for stochastic differential equations requires continuity properties of the coefficients, we shall assume in the following that $b(t, z)$ and $\rho(t, z)$ are continuous in z . The main result for e-consistent diffusion processes is divided into the two following theorems. The first one only requires consistency with $EP(K, n)$.

Theorem 3.6.1. *Let $K \in \mathbb{N}$, $n = (n_1, \dots, n_K) \in \mathbb{N}_0^K$, $(\mathcal{F}_t) = (\mathcal{F}_t^W)$, and let the diffusion Z , b and ρ be as above. If Z is e-consistent with $BEP(K, n)$ or with $EP(K, n)$, then necessarily the exponents are constant,*

$$Z^{i, n_i+1} \equiv z_0^{i, n_i+1},$$

for all $1 \leq i \leq K$.

Proof. The significant difference between this proof and that of Theorem 3.2.1 is that now the diffusion matrix a and the drift b depend continuously on z .

First, observe that the following sets of singular values,

$$\mathcal{M} := \bigcup_{i=1}^K \{z \in \mathbb{R}^m \mid p_i(z) = 0 \text{ or } p_{[i]}(z) = 0\}$$

and

$$\mathcal{N} := \{z \in \mathbb{R}^m \mid z_{i, n_i+1} = z_{j, n_j+1} + z_{k, n_k+1} \text{ for some } 1 \leq i, j, k \leq K\},$$

are contained in a finite union of hyperplanes of $\mathbb{R}^m$ (see (3.14)). Hence $(\mathcal{M} \cup \mathcal{N}) \subset \mathbb{R}^m$ has Lebesgue measure zero. Thus the topological closure of $\mathcal{H} := \mathbb{R}^m \setminus (\mathcal{M} \cup \mathcal{N})$ is $\mathbb{R}^m$.

Now let Z be the diffusion in (3.34) which is e-consistent with either $BEP(K, n)$ or $EP(K, n)$. A closer look at the proof of Lemma 3.5.4 shows that the boundedness assumption $z \in \mathcal{Z}$ was not used for the regular case $z \in \mathcal{H}$ (see (3.30)). Combining this with (3.16), (3.17) and Remark 3.2.1, we conclude that, for any $1 \leq i \leq K$ and $\lambda \in \mathbb{N}$,

$$b^{i, n_i+1}(t, Z_t(\omega)) = \rho^{i, n_i+1; \lambda}(t, Z_t(\omega)) = 0, \quad \text{for } (t, \omega) \in \{Z \in \mathcal{H}\} \setminus N,$$

where N is a predictable $dt \otimes d\mathbb{P}$ -nullset. By the very definition of the product measure,

$$0 = \int_N 1 \, dt \otimes d\mathbb{P} = \int_{\mathbb{R}_+} \mathbb{P}[N_t] \, dt,$$

where $N_t := \{\omega \mid (t, \omega) \in N\} \in \mathcal{F}$. Consequently $\mathbb{P}[N_t] = 0$ for almost every $t \in \mathbb{R}_+$. Hence, by continuity of $b(t, \cdot)$ and $\rho(t, \cdot)$,

$$b^{i, n_i+1}(t, \cdot) = \rho^{i, n_i+1; \lambda}(t, \cdot) = 0 \tag{3.35}$$

on $\text{supp}(Z_t) \cap \mathcal{H}$, for almost every $t \in \mathbb{R}_+$. Here $\text{supp}(Z_t)$ denotes the support of the distribution of Z_t , which is by definition the smallest closed set $A \subset \mathbb{R}^m$ with $\mathbb{P}[Z_t \in A] = 1$. Thus, again by continuity of $b(t, \cdot)$ and $\rho(t, \cdot)$, equality (3.35) holds for almost every $t \in \mathbb{R}_+$ on the closure of $\text{supp}(Z_t) \cap \mathcal{H}$, which is $\text{supp}(Z_t)$. Hence we may replace the functions $b^{i, n_i+1}(t, \cdot)$ and $\rho^{i, n_i+1; \lambda}(t, \cdot)$ by zero for almost every t without changing the diffusion Z , whence the assertion follows. $\square$

The sum of two real-valued diffusion processes with coefficients continuous in some argument is again a real-valued diffusion with coefficients continuous in that argument. Consequently, we may assume that the exponents z_0^{i,n_i+1} of the above e-consistent diffusion are mutually distinct. Since otherwise we add the corresponding polynomials to obtain in a canonical way an $\mathbb{R}^{\tilde{m}}$ -valued diffusion $\tilde{Z}$ which is e-consistent with $BEP(\tilde{K}, \tilde{n})$ or $EP(\tilde{K}, \tilde{n})$, for some $\tilde{K} < K$, $\tilde{m} < m$ and some $\tilde{n} \in \mathbb{N}_0^{\tilde{K}}$. Clearly $\tilde{Z}$ provides the same interest rate model as Z and its coefficients are continuous in z .

For the second theorem we have to require e-consistency with $BEP(K, n)$. After a reparametrization if necessary, we may thus assume that

$$0 \leq z_0^{1,n_1+1} < \dots < z_0^{K,n_K+1},$$

see (3.12). The continuation of Theorem 3.6.1 is now as follows:

Theorem 3.6.2. *If Z is e-consistent with $BEP(K, n)$, then it is non-trivial only if there exists a pair of indices $1 \leq i < j \leq K$, such that*

$$2z_0^{i,n_i+1} = z_0^{j,n_j+1}.$$

Proof. If there is no pair of indices $1 \leq i < j \leq K$ such that $2z_0^{i,n_i+1} = z_0^{j,n_j+1}$, then $\mathcal{D}' = \mathbb{R}_+ \times \Omega$. But then Z is deterministic by the second part of Theorem 3.2.1. $\square$

The message of Theorem 3.6.1 is the following: there is no possibility of modelling the term structure of interest rates by exponential-polynomial families with varying exponents driven by diffusion processes. From this point of view there is no use for daily estimations of the exponents of exponential-polynomial type functions, such as G_{NS} or G_S . Once the exponents are chosen, they have to be kept constant. Furthermore, there is a strong restriction on this choice by Theorem 3.6.2. It will be shown in the next section what this means for G_{NS} and G_S in particular.

Remark 3.6.1. The boundedness assumption in Theorem 3.6.2 – that is, e-consistency with $BEP(K, n)$ – is essential for the strong (negative) result to be valid. It can easily be checked that $G(x, z) = z_0 + z_1 x \in EP(1, 1)$ allows for a non-trivial consistent diffusion process, see [25, Section 8].

Remark 3.6.2. The choice of an infinite time horizon for traded bonds is not a restriction, see (3.2). Indeed, we can limit our considerations to bonds $P(t, T)$ which mature within a given finite time interval $[0, T^*]$. Consequently, the HJM drift condition (3.4) can only be deduced for $x \in [0, T^* - t]$, for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in [0, T^*] \times \Omega$. But the functions appearing in (3.4) are analytic in x . Hence, whenever $t < T^*$, relation (3.4) extends to all $x \in \mathbb{R}_+$. All conclusions on e-consistent Itô processes $(Z_t)_{0 \leq t \leq T^*}$ can now be drawn as before.

3.7 Applications

In this section we apply the results on e-consistent diffusion processes to the Nelson–Siegel and Svensson families.

3.7.1 The Nelson–Siegel Family

$$G_{NS}(x, z) = z_1 + (z_2 + z_3x)e^{-z_4x}$$

In view of Theorem 3.6.1 we have $z_4 > 0$. Hence it is immediate from Theorem 3.6.2 that there is no non-trivial e-consistent diffusion. This result has already been obtained in [26] for e-consistent Itô processes.

3.7.2 The Svensson Family

$$G_S(x, z) = z_1 + (z_2 + z_3x)e^{-z_5x} + z_4xe^{-z_6x}$$

By Theorems 3.6.1 and 3.6.2 there remain the two choices

- i) $2z_6 = z_5 > 0$,
- ii) $2z_5 = z_6 > 0$.

We shall identify the e-consistent diffusion process $Z = (Z^1, \dots, Z^6)$ in both cases. Let $\mathbb{Q}$ be an ELMM. Under $\mathbb{Q}$ the diffusion Z transforms into a consistent one. Now we proceed as in the proof of Theorem 3.2.1. The expansion (3.24) is given by

$$\begin{aligned} Q_1(x) + Q_2(x)e^{-z_5x} + Q_3(x)e^{-z_6x} \\ + Q_4(x)e^{-2z_5x} + Q_5(x)e^{-(z_5+z_6)x} + Q_6(x)e^{-2z_6x} = 0, \end{aligned}$$

for polynomials $Q_1, \dots, Q_6$. Explicitly,

$$\begin{aligned} Q_1(x) &= -a_{1,1}x + \dots \\ Q_2(x) &= -a_{1,3}x^2 + \dots \\ Q_3(x) &= -a_{1,4}x^2 + \dots \\ Q_4(x) &= \frac{a_{3,3}}{z_5}x^2 + \dots \\ Q_6(x) &= \frac{a_{4,4}}{z_6}x^2 + \dots \end{aligned}$$

where $\dots$ denotes terms of lower order in x . Hence $a_{1,1} = 0$ in any case. By the usual arguments (the matrix a is nonnegative definite) the degree of Q_2 and Q_3 reduces to at most 1. Thus in both cases i) and ii) it follows that $a_{3,3} = a_{4,4} = 0$. We are left with

$$\begin{aligned}
Q_1(x) &= b_1, \\
Q_2(x) &= (b_3 + z_3 z_5)x + b_2 - z_3 - \frac{a_{2,2}}{z_5} + z_2 z_5, \\
Q_3(x) &= (b_4 + z_4 z_6)x - z_4, \\
Q_4(x) &= \frac{a_{2,2}}{z_5},
\end{aligned} \tag{3.36}$$

while $Q_5 = Q_6 = 0$. Since in case i) Q_4 must be 0, we have $a_{2,2} = 0$ and Z is deterministic. We conclude that there is no non-trivial e-consistent diffusion in case i).

In case ii) the condition $Q_3 + Q_4 = 0$ leads to

$$a_{2,2} = z_4 z_5. \tag{3.37}$$

Hence there is a possibility of a non-deterministic consistent diffusion Z . We derive from (3.36) and (3.37)

$$\begin{aligned}
b_1 &= 0, \\
b_2 &= z_3 + z_4 - z_5 z_2, \\
b_3 &= -z_5 z_3, \\
b_4 &= -2z_5 z_4.
\end{aligned}$$

Therefore the dynamics of $Z^1, Z^3, \dots, Z^6$ are deterministic. In particular,

$$\begin{aligned}
Z_t^1 &\equiv z_0^1, \\
Z_t^3 &= z_0^3 e^{-z_0^5 t}, \\
Z_t^4 &= z_0^4 e^{-2z_0^5 t},
\end{aligned} \tag{3.38}$$

while $Z_t^5 \equiv z_0^5$ and $Z_t^6 \equiv 2z_0^5$. Denoting by $\tilde{W}$ the Girsanov transform of W , we have under the ELMM $\mathbb{Q}$

$$Z_t^2 = z_0^2 + \int_0^t (\Phi(s) - z_0^5 Z_s^2) ds + \sum_{\lambda \in \mathbb{N}} \int_0^t \rho^{2,\lambda}(s) d\tilde{\beta}_s^\lambda, \tag{3.39}$$

where $\Phi(t)$ and $\rho^{2,\lambda}(t)$ are deterministic functions in t , namely

$$\Phi(t) := z_0^3 e^{-z_0^5 t} + z_0^4 e^{-2z_0^5 t}$$

and

$$\sum_{\lambda \in \mathbb{N}} (\rho^{2,\lambda}(t))^2 = z_0^4 z_0^5 e^{-2z_0^5 t}.$$

By Lévy's characterization theorem, see [49, Theorem (3.6), Chap. IV], the real-valued process

$$\beta_t^* := \sum_{\lambda \in \mathbb{N}} \int_0^t \frac{\rho^{2,\lambda}(s)}{\sqrt{z_0^4 z_0^5} e^{-z_0^5 s}} d\tilde{\beta}_s^\lambda, \quad t \in \mathbb{R}_+,$$

is an $(\mathcal{F}_t)$ -Brownian motion under $\mathbb{Q}$. Hence the corresponding short rates $R(t) = G_S(0, Z_t) = z_0^1 + Z_t^2$ satisfy

$$dR(t) = (\phi(t) - z_0^5 R(t)) dt + \tilde{\rho}(t) d\beta_t^*,$$

where $\phi(t) := \Phi(t) + z_0^1 z_0^5$ and $\tilde{\rho}(t) := \sqrt{z_0^4 z_0^5} e^{-z_0^5 t}$. This is just the generalized Vasicek model. It can be easily given a closed-form solution for $R(t)$, see [47, p. 293].

Summarizing case ii) we have found a non-trivial e-consistent diffusion process, which is identified by (3.38) and (3.39). Actually Φ has to be replaced by a predictable process $\tilde{\Phi}$ due to the change of measure. Nevertheless, this is just a one-factor model. The corresponding interest rate model is the generalized Vasicek short rate model. This is very unsatisfactory since Svensson-type functions $G_S(x, z)$ have six factors $z_1, \dots, z_6$ which are observed. And it is evident that just one of them, z_2 , can be chosen to be non-deterministic.

3.8 Conclusions

Bounded exponential-polynomial families such as the Nelson–Siegel and Svensson families may be well suited for daily estimations of the forward rate curve. They are problematic when used for arbitrage-free interest rate modelling by diffusion processes. This is because the exponents have to be kept constant and, moreover, this choice is very restricted. We have shown for the Nelson–Siegel family in particular that there exists no non-trivial e-consistent diffusion process. However, there is a choice for the Svensson family, albeit still a very limited one, since all parameters but one have to be kept either constant or deterministic.

For further applications of the results from this chapter we refer to [30].

4. The HJM Methodology Revisited

This chapter is devoted to a detailed discussion and extension of the seminal paper by HJM [35]. The HJM framework basically unifies all continuous interest rate models. In particular, it is easy to see that the factor models in Chapter 3 are part of it. HJM represent each particular forward rate process $(f(t, T))_{t \in [0, T]}$ as an Itô process based on a fixed finite dimensional Brownian motion. We follow their terminology but will allow for an infinite dimensional underlying Brownian motion.

4.1 Term Structure Movements

Let $W = (\beta^j)_{j \in \mathbb{N}}$ denote an infinite dimensional $(\mathcal{F}_t)$ -Brownian motion and define the triangle subset of $\mathbb{R}^2$

$$\Delta^2 := \{(t, T) \in \mathbb{R}^2 \mid 0 \leq t \leq T\}.$$

We recapture the HJM setup [35, Condition C.1] for the term structure movements. Let $\alpha(t, T, \omega)$ and $(\sigma^j(t, T, \omega))_{j \in \mathbb{N}}$ be measurable mappings from $(\Delta^2 \times \Omega, \mathcal{B}(\Delta^2) \otimes \mathcal{F})$ into $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $(\ell^2, \mathcal{B}(\ell^2))$, respectively. We assume that for each $T \in \mathbb{R}_+$, $\sigma(\cdot, T) \in \mathcal{L}_T^{\text{loc}}(\mathbb{R})$ and $(\alpha(t, T))_{t \in [0, T]}$ is predictable such that $\alpha(\cdot, T) \in L^1([0, T])$, $\mathbb{P}$ -a.s. Then for fixed, but arbitrary $T \in \mathbb{R}_+$, the forward rate for date T evolves as the Itô process

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) ds + \sum_{j \in \mathbb{N}} \int_0^t \sigma^j(s, T) d\beta_s^j, \quad t \in [0, T]. \quad (4.1)$$

Here $f(0, \cdot) \in L_{\text{loc}}^1(\mathbb{R}_+)$ is a non-random initial forward curve. In the classical HJM framework we have $\sigma^j \equiv 0$ for $j > d$, for some $d \in \mathbb{N}$.

We emphasize that (4.1) represents a system of infinitely many Itô processes indexed by $T \in \mathbb{R}_+$. In what follows we tend to transform this into *one* infinite dimensional process.

HJM give sufficient technical conditions (see [35, Conditions C.2 and C.3]) such that the bond prices $P(t, T)$ and the savings account $B(t)$ follow Itô processes. We shall reformulate them subsequently within a functional analytic framework.

4.2 The Musiela Parametrization

In Section 1.3 we have already encountered the advantages of the Musiela [46] parametrization $r_t(x) = f(t, x+t)$ where $x \geq 0$ denotes time to maturity. We now shall see how the HJM dynamics (4.1) can be expressed in the language of stochastic equations in infinite dimensions introduced in Section 2.4.

Let $\{S(t) \mid t \in \mathbb{R}_+\}$ denote the semigroup of right shifts which is defined by $S(t)f(x) = f(x+t)$, for any function $f: \mathbb{R}_+ \rightarrow \mathbb{R}$.

Fix $t, x \in \mathbb{R}_+$. Then (4.1) can be simply rewritten, for T replaced by $x+t$,

$$\begin{aligned} f(t, x+t) &= S(t)f(0, x) + \int_0^t S(t-s)\alpha(s, x+s) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t S(t-s)\sigma^j(s, x+s) d\beta_s^j \end{aligned} \quad (4.2)$$

where $S(t)$ acts on $f(0, x)$, $\alpha(s, x+s)$ and $\sigma^j(s, x+s)$ as functions in x . We shall work out in an axiomatic way the minimal requirements on a Hilbert space H such that (4.2) can be given a meaning when

$$r_t := f(t, \cdot + t), \quad t \in \mathbb{R}_+$$

is considered as an H -valued process. Also the HJM no-arbitrage condition [35, Condition C.4] has to be recaptured. In particular, this means that $(P(t, T))_{t \in [0, T]}$ and $(B(t))_{t \in \mathbb{R}_+}$ have to follow Itô processes.

Clearly $H \subset L^1_{\text{loc}}(\mathbb{R}_+)$, by (1.1). Observe that (4.2) is a pointwise equality for all $x \in \mathbb{R}_+$. Hence pointwise evaluation has to be well specified for $h \in H$. Moreover, we have to interchange integration and pointwise evaluation since x appears under the integral sign. Accordingly, we assume

(H1) The functions $h \in H$ are continuous¹ and the pointwise evaluation $\mathcal{J}_x(h) := h(x)$ is a continuous linear functional² on H , for all $x \in \mathbb{R}_+$.

Indeed, it is not very restrictive to assume that the forward curves $f(t, T)$ given by (4.1) are continuous in $T \in [t, \infty)$, see [35, Lemma 2].

Here is an immediate consequence of assumption **(H1)**.

Lemma 4.2.1. *For any $u \in \mathbb{R}_+$ there exists a number $k(u)$ such that*

$$\|\mathcal{J}_x\|_H \leq k(u), \quad \forall x \in [0, u].$$

Furthermore, $(x, h) \mapsto \mathcal{J}_x(h) : \mathbb{R}_+ \times H \rightarrow \mathbb{R}$ is jointly continuous.

¹ That is, for any $h \in H$ there exists a continuous representative. We shall systematically replace the equivalence class h by its continuous representative which we still denote by h .

² Hence $\mathcal{J}_x$ is element of H which we identify with its dual here and subsequently.

Proof. Let $u \in \mathbb{R}_+$. By **(H1)**, $\sup_{x \in [0, u]} |\mathcal{J}_x(h)| < \infty$ for all $h \in H$. Now the Banach–Steinhaus theorem [50, Theorem 2.6] yields the existence of $k(u)$.

Clearly, $x \mapsto \mathcal{J}_x : \mathbb{R}_+ \rightarrow H$ is weakly continuous; that is, continuous with respect to the weak topology on H . The second assertion of the lemma follows by considering

$$|\mathcal{J}_{x_n}(h_n) - \mathcal{J}_x(h)| \leq \|\mathcal{J}_{x_n}\|_H \|h_n - h\|_H + |\mathcal{J}_{x_n}(h) - \mathcal{J}_x(h)|$$

and using the above estimate. $\square$

In view of Section 2.4 we have to require furthermore

(H2) The semigroup $\{S(t) \mid t \in \mathbb{R}_+\}$ is strongly continuous in H with infinitesimal generator denoted by A .

For any $h \in D(A)$ we have $\frac{d}{dt}S(t)h = S(t)Ah$ in H for all $t \in \mathbb{R}_+$, see e.g. [50, Theorem 13.35]. By **(H1)** therefore $\frac{d}{dt}h(t) = Ah(t)$ for all $t \in \mathbb{R}_+$. Hence $Ah = h' \in H$. But this implies $h \in C^1(\mathbb{R}_+)$. We have thus shown

Lemma 4.2.2. *Under **(H1)**–**(H2)** we have*

$$D(A) \subset \{h \in H \cap C^1(\mathbb{R}_+) \mid h' \in H\}.$$

For the coefficients in (4.2) we now formulate the analogon to [35, Condition C.1]. We write $\alpha_t(\omega) := \alpha(t, \omega, \cdot + t)$ and $\sigma = (\sigma^j)_{j \in \mathbb{N}}$ where $\sigma_t^j(\omega) := \sigma^j(t, \omega, \cdot + t)$.

(C1) The initial forward curve $r_0 = f(0, \cdot)$ lies in H .

(C2) The processes α and σ are H -, resp. $L_2^0(H)$ -valued predictable and

$$\mathbb{P} \left[\int_0^t \left(\|\alpha_s\|_H + \|\sigma_s\|_{L_2^0(H)}^2 \right) ds < \infty \right] = 1, \quad \forall t \in \mathbb{R}_+.$$

Now (4.2) reads, for any $t, x \in \mathbb{R}_+$,

$$\begin{aligned} \mathcal{J}_x(r_t) &= \mathcal{J}_x(S(t)r_0) + \int_0^t \mathcal{J}_x(S(t-s)\alpha_s) ds + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{J}_x(S(t-s)\sigma_s^j) d\beta_s^j \\ &= \mathcal{J}_x \left(S(t)r_0 + \int_0^t S(t-s)\alpha_s ds + \sum_{j \in \mathbb{N}} \int_0^t S(t-s)\sigma_s^j d\beta_s^j \right), \quad \mathbb{P}\text{-a.s.} \end{aligned} \tag{4.3}$$

see (2.12). For t fixed, the exceptional $\mathbb{P}$ -nullset in equality (4.3) depends on x . But according to **(H1)** both sides are continuous in x . A standard argument yields equality $\mathbb{P}$ -a.s. simultaneously for all $x \in \mathbb{R}_+$. Pointwise equality of two functions yields identity of the functions, hence

$$r_t = S(t)r_0 + \int_0^t S(t-s)\alpha_s ds + \sum_{j \in \mathbb{N}} \int_0^t S(t-s)\sigma_s^j d\beta_s^j, \quad \mathbb{P}\text{-a.s.} \tag{4.4}$$

By Lemma 2.4.1 the right hand side of (4.4) has a predictable version, which we still denote by r_t . Consequently, r is a mild solution to the stochastic equation

$$\begin{cases} dr_t = (Ar_t + \alpha_t) dt + \sum_{j \in \mathbb{N}} \sigma_t^j d\beta_t^j \\ r_0 = f(0, \cdot). \end{cases} \quad (4.5)$$

We make an additional assumption.

(C3) There exists a continuous version of r , still denoted by r .

Condition **(C3)** is satisfied if $S(t)$ forms a contraction semigroup in H , see [19, Theorem 6.10], or if we require $\sigma \in \mathcal{L}^2(H)$, see [19, Proposition 7.3]. Notice, however, that r is *not* an H -valued Itô process in general.

Remark 4.2.1. Combining **(C2)**, **(C3)** and Lemma 4.2.1 we conclude that the random fields $r_t(\omega, x)$, $\alpha_t(\omega, x)$ and $\sigma_t^j(\omega, x)$ are $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+)$ -measurable and $r_t(\omega, x)$ is jointly continuous in (t, x) for $\mathbb{P}$ -a.e. ω .

So far we have reformulated the classical HJM setup in the spirit of Musiela [46] by putting additional assumptions **(C1)** and **(C2)** on the coefficients (reflecting essentially [35, Conditions C.1-C.3]). Moreover, we allow for an infinite dimensional driving Brownian motion. The process r , defined as the continuous mild solution to (4.5), is related to $f(t, T)$ from (4.1) by

$$r_t(x) = f(t, x+t) \quad \forall x \in \mathbb{R}_+, \quad \mathbb{P}\text{-a.s.} \quad \forall t \in \mathbb{R}_+.$$

From now on, the bond prices and the savings account are given by

$$\begin{aligned} P(t, T) &= \exp \left(- \int_0^{T-t} r_t(x) dx \right), \quad (t, T) \in \Delta^2 \\ B(t) &= \exp \left(\int_0^t r_s(0) ds \right), \quad t \in \mathbb{R}_+. \end{aligned}$$

4.3 Arbitrage-free Term Structure Movements

In this section we derive the dynamics of the discounted bond prices. Since r is not an H -valued Itô process, we cannot use Itô's formula directly but we rather have to deal with the random field $(r_t(x))_{t,x \in \mathbb{R}_+}$ and apply the stochastic Fubini theorem. We essentially adapt the technique from [35]. The notion of an (E)LMM ((equivalent) local martingale measure) is exactly as in Definition 3.1.1. Sufficient conditions for the existence of an ELMM are provided.

First we introduce the linear functional $\mathcal{I}_u$ on H defined by

$$\mathcal{I}_u(h) := \int_0^u h(x) dx, \quad u \in \mathbb{R}_+.$$

There is an immediate consequence of assumption **(H1)**.

Lemma 4.3.1. *The linear functional $\mathcal{I}_u$ is continuous on H . In particular,*

$$\|\mathcal{I}_u\|_H \leq uk(u), \quad \forall u \in \mathbb{R}_+$$

where $k(u)$ is the constant from Lemma 4.2.1.

Moreover, the mapping $(u, h) \mapsto \mathcal{I}_u(h) : \mathbb{R}_+ \times H \rightarrow \mathbb{R}$ is jointly continuous.

Proof. The estimate is trivial. The second assertion follows as in the proof of Lemma 4.2.1 by the weak continuity of $u \mapsto \mathcal{I}_u : \mathbb{R}_+ \rightarrow H$. $\square$

It follows from (C3) and Lemma 4.3.1 that $P(t, T) = \exp(-\mathcal{I}_{T-t}(r_t))$ is $\mathcal{F}_t$ -measurable and jointly continuous in $(t, T) \in \Delta^2$.

Fix $(t, T) \in \Delta^2$. Then we have by (4.4), Lemma 4.3.1 and (2.12)

$$\begin{aligned} I &:= -\log P(t, T) \\ &= \mathcal{I}_{T-t}(S(t)r_0) + \int_0^t \mathcal{I}_{T-t}(S(t-s)\alpha_s) ds + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_{T-t}(S(t-s)\sigma_s^j) d\beta_s^j \\ &= \mathcal{I}_T(r_0) - \mathcal{I}_t(r_0) + \int_0^t (\mathcal{I}_{T-s}(\alpha_s) - \mathcal{I}_{t-s}(\alpha_s)) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t (\mathcal{I}_{T-s}(\sigma_s^j) - \mathcal{I}_{t-s}(\sigma_s^j)) d\beta_s^j, \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

where we took into account the obvious relation $\mathcal{I}_u \circ S(t) = \mathcal{I}_{t+u} - \mathcal{I}_t$. The processes $(\mathcal{I}_{T-s}(\sigma_s^j))_{s \in [0, T]}$ are predictable and

$$\sum_{j \in \mathbb{N}} |\mathcal{I}_{T-s}(\sigma_s^j(\omega))|^2 \leq (Tk(T))^2 \sum_{j \in \mathbb{N}} \|\sigma_s^j(\omega)\|_H^2 < \infty, \quad \forall (s, \omega) \in [0, T] \times \Omega.$$

Hence $(\mathcal{I}_{T-s} \circ \sigma_s)_{s \in [0, T]}$ is ℓ^2 -valued predictable. Similarly

$$\mathbb{P} \left[\int_0^T \sum_{j \in \mathbb{N}} |\mathcal{I}_{T-s}(\sigma_s^j)|^2 ds < \infty \right] = 1,$$

thus $(\mathcal{I}_{T-s} \circ \sigma_s) \in \mathcal{L}_T^{\text{loc}}(\mathbb{R})$. Replacing T by t , also $(\mathcal{I}_{t-s} \circ \sigma_s)_{s \in [0, t]} \in \mathcal{L}_t^{\text{loc}}(\mathbb{R})$. By linearity we can thus split up the integrals and write

$$I = I_1 - I_2, \quad \mathbb{P}\text{-a.s.}$$

where

$$\begin{aligned} I_1 &:= \mathcal{I}_T(r_0) + \int_0^t \mathcal{I}_{T-s}(\alpha_s) ds + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_{T-s}(\sigma_s^j) d\beta_s^j \\ I_2 &:= \mathcal{I}_t(r_0) + \int_0^t \mathcal{I}_{t-s}(\alpha_s) ds + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_{t-s}(\sigma_s^j) d\beta_s^j. \end{aligned}$$

The integrals in I_2 need to be transformed. We introduce the mappings

$$\tilde{\sigma}_s^j(\omega, u) := \begin{cases} \sigma_s^j(\omega, u - s), & \text{if } s \leq u \\ 0, & \text{otherwise.} \end{cases}$$

Substituting $u = x + s$, we have

$$\mathcal{I}_{t-s}(\sigma_s^j) = \int_0^{t-s} \sigma_s^j(x) dx = \int_s^t \sigma_s^j(u - s) du = \int_0^t \tilde{\sigma}_s^j(u) du.$$

In view of Remark 4.2.1, $\tilde{\sigma} = (\tilde{\sigma}^j)_{j \in \mathbb{N}}$ is measurable from $(\mathbb{R}_+ \times \Omega \times \mathbb{R}_+, \mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+))$ into $(\ell^2, \mathcal{B}(\ell^2))$. Moreover, by Lemma 4.2.1 and (C2)

$$\begin{aligned} \int_0^t \int_0^t \|\tilde{\sigma}_s(u)\|_{\ell^2}^2 du ds &= \int_0^t \int_0^t \sum_{j \in \mathbb{N}} |\tilde{\sigma}_s^j(u)|^2 du ds \\ &= \int_0^t \sum_{j \in \mathbb{N}} \left(\int_0^{t-s} |\sigma_s^j(x)|^2 dx \right) ds \\ &\leq t(k(t))^2 \int_0^t \sum_{j \in \mathbb{N}} \|\sigma_s^j\|_H^2 ds < \infty, \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

Consequently, the stochastic Fubini theorem 2.3.2 applies and, by (2.10)–(2.11),

$$\begin{aligned} \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_{t-s}(\sigma_s^j) d\beta_s^j &= \sum_{j \in \mathbb{N}} \int_0^t \left(\int_0^t \tilde{\sigma}_s^j(u) du \right) d\beta_s^j \\ &= \int_0^t \left(\sum_{j \in \mathbb{N}} \int_0^t \tilde{\sigma}_s^j(u) d\beta_s^j \right) du \\ &= \int_0^t \left(\sum_{j \in \mathbb{N}} \int_0^u \sigma_s^j(u - s) d\beta_s^j \right) du, \quad \mathbb{P}\text{-a.s.} \end{aligned} \tag{4.6}$$

Using the ordinary Fubini theorem we derive similarly

$$\int_0^t \mathcal{I}_{t-s}(\alpha_s) ds = \int_0^t \left(\int_0^u \alpha_s(u - s) ds \right) du, \quad \mathbb{P}\text{-a.s.} \tag{4.7}$$

Combining (4.6) and (4.7) we can write, by (2.12),

$$\begin{aligned} I_2 &= \int_0^t \mathcal{J}_0 \left(S(u)r_0 + \int_0^u S(u - s)\alpha_s ds + \sum_{j \in \mathbb{N}} \int_0^u S(u - s)\sigma_s^j d\beta_s^j \right) du \\ &= \int_0^t r_u(0) du, \quad \mathbb{P}\text{-a.s.} \end{aligned}$$

Taking into account $\mathcal{I}_T(r_0) = -\log P(0, T)$ we arrive at the following representation

$$\begin{aligned} \log P(t, T) &= \log P(0, T) + \int_0^t (r_s(0) - \mathcal{I}_{T-s}(\alpha_s)) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t (-\mathcal{I}_{T-s}(\sigma_s^j)) d\beta_s^j, \quad \mathbb{P}\text{-a.s.} \quad \forall (t, T) \in \Delta^2. \end{aligned}$$

Since $P(t, T)$ is jointly continuous in $(t, T) \in \Delta^2$, the $\mathbb{P}$ -nullset can be chosen for each $T \in \mathbb{R}_+$ independently of $t \in [0, T]$. Whence $(\log P(t, T))_{t \in [0, T]}$ is a real-valued Itô process. Accordingly, so is $(\log P(t, T) - \log B(t))_{t \in [0, T]}$. Applying Itô's formula to e^x gives for the bond price process

$$\begin{aligned} P(t, T) &= P(0, T) + \int_0^t P(s, T) \left(r_s(0) - \mathcal{I}_{T-s}(\alpha_s) + \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_{T-s}(\sigma_s^j))^2 \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t P(s, T) (-\mathcal{I}_{T-s}(\sigma_s^j)) d\beta_s^j, \quad t \in [0, T] \end{aligned} \tag{4.8}$$

and for the discounted bond price process $Z(t, T) := P(t, T)/B(t)$

$$\begin{aligned} Z(t, T) &= P(0, T) + \int_0^t Z(s, T) \left(-\mathcal{I}_{T-s}(\alpha_s) + \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_{T-s}(\sigma_s^j))^2 \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t Z(s, T) (-\mathcal{I}_{T-s}(\sigma_s^j)) d\beta_s^j, \quad t \in [0, T]. \end{aligned} \tag{4.9}$$

The following mapping will play an important role. For any continuous function f on $\mathbb{R}_+$ we define the continuous function $\mathcal{S}f : \mathbb{R}_+ \rightarrow \mathbb{R}$ by

$$(\mathcal{S}f)(x) := f(x) \int_0^x f(\eta) d\eta, \quad x \in \mathbb{R}_+. \tag{4.10}$$

Here is an elementary result on $\mathcal{S}$.

Lemma 4.3.2. *Let $\Phi = (\Phi^j)_{j \in \mathbb{N}} \in L_2^0(H)$. Then $f(x) := \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_x(\Phi^j))^2$ is a continuously differentiable function in $x \in \mathbb{R}_+$ with*

$$f'(x) = \sum_{j \in \mathbb{N}} (\mathcal{S}\Phi^j)(x).$$

Both series converge uniformly in x on compacts.

Proof. Set $f_n(x) := \frac{1}{2} \sum_{j=1}^n (\mathcal{I}_x(\Phi^j))^2$ which is obviously continuously differentiable in $x \in \mathbb{R}_+$ with

$$f'_n(x) = \sum_{j=1}^n (\mathcal{S}\Phi^j)(x).$$

Moreover, for $m < n \in \mathbb{N}$

$$|f_n(x) - f_m(x)| + |f'_n(x) - f'_m(x)| \leq (\|\mathcal{I}_x\|_H^2 + \|\mathcal{J}_x\|_H \|\mathcal{I}_x\|_H) \sum_{j=m+1}^n \|\Phi^j\|_H^2$$

which tends to zero for $m, n \rightarrow \infty$ uniformly in x on compacts, by Lemmas 4.2.1 and 4.3.1. Consequently, the function $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ exists and is continuously differentiable in $x \in \mathbb{R}_+$ with $f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$. $\square$

Below we will consider a particular Girsanov transformation from $\mathbb{P}$ to some ELMM $\mathbb{Q}$ which only affects a change in the drift α for the $\mathbb{Q}$ -dynamics of r . So let us pretend for the moment that $\mathbb{P}$ itself is a local martingale measure. Then the HJM drift condition [35, (18)] reads as follows.

Lemma 4.3.3. *The discounted bond price process $(Z(t, T))_{t \in [0, T]}$ follows a local martingale, for all $T \in \mathbb{R}_+$, if and only if*

$$\alpha = \sum_{j \in \mathbb{N}} \mathcal{S}\sigma^j, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (4.11)$$

Proof. Let $T \in \mathbb{R}_+$. By [49, Proposition (1.2)], $(Z(t, T))_{t \in [0, T]}$ is a local martingale if and only if the integral with respect to ds in (4.9) is indistinguishable from zero. Since $Z(s, T) > 0$, this is equivalent to

$$\mathcal{I}_{T-t}(\alpha_t) = \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_{T-t}(\sigma_t^j))^2, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (4.12)$$

see the proof of (3.6). The $dt \otimes d\mathbb{P}$ -nullset depends on T . But by Lemma 4.3.2 the right hand side of (4.12) is continuously differentiable in $T \geq t$ and so is obviously the left hand side. By a standard argument we find a $dt \otimes d\mathbb{P}$ -nullset $\mathcal{N} \in \mathcal{P}$ with the property that

$$\mathcal{I}_x(\alpha_t(\omega)) = \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_x(\sigma_t^j(\omega)))^2, \quad \forall (t, \omega, x) \in \mathcal{N}^c \times \mathbb{R}_+.$$

Differentiation yields the assertion. $\square$

Observe that Lemma 4.3.3 imposes through (4.11) implicitly a measurability and integrability condition on $\sum_{j \in \mathbb{N}} \mathcal{S}\sigma^j$. Their validity is guaranteed by the following pair of assumptions.

(C4) The processes $\mathcal{S}\sigma^j$ are H -valued predictable.

(H3) There exists a constant K such that

$$\|Sh\|_H \leq K\|h\|_H^2$$

for all $h \in H$ with $Sh \in H$.

In general, $\mathbb{P}$ is considered as physical measure. Accordingly, the conditions of Lemma 4.3.3 are not satisfied under $\mathbb{P}$. The next condition assures the existence of an ELMM, see [35, Condition C.4].

(C5) There exists an ℓ^2 -valued predictable process $\gamma = (\gamma^j)_{j \in \mathbb{N}}$ satisfying Novikov's condition (2.18) and

$$\sum_{j \in \mathbb{N}} \gamma^j \sigma^j = \sum_{j \in \mathbb{N}} \mathcal{S}\sigma^j - \alpha, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (4.13)$$

By Hölder's inequality and Lemma 4.2.1, the series

$$\mathcal{J}_x \left(\sum_{j \in \mathbb{N}} \gamma_t^j \sigma_t^j \right) = \sum_{j \in \mathbb{N}} \gamma_t^j \mathcal{J}_x(\sigma_t^j)$$

converges uniformly in x on compacts. Taking into account Lemma 4.3.2 and integrating, we see that (4.13) is equivalent to

$$\sum_{j \in \mathbb{N}} \gamma_t^j \mathcal{I}_{T-t}(\sigma_t^j) = \frac{1}{2} \sum_{j \in \mathbb{N}} \left(\mathcal{I}_{T-t}(\sigma_t^j) \right)^2 - \mathcal{I}_{T-t}(\alpha_t), \quad \forall T \geq t, \quad dt \otimes d\mathbb{P}\text{-a.s.} \quad (4.14)$$

There is an intuitive interpretation of $-\gamma$ as *market price of risk*, by considering the bond price dynamics (4.8): after localization, the right hand side of (4.14) is, roughly speaking, the difference between the

$$\text{"average return } \mathbb{E} \left[\frac{dP(t, T)}{P(t, T)} \right] \text{"}$$

of the bond and the "risk-less rate" $r_t(0)$. Whence the interpretation of $-\gamma_t^j$ as market price of risk in units of the volatility, $-\mathcal{I}_{T-t}(\sigma_t^j)$, for the j -th noise factor. Whence also the expression *risk neutral measure* for $\mathbb{Q}$.

Condition (C5) allows for Lemma 2.3.2. Hence we can define $\mathbb{Q} \sim \mathbb{P}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \mathcal{E}(\gamma \cdot W)_\infty,$$

recall the definition of the stochastic exponential (2.17). Girsanov's theorem, see Proposition 2.3.1, yields that

$$\tilde{W}_t = W_t - \int_0^t \gamma_s ds, \quad t \in \mathbb{R}_+$$

is a Wiener process for the measure $\mathbb{Q}$. The dynamics of r change accordingly.

Proposition 4.3.1. *Under the above assumptions, r satisfies*

$$r_t = S(t)r_0 + \int_0^t \left(S(t-s) \sum_{j \in \mathbb{N}} S \sigma_s^j \right) ds + \sum_{j \in \mathbb{N}} \int_0^t S(t-s) \sigma_s^j d\tilde{\beta}_s^j \quad (4.15)$$

with respect to $\mathbb{Q}$. Consequently, $\mathbb{Q}$ is an ELMM.

Proof. The statement on (r_t) follows by combining (2.19), (4.4) and (4.13). Lemma 4.3.3 yields the assertion on $\mathbb{Q}$. $\square$

4.4 Contingent Claim Valuation

This section sketches how to value contingent claims in the preceding bond market. Yet we do not address here the question of completeness, which essentially is equivalent to uniqueness of the ELMM $\mathbb{Q}$. Since the underlying noise is an infinite dimensional Brownian motion, the notion of infinite dimensional admissible trading strategies were convenient. This, however, is beyond the scope of these notes and will be exploited elsewhere. For a treatment of the finite dimensional case we refer the reader to [47, Section 10.1]. Measure-valued trading strategies are introduced in [10].

We content ourself in postulating that $\mathbb{Q}$ is *the* risk neutral measure (equivalently, $-\gamma$ is *the* market price of risk) and all prices are computed as expectations under $\mathbb{Q}$. This is justified at least if all discounted bond price processes $(Z(t, T))_{t \in [0, T]}$ follow true $\mathbb{Q}$ -martingales. Since then any attainable claim (if ever defined in a reasonable way) will admit a unique arbitrage price. It is determined by the replicating strategy.

The subsequent introduction of the forward measure, the forward LIBOR rates and caplets is preliminary for the discussion of the popular BGM model in Section 5.6.

4.4.1 When Is $Z(\cdot, T)$ a True $\mathbb{Q}$ -Martingale?

We can give a sufficient condition for the discounted bond price process to follow a true $\mathbb{Q}$ -martingale. The dynamics (4.9) read under $\mathbb{Q}$

$$Z(t, T) = P(0, T) + \sum_{j \in \mathbb{N}} \int_0^t Z(s, T) (-\mathcal{I}_{T-s}(\sigma_s^j)) d\tilde{\beta}_s^j, \quad t \in [0, T],$$

see (4.14). Itô's formula yields the representation as the stochastic exponential

$$Z(t, T) = P(0, T) \mathcal{E} \left(\sum_{j \in \mathbb{N}} \int_0^t (-\mathcal{I}_{T-s}(\sigma_s^j)) d\tilde{\beta}_s^j \right)_t. \quad (4.16)$$

But now we can use Novikov's criterion, see Lemma 2.3.2, to derive

Lemma 4.4.1. *If*

$$\mathbb{E}_{\mathbb{Q}} \left[\exp \left(\frac{1}{2} \int_0^T \|\mathcal{I}_{T-t} \circ \sigma_t\|_{\ell^2}^2 dt \right) \right] < \infty$$

then $(Z(t, T))_{t \in [0, T]}$ is a $\mathbb{Q}$ -martingale.

Another sufficient criterion is positivity of the forward rates. Indeed, if $r_t \geq 0$ for all $t \in [0, T]$ then $0 \leq Z(t, T) \leq 1$ for all $t \in [0, T]$. Moreover, a bounded local martingale is a uniformly integrable martingale. Positivity of r is related to stochastic invariance of the convex cone of non-negative functions h in H under the dynamics (4.5). We do not address this problem here and refer the reader to [43].

4.4.2 The Forward Measure

For pricing purposes it is convenient to use a technique called “change of numéraire”, see [21] for a thorough treatment. We give an application for pricing bond options.

Assume that $(Z(t, T))_{t \in [0, T]}$ is a $\mathbb{Q}$ -martingale. We can define the measure $\mathbb{Q}^T \sim \mathbb{Q}$ on $(\Omega, \mathcal{F}_T)$ by

$$\frac{d\mathbb{Q}^T}{d\mathbb{Q}} = \frac{1}{P(0, T)B(T)}.$$

By (4.16) we then have

$$\mathbb{E}_{\mathbb{Q}} \left[\frac{d\mathbb{Q}^T}{d\mathbb{Q}} \mid \mathcal{F}_t \right] = \frac{Z(t, T)}{P(0, T)} = \mathcal{E} \left(\sum_{j \in \mathbb{N}} \int_0^{\cdot} (-\mathcal{I}_{T-s}(\sigma_s^j)) d\tilde{\beta}_s^j \right)_t, \quad t \in [0, T]. \quad (4.17)$$

Definition 4.4.1. *The measure $\mathbb{Q}^T$ is called the T -forward measure.*

A claim X due at time T is in $L^1(\Omega, \mathcal{F}_T, \mathbb{Q}^T)$ if and only if $X/B(T) \in L^1(\Omega, \mathcal{F}_T, \mathbb{Q})$. Bayes’ rule, see e.g. [39, Lemma 5.3, Chap. 3], yields the equality

$$P(t, T) \mathbb{E}_{\mathbb{Q}^T}[X \mid \mathcal{F}_t] = B(t) \mathbb{E}_{\mathbb{Q}} \left[\frac{X}{B(T)} \mid \mathcal{F}_t \right], \quad \mathbb{Q}\text{-a.s.} \quad \forall t \in [0, T]. \quad (4.18)$$

On the right hand side stands the time t price of the claim X . This is a very useful relation, since the left hand side allows for closed form expressions if X has a nice distribution under $\mathbb{Q}^T$. We will later use it for pricing a cap.

4.4.3 Forward LIBOR Rates

Let us introduce an important $\mathbb{Q}^T$ -local martingale. We will use the notation $\mathcal{I}_u^v := \mathcal{I}_v - \mathcal{I}_u$ for $(u, v) \in \Delta^2$. Fix $\delta > 0$.

Definition 4.4.2. *The forward δ -period³ LIBOR rate for the future date T prevailing at time t is*

$$L(t, T) := \frac{1}{\delta} \left(e^{\mathcal{I}_{T-t}^{T+\delta-t}(r_t)} - 1 \right), \quad (t, T) \in \Delta^2.$$

We can re-express $L(t, T)$ directly in term of bond prices

$$1 + \delta L(t, T) = \frac{P(t, T)}{P(t, T + \delta)}.$$

Whence the meaning of $L(t, T)$ as forward average simple rate for a loan over the future time period $[T, T + \delta]$.

Let $T \in \mathbb{R}_+$ and write $\ell(t, T) := \mathcal{I}_{T-t}^{T+\delta-t}(r_t)$. Combining Lemma 4.3.1 and (4.15) we conclude that $(\ell(t, T))_{t \in [0, T]}$ is a real-valued Itô process and

$$\begin{aligned} \ell(t, T) &= \ell(0, T) + \int_0^t \mathcal{I}_{T-t}^{T+\delta-t} \left(S(t-s) \sum_{j \in \mathbb{N}} S \sigma_s^j \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_{T-t}^{T+\delta-t} (S(t-s) \sigma_s^j) d\tilde{\beta}_s^j \\ &= \ell(0, T) + \frac{1}{2} \int_0^t \sum_{j \in \mathbb{N}} ((\mathcal{I}_{T+\delta-s}(\sigma_s^j))^2 - (\mathcal{I}_{T-s}(\sigma_s^j))^2) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_{T-s}^{T+\delta-s}(\sigma_s^j) d\tilde{\beta}_s^j, \quad t \in [0, T]. \end{aligned}$$

We have used the relation $\mathcal{I}_{T-t}^{T+\delta-t} \circ S(t-s) = \mathcal{I}_{T-s}^{T+\delta-s}$ and Lemma 4.3.2. Itô's formula yields

$$\begin{aligned} L(t, T) &= L(0, T) \\ &\quad + \frac{1}{2\delta} \int_0^t e^{\ell(s, T)} \sum_{j \in \mathbb{N}} ((\mathcal{I}_{T+\delta-s}(\sigma_s^j))^2 - (\mathcal{I}_{T-s}(\sigma_s^j))^2 \\ &\quad \quad \quad + (\mathcal{I}_{T-s}^{T+\delta-s}(\sigma_s^j))^2) ds \\ &\quad + \sum_{j \in \mathbb{N}} \frac{1}{\delta} \int_0^t e^{\ell(s, T)} \mathcal{I}_{T-s}^{T+\delta-s}(\sigma_s^j) d\tilde{\beta}_s^j \end{aligned}$$

Hence

³ Typically, δ stands for 3 or 6 months.

$$\begin{aligned}
L(t, T) = L(0, T) + \frac{1}{\delta} \int_0^t e^{\ell(s, T)} \sum_{j \in \mathbb{N}} (\mathcal{I}_{T-s}^{T+\delta-s}(\sigma_s^j) \mathcal{I}_{T+\delta-s}(\sigma_s^j)) ds \\
+ \sum_{j \in \mathbb{N}} \frac{1}{\delta} \int_0^t e^{\ell(s, T)} \mathcal{I}_{T-s}^{T+\delta-s}(\sigma_s^j) d\tilde{\beta}_s^j, \quad t \in [0, T].
\end{aligned}$$

Now assume that $(Z(t, T + \delta))_{t \in [0, T+\delta]}$ is a $\mathbb{Q}$ -martingale. Using (4.17) and Girsanov's theorem 2.3.3 we conclude that

$$\beta_t^{(T+\delta), j} = \tilde{\beta}_t^j + \int_0^t \mathcal{I}_{T+\delta-s}(\sigma_s^j) ds, \quad t \in [0, T + \delta], \quad j \in \mathbb{N}$$

are independent $(\mathcal{F}_t)_{t \in [0, T+\delta]}$ -Brownian motions under $\mathbb{Q}^{T+\delta}$. Consequently, the process $(L(t, T))_{t \in [0, T]}$ is a $\mathbb{Q}^{T+\delta}$ -local martingale with representation

$$L(t, T) = L(0, T) + \sum_{j \in \mathbb{N}} \frac{1}{\delta} \int_0^t e^{\ell(s, T)} \mathcal{I}_{T-s}^{T+\delta-s}(\sigma_s^j) d\beta_s^{(T+\delta), j}, \quad t \in [0, T]. \quad (4.19)$$

4.4.4 Caps

For a thorough introduction to interest rate derivatives, such as swaps, caps, floors and swaptions, we refer to [47, Chap. 16]. We restrict our consideration to the following basic instrument.

A *caplet* with reset date T and settlement date $T + \delta$ pays the holder the difference between the LIBOR $L(T, T)$ and the strike rate κ . Its cash-flow at time $T + \delta$ is

$$\delta(L(T, T) - \kappa)^+.$$

A *cap* is a strip of caplets. If we can price caplets, we can price caps. The arbitrage price of the caplet at time $t \leq T$ equals, see [47, p. 391],

$$\begin{aligned}
Cpl(t) &= B(t) \mathbb{E}_{\mathbb{Q}} \left[\frac{1}{B(T + \delta)} \delta(L(T, T) - \kappa)^+ \mid \mathcal{F}_t \right] \\
&= P(t, T + \delta) \mathbb{E}_{\mathbb{Q}^{T+\delta}} [\delta(L(T, T) - \kappa)^+ \mid \mathcal{F}_t],
\end{aligned}$$

where we have used (4.18).

Suppose that there exists a deterministic ℓ^2 -valued measurable function $\pi(\cdot, T) = (\pi^j(\cdot, T))_{j \in \mathbb{N}}$ with $\int_0^T \|\pi(t, T)\|_{\ell^2}^2 dt < \infty$ and such that

$$\mathcal{I}_{T-t}^{T+\delta-t}(\sigma_t^j) = (1 - e^{-\ell(t, T)}) \pi^j(t, T), \quad dt \otimes d\mathbb{Q}\text{-a.s. on } [0, T] \times \Omega. \quad (4.20)$$

Then (4.19) reads

$$L(t, T) = L(0, T) + \sum_{j \in \mathbb{N}} \int_0^t L(s, T) \pi^j(s, T) d\beta_s^{(T+\delta), j}, \quad t \in [0, T].$$

This equation is uniquely solved by the stochastic exponential

$$L(t, T) = L(0, T) \mathcal{E} \left(\sum_{j \in \mathbb{N}} \int_0^{\cdot} \pi^j(s, T) d\beta_s^{(T+\delta), j} \right)_t.$$

By [19, Lemma 10.15] there exists a real-valued standard $(\mathcal{F}_t)_{t \in [0, T]}$ -Brownian motion $(\beta_t^0)_{t \in [0, T]}$ under $\mathbb{Q}^{T+\delta}$ such that

$$\sum_{j \in \mathbb{N}} \int_0^t \pi^j(s, T) d\beta_s^{(T+\delta), j} = \int_0^t \|\pi(s, T)\|_{\ell^2}^2 d\beta_s^0, \quad t \in [0, T].$$

Accordingly, $L(T, T)$ is lognormally distributed under $\mathbb{Q}^{T+\delta}$, and the following pricing formula holds, see [47, Lemma 16.3.1].

Lemma 4.4.2. *Under the above assumptions, the time t price of the caplet equals*

$$Cpl(t) = \delta P(t, T + \delta) (L(t, T) N(e_1(t, T)) - \kappa N(e_2(t, T))) \quad (4.21)$$

where

$$e_{1,2} := \frac{\log \left(\frac{L(t, T)}{\kappa} \right) \pm \frac{1}{2} v_0^2(t, T)}{v_0(t, T)}$$

$$v_0^2(t, T) := \int_t^T \|\pi(s, T)\|_{\ell^2}^2 ds.$$

and N stands for the standard Gaussian cumulative distribution function.

BGM [13] present a method for modelling the forward LIBOR rates such that (4.20) holds for all $T \in \mathbb{R}_+$. We will recapture their approach in the present setting in Section 5.6.

4.5 What Is a Model?

Up to now we have analyzed the stochastic behavior of a single generic forward curve process r within the HJM framework. For modelling purposes we rather want the stochastic evolution being described by a dynamical system like (4.5) with state dependent coefficients $\alpha_t(\omega) = \alpha(t, \omega, r_t(\omega))$ and $\sigma_t(\omega) = \sigma(t, \omega, r_t(\omega))$, which allows for arbitrary initial curves r_0 .

From the last section it seems plausible to specify as model ingredients the volatility structure σ and a market price of risk vector $-\gamma$. We then let the drift α being defined by (4.13). Proposition 4.3.1 accordingly assures the existence of an ELMM $\mathbb{Q}$.

As it can be seen from (4.15), γ does not appear in the $\mathbb{Q}$ -dynamics of r . Thus, instead of solving (4.5) directly we rather start with the Wiener process $\tilde{W}$ under $\mathbb{Q}$, solve (4.15) and transform $\tilde{W} \rightsquigarrow W$, by Girsanov's theorem, to have γ back in the drift for the $\mathbb{P}$ -dynamics of r . Compare with [35, Lemma 1]. That way, however, the measure $\mathbb{P}$ and the Wiener process W will depend on r implicitly by $\gamma_t(\omega) = \gamma(t, \omega, r_t(\omega))$. Yet W will be adapted to $(\mathcal{F}_t)$. Which is convenient since in this chapter we never have needed the filtration $(\mathcal{F}_t)$ being generated by W .

Henceforth we are given a complete filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathbb{R}_+}, \mathbb{Q})$$

satisfying the usual conditions (see at the beginning of Chap. 2, in particular $\mathcal{F} = \mathcal{F}_\infty$). The measure $\mathbb{Q}$ will be considered as the risk neutral measure. Let $\tilde{W} = (\tilde{\beta}^j)_{j \in \mathbb{N}}$ be an infinite dimensional $(\mathcal{F}_t)$ -Brownian motion under $\mathbb{Q}$. The random mappings $\sigma(t, \omega, h)$ (volatility structure) and $-\gamma(t, \omega, h)$ (market price of risk) are specified in accordance with the following assumptions.

- (D1) The mappings σ and γ are measurable from $(\mathbb{R}_+ \times \Omega \times H, \mathcal{P} \otimes \mathcal{B}(H))$ into $(L_2^0(H), \mathcal{B}(L_2^0(H)))$, resp. $(\ell^2, \mathcal{B}(\ell^2))$.
- (D2) The mappings $\mathcal{S}\sigma^j$ are measurable from $(\mathbb{R}_+ \times \Omega \times H, \mathcal{P} \otimes \mathcal{B}(H))$ into $(H, \mathcal{B}(H))$.
- (D3) There exists a function $\Gamma \in L^2(\mathbb{R}_+)$ such that

$$\|\gamma(t, \omega, h)\|_{\ell^2} \leq \Gamma(t), \quad \forall (t, \omega, h) \in \mathbb{R}_+ \times \Omega \times H.$$

Observe that assumptions (D1)–(D2) together with (H3) imply that

$$\alpha_{HJM}(t, \omega, h) := \sum_{j \in \mathbb{N}} \mathcal{S}\sigma^j(t, \omega, h) \quad (4.22)$$

is a measurable mapping from $(\mathbb{R}_+ \times \Omega \times H, \mathcal{P} \otimes \mathcal{B}(H))$ into $(H, \mathcal{B}(H))$.

Definition 4.5.1. *By the HJMM equation (HJM equation in the Musiela parametrization) we mean the following stochastic equation⁴ in H*

$$\begin{cases} dr_t = (Ar_t + \alpha_{HJM}(t, r_t)) dt + \sigma(t, r_t) d\tilde{W}_t \\ r_0 = h_0. \end{cases} \quad (4.23)$$

Definition 4.5.2. *Let σ, γ satisfy (D1)–(D3) and let $I \subset H$ be a set of initial forward curves. We call (σ, γ, I) a (local) HJM model in H if for every space time initial point $(t_0, h_0) \in \mathbb{R}_+ \times I$ there exists a unique continuous (local) mild solution $r = r^{(t_0, h_0)}$ to the time t_0 -shifted⁵ HJMM equation (4.23).*

⁴ See Remark 2.4.1 for notation.

⁵ recall (2.25)

This definition is justified by the following result.

Theorem 4.5.1. *Let (σ, γ, I) be an HJM model. Then any space time initial point $(t_0, h_0) \in \mathbb{R}_+ \times I$ specifies a measure $\mathbb{P} \sim \mathbb{Q}$ and a Wiener process W with respect to $(\Omega, \mathcal{F}, (\mathcal{F}_t^{(t_0)})_{t \in \mathbb{R}_+}, \mathbb{P})$ such that (C1)–(C5) hold for*

$$\alpha_t(\omega) = \alpha_{HJM}(t_0 + t, \omega, r_t(\omega)) - \sum_{j \in \mathbb{N}} \gamma^j(t_0 + t, \omega, r_t(\omega)) \sigma^j(t_0 + t, \omega, r_t(\omega))$$

$$\sigma_t(\omega) = \sigma(t_0 + t, \omega, r_t(\omega))$$

$$\gamma_t(\omega) = \gamma(t_0 + t, \omega, r_t(\omega))$$

where $r = r^{(t_0, h_0)}$. Accordingly, $\mathbb{Q}$ is an ELMM for this setup.

Proof. Let $r = r^{(t_0, h_0)}$ be the continuous mild solution to the time t_0 -shifted version of (4.23). From (D3) we deduce that

$$\int_{\mathbb{R}_+} \| -\gamma(t_0 + s, \omega, r_s(\omega)) \|_{\ell^2}^2 ds \leq \| \Gamma \|_{L^2(\mathbb{R}_+)}^2 < \infty, \quad \forall \omega \in \Omega. \quad (4.24)$$

Hence Lemma 2.3.2 applies and we can define the measure $\mathbb{P} \sim \mathbb{Q}$ by

$$\frac{d\mathbb{P}}{d\mathbb{Q}} = \mathcal{E}(-\gamma \cdot \tilde{W}^{(t_0)})_\infty \quad (4.25)$$

(notice that $\mathcal{F} = \mathcal{F}_\infty = \mathcal{F}_\infty^{(t_0)}$). According to Girsanov's theorem, see Proposition 2.3.1,

$$W_t = \tilde{W}_t^{(t_0)} + \int_0^t \gamma_s ds$$

is a Wiener process under $\mathbb{P}$.

By (H3) and Hölder's inequality

$$\| \alpha_t \|_H \leq K \sum_{j \in \mathbb{N}} \| \sigma_t^j \|_H^2 + \| \gamma_t \|_{\ell^2} \| \sigma_t \|_{L_2^0(H)} \leq (K + \frac{1}{2}) \| \sigma_t \|_{L_2^0(H)}^2 + \frac{1}{2} \| \gamma_t \|_{\ell^2}^2.$$

Now (4.24) yields (C2), since $(\sigma_t)_{t \in \mathbb{R}_+} \in \mathcal{L}^{\text{loc}}(H)$ by the very definition of a mild solution. Using (2.19), the process r is seen to be the continuous mild solution of (4.5) with $f(0, \cdot)$ replaced by h_0 , whence (C3). Conditions (C4)–(C5) are implied by (D1)–(D3) and $\mathbb{Q}$ clearly is an ELMM for this setup. $\square$

Remark 4.5.1. Theorem 4.5.1 points out that a *mild* solution r to the HJMM equation (4.23) is the right concept for the HJM framework. This is also in accordance with Remark 2.4.4. It is therefore an essential request to derive the results from [8] without assuming that r is a (local) strong solution, see [8, Assumption 2.1]

Remark 4.5.2. Local HJM models are useless for pricing financial instruments. Yet in some cases we only can assert local existence and uniqueness for the HJMM equation (4.23), which is still adapted for the consistency analysis in Chapter 7. Therefore we also included the local case in Definition 4.5.2.

The specification of the initial forward curve set I is convenient and allows to incorporate also the following classical examples in our framework. Let $-\gamma$ be any market price of risk satisfying (D1) and (D3). See Sections 7.4.1–7.4.2 for terminology and results.

Example 4.5.1 (CIR Model). Here $I = \{g_0 + yg_1 \mid y > 0\}$ where g_0, g_1 are given functions, $\sigma^1 = \sigma^1(h) = \sqrt{\alpha \mathcal{J}_0(h)} g_1$ for all $h \in I$ and $\sigma^j \equiv 0$ for $j \geq 2$. Then (σ, γ, I) is a local HJM model.

Example 4.5.2 (Vasicek Model). The volatility is constant, $\sigma^1 \equiv \sqrt{a} g_1$, where $g_1(x) = e^{\beta x}$ and $\sigma^j \equiv 0$ for $j \geq 2$. Moreover, $I = \{g_0 + yg_1 \mid y \in \mathbb{R}\}$ and g_0 is a given function. Then (σ, γ, I) is an HJM model.

5. The Forward Curve Spaces H_w

It is desirable to find easy verifiable criterions for the volatility coefficient σ , introduced in the preceding section, to provide a (local) HJM model. We shall approach this by the classical Lipschitz condition which we presented in Section 2.4.2. But if $\sigma(t, \omega, h)$ is locally bounded and locally Lipschitz continuous in h , what can we say about $\alpha_{HJM}(t, \omega, h)$ (see (4.22))? In this chapter we introduce a class of Hilbert spaces H_w which comply with the following requirements:

- H_w satisfies assumptions **(H1)**–**(H3)** from Sections 4.2–4.3.
- H_w is coherent with economical aspects of the forward curve $x \mapsto r_t(x)$.
- H_w is convenient for deriving (local) existence and uniqueness results for the HJMM equation (4.23).

We then examine different types of volatility specifications and give concrete examples.

5.1 Definition of H_w

Since in practice the forward curve is obtained by smoothing data points using smooth fitting methods (see Section 1.2) it is reasonable to assume

$$\int_{\mathbb{R}_+} |r'_t(x)|^2 dx < \infty.$$

Moreover, the curve flattens for large time to maturity x . There is no reason to believe that the forward rate for an instantaneous loan that begins in 10 years differs much from one which begins one day later. We take this into account by penalizing irregularities of $r_t(x)$ for large x by some increasing weighting function $w(x) \geq 1$, that is,

$$\int_{\mathbb{R}_+} |r'_t(x)|^2 w(x) dx < \infty.$$

However, this does not define a norm yet since constant functions are not distinguished. So we add the square of the short rate $|r_t(0)|^2$.

We shall give a mathematically rigorous approach. Let us recall a few facts from real analysis (see e.g. [15, Section VIII.2]). Let $h \in L^1_{\text{loc}}(\mathbb{R}_+)$. The weak derivative $h' \in L^1_{\text{loc}}(\mathbb{R}_+)$ of h , if it exists, is uniquely specified by the property

$$\int_{\mathbb{R}_+} h(x)\varphi'(x) dx = - \int_{\mathbb{R}_+} h'(x)\varphi(x) dx, \quad \forall \varphi \in C_c^1((0, \infty)).$$

If h has a weak derivative h' then there exists an absolutely continuous representative of h , still denoted by h , such that

$$h(x) - h(y) = \int_y^x h'(u) du, \quad \forall x, y \in \mathbb{R}_+. \quad (5.1)$$

Accordingly, the following definition makes sense.

Definition 5.1.1. Let $w : \mathbb{R}_+ \rightarrow [1, \infty)$ be a non-decreasing C^1 -function such that

$$w^{-\frac{1}{3}} \in L^1(\mathbb{R}_+). \quad (5.2)$$

We write

$$\|h\|_w^2 := |h(0)|^2 + \int_{\mathbb{R}_+} |h'(x)|^2 w(x) dx$$

and define

$$H_w := \{h \in L^1_{\text{loc}}(\mathbb{R}_+) \mid \exists h' \in L^1_{\text{loc}}(\mathbb{R}_+) \text{ and } \|h\|_w < \infty\}.$$

The choice of H_w is established by the next theorem.

Theorem 5.1.1. The set H_w equipped with $\|\cdot\|_w$ forms a separable Hilbert space meeting (H1)–(H3).

Before proving the theorem we give two guiding examples of admissible weighting functions w which satisfy condition (5.2).

Example 5.1.1. $w(x) = e^{\alpha x}$, for $\alpha > 0$.

Example 5.1.2. $w(x) = (1+x)^\alpha$, for $\alpha > 3$.

Proof. It is clear that $\|\cdot\|_w$ is a norm. Consider the separable Hilbert space $\mathbb{R} \times L^2(\mathbb{R}_+)$ equipped with the norm $(|\cdot|^2 + \|\cdot\|_{L^2(\mathbb{R}_+)}^2)^{\frac{1}{2}}$. Then the linear operator $T : H_w \rightarrow \mathbb{R} \times L^2(\mathbb{R}_+)$ given by

$$Th = \left(h(0), h'w^{\frac{1}{2}} \right), \quad h \in H_w$$

is an isometry with inverse

$$(T^{-1}(u, f))(x) = u + \int_0^x f(\eta)w^{-\frac{1}{2}}(\eta) d\eta, \quad (u, f) \in \mathbb{R} \times L^2(\mathbb{R}_+).$$

Hence H_w is a separable Hilbert space.

We claim that

$$\mathcal{D}_0 := \{f \in C^2(\mathbb{R}_+) \mid f' \in C_c^1(\mathbb{R}_+)\}$$

is dense in H_w . Indeed, $C_c^1(\mathbb{R}_+)$ is dense in $L^2(\mathbb{R}_+)$, see [15, Corollaire IV.23]. Fix $h \in H_w$ and let (f_n) be a approximating sequence of $h'w^{\frac{1}{2}}$ in $L^2(\mathbb{R}_+)$ with $f_n \in C_c^1(\mathbb{R}_+)$ for all $n \in \mathbb{N}$. Write $h_n := T^{-1}(h(0), f_n)$. Obviously, $h_n \in \mathcal{D}_0$ and $h_n \rightarrow h$ in H_w . Whence the claim.

Since w^{-1} is bounded, (5.2) implies in particular $w^{-1} \in L^1(\mathbb{R}_+)$. The subsequent Sobolev-type inequalities (5.4)–(5.8) are crucial. Throughout, h denotes a function in H_w . The constants C_1 – C_4 are universal and depend only on w .

First we prove

$$\|h'\|_{L^1(\mathbb{R}_+)} \leq C_1 \|h\|_w. \quad (5.3)$$

This is established by Hölder's inequality

$$\begin{aligned} \int_{\mathbb{R}_+} |h'(x)| dx &= \int_{\mathbb{R}_+} |h'(x)| w^{\frac{1}{2}}(x) w^{-\frac{1}{2}}(x) dx \\ &\leq \left(\int_{\mathbb{R}_+} |h'(x)|^2 w(x) dx \right)^{\frac{1}{2}} \left(\int_{\mathbb{R}_+} w^{-1}(x) dx \right)^{\frac{1}{2}} \\ &\leq \|h\|_w \|w^{-1}\|_{L^1(\mathbb{R}_+)}^{\frac{1}{2}}. \end{aligned}$$

An important consequence of (5.3) is that $h(x)$ converges to a limit $h(\infty) \in \mathbb{R}$ for $x \rightarrow \infty$, see (5.1). This is of course a desirable property for forward curves.

For $C_2 = 1 + C_1$ it follows easily from (5.1) and (5.3) that

$$\|h\|_{L^\infty(\mathbb{R}_+)} \leq C_2 \|h\|_w. \quad (5.4)$$

Now write

$$\Pi(x) := \int_x^\infty w^{-1}(\eta) d\eta.$$

Then we have, by monotonicity of w ,

$$\Pi(x) \leq w^{-\frac{2}{3}}(x) \int_x^\infty w^{-\frac{1}{3}}(\eta) d\eta \leq w^{-\frac{2}{3}}(x) \|w^{-\frac{1}{3}}\|_{L^1(\mathbb{R}_+)}, \quad x \in \mathbb{R}_+$$

and hence, by (5.2),

$$\|\Pi^{\frac{3}{2}} w\|_{L^\infty(\mathbb{R}_+)} \leq \|w^{-\frac{1}{3}}\|_{L^1(\mathbb{R}_+)} < \infty. \quad (5.5)$$

Moreover,

$$\|\Pi^{\frac{1}{2}}\|_{L^1(\mathbb{R}_+)} \leq \|w^{-\frac{1}{3}}\|_{L^1(\mathbb{R}_+)}^2 < \infty. \quad (5.6)$$

By the same method as above we derive

$$\begin{aligned} \int_{\mathbb{R}_+} |h(x) - h(\infty)| dx &= \int_{\mathbb{R}_+} \left| \int_x^\infty h'(\eta) w^{\frac{1}{2}}(\eta) w^{-\frac{1}{2}}(\eta) d\eta \right| dx \\ &\leq \int_{\mathbb{R}_+} \left(\int_x^\infty |h'(\eta)|^2 w(\eta) d\eta \right)^{\frac{1}{2}} \Pi^{\frac{1}{2}}(x) dx \\ &\leq \|h\|_w \|\Pi^{\frac{1}{2}}\|_{L^1(\mathbb{R}_+)}. \end{aligned}$$

Hence, by (5.6),

$$\|h - h(\infty)\|_{L^1(\mathbb{R}_+)} \leq C_3 \|h\|_w. \quad (5.7)$$

Finally we compute, using Hölder's inequality again,

$$\begin{aligned} \int_{\mathbb{R}_+} |h(x) - h(\infty)|^4 w(x) dx &= \int_{\mathbb{R}_+} \left| \int_x^\infty h'(\eta) w^{\frac{1}{2}}(\eta) w^{-\frac{1}{2}}(\eta) d\eta \right|^4 w(x) dx \\ &\leq \int_{\mathbb{R}_+} \left(\int_{\mathbb{R}_+} |h'(\eta)|^2 w(\eta) d\eta \right)^2 \Pi^2(x) w(x) dx \\ &\leq \|h\|_w^4 \|\Pi^{\frac{3}{2}} w\|_{L^\infty(\mathbb{R}_+)} \|\Pi^{\frac{1}{2}}\|_{L^1(\mathbb{R}_+)}. \end{aligned}$$

Therefore, by (5.5)–(5.6),

$$\|(h - h(\infty))^4 w\|_{L^1(\mathbb{R}_+)} \leq C_4 \|h\|_w^4. \quad (5.8)$$

Now we can prove **(H1)**–**(H3)**. Estimate (5.4) yields continuity of the linear functional $\mathcal{J}_x$ for all $x \in \mathbb{R}_+$. Since by the preceding remarks, see (5.1), H_w consists of continuous functions, **(H1)** is established.

Let $h \in H_w$ and $\varphi \in C_c^1((0, \infty))$. Then the shift semigroup $S(t)$ satisfies

$$\begin{aligned} \int_{\mathbb{R}_+} S(t)h(x)\varphi'(x) dx &= \int_{\mathbb{R}_+} h(x)\varphi'(x-t) dx = - \int_{\mathbb{R}_+} h'(x)\varphi(x-t) dx \\ &= - \int_{\mathbb{R}_+} h'(x+t)\varphi(x) dx \end{aligned} \quad (5.9)$$

since $\varphi(\cdot - t) \in C_c^1((t, \infty))$. Consequently, $(S(t)h)'$ exists and equals $S(t)h'$. In view of (5.4) it follows

$$\|S(t)h\|_w^2 = |h(t)|^2 + \int_{\mathbb{R}_+} |h'(x+t)|^2 w(x) dx \leq (C_2^2 + 1) \|h\|_w^2. \quad (5.10)$$

We have used the monotonicity of w . Hence $S(t)h \in H_w$ and $S(t)$ is bounded for all $t \in \mathbb{R}_+$.

We have to show strong continuity of $S(t)$. We start with the following observation. By (5.1)

$$h(x+t) - h(x) = t \int_0^1 h'(x+st) ds, \quad h \in H_w. \quad (5.11)$$

Let $g \in \mathcal{D}_0$. Since $g' \in H_w$, (5.10) and (5.11) yield

$$\begin{aligned} \|S(t)g - g\|_w^2 &= |g(t) - g(0)|^2 + \int_{\mathbb{R}_+} |g'(x+t) - g'(x)|^2 w(x) dx \\ &\leq |g(t) - g(0)|^2 + t^2 \int_0^1 \int_{\mathbb{R}_+} |g''(x+st)|^2 w(x) dx ds \\ &\leq |g(t) - g(0)|^2 + t^2 \int_0^1 \|S(st)g'\|_w^2 ds \rightarrow 0 \quad \text{for } t \rightarrow 0. \end{aligned}$$

Hence $S(t)$ is strongly continuous on $\mathcal{D}_0$. But for any $h \in H_w$ and $\varepsilon > 0$ there exists $g \in \mathcal{D}_0$ with $\|h - g\|_w \leq \frac{\varepsilon}{4(C_2+1)}$. Combining this with (5.10) yields

$$\begin{aligned} \|S(t)h - h\|_w &\leq \|S(t)(h - g)\|_w + \|S(t)g - g\|_w + \|g - h\|_w \\ &\leq (C_2 + 1) \frac{\varepsilon}{4(C_2 + 1)} + \|S(t)g - g\|_w + \frac{\varepsilon}{4(C_2 + 1)} \leq \varepsilon \end{aligned}$$

for t small enough. We conclude that $S(t)$ is strongly continuous on H_w , whence the validity of **(H2)**.

It remains to prove the estimate in **(H3)**. Let $h \in H_w$. Similarly to (5.9) one shows that $(Sh)'$ exists with

$$(Sh)'(x) = h'(x) \int_0^x h(\eta) d\eta + h^2(x).$$

We define

$$H_w^0 := \{h \in H_w \mid h(\infty) = 0\}. \quad (5.12)$$

By (5.4), H_w^0 is a closed subspace of H_w . We claim that

$$Sh \in H_w \iff h \in H_w^0. \quad (5.13)$$

Assume that $h(\infty) > 0$. Then there exists $x_0 \in \mathbb{R}_+$ such that $h(x) \geq \frac{h(\infty)}{2}$ for all $x \geq x_0$. But then

$$Sh(x) \geq h(x) \int_0^{x_0} h(\eta) d\eta + \left(\frac{h(\infty)}{2}\right)^2 (x - x_0), \quad \forall x \geq x_0$$

which explodes for $x \rightarrow \infty$. The same reasoning applies if $h(\infty) < 0$. Whence the necessity in (5.13).

Conversely, assume that $h(\infty) = 0$. Using (5.7) and (5.8) we estimate

$$\begin{aligned} \|Sh\|_w^2 &= \int_{\mathbb{R}_+} \left| h'(x) \int_0^x h(\eta) d\eta + h^2(x) \right|^2 w(x) dx \\ &\leq 2 \int_{\mathbb{R}_+} \left\{ |h'(x)|^2 \left(\int_0^x |h(\eta)| d\eta \right)^2 + h^4(x) \right\} w(x) dx \\ &\leq 2(\|h\|_{L^1(\mathbb{R}_+)}^2 \|h\|_w^2 + C_4 \|h\|_w^4) \leq 2(C_3^2 + C_4) \|h\|_w^4. \end{aligned}$$

Hence $Sh \in H_w$ and (5.13) is proved. Moreover, we conclude that (H3) holds with $K = \sqrt{2(C_3^2 + C_4)}$. $\square$

Remark 5.1.1. It is not difficult to see that (H1) and (H2) hold without assumption (5.2). Yet it makes the proof more handy.

By the preceding proof we can identify the full generator A of $S(t)$ in H_w .

Corollary 5.1.1. *We have $D(A) = \{h \in H_w \mid h' \in H_w\}$ and $Ah = h'$.*

Proof. Write $\mathcal{D} := \{h \in H_w \mid h' \in H_w\}$. By Lemma 4.2.2, $D(A) \subset \mathcal{D}$. Conversely, let $h \in \mathcal{D}$. Applying (5.11) we get

$$\frac{h'(x+t) - h'(x)}{t} - h''(x) = \int_0^1 (h''(x+st) - h''(x)) ds.$$

Consequently,

$$\begin{aligned} n(t) &:= \int_{\mathbb{R}_+} \left| \frac{h'(x+t) - h'(x)}{t} - h''(x) \right|^2 w(x) dx \\ &\leq \int_0^1 \int_{\mathbb{R}_+} |h''(x+st) - h''(x)|^2 w(x) dx ds \\ &\leq \int_0^1 \|S(st)h' - h'\|_w^2 ds \rightarrow 0 \quad \text{for } t \rightarrow 0 \end{aligned}$$

by dominated convergence, see (5.10). Since h' is continuous on $\mathbb{R}_+$, see (5.1), we conclude

$$\left\| \frac{S(t)h - h}{t} - h' \right\|_w^2 = \left| \frac{h(t) - h(0)}{t} - h'(0) \right|^2 + n(t) \rightarrow 0 \quad \text{for } t \rightarrow 0,$$

whence $h \in D(A)$. $\square$

Most important for the HJMM equation is the following result.

Corollary 5.1.2. *The mapping $\mathcal{S} : H_w^0 \rightarrow H_w^0$ is locally Lipschitz continuous, that is*

$$\|\mathcal{S}g - \mathcal{S}h\|_w \leq C_5(\|g\|_w + \|h\|_w) \|g - h\|_w, \quad \forall g, h \in H_w^0$$

where the constant C_5 only depends on w .

Proof. In view of (5.13) and (5.7) it is immediate that $Sh \in H_w^0$ for all $h \in H_w^0$. Let $g, h \in H_w^0$. Using Hölder's inequality and (5.7)–(5.8) we calculate

$$\begin{aligned}
\|Sg - Sh\|_w^2 &= \int_{\mathbb{R}_+} \left| g'(x) \int_0^x g(\eta) d\eta - h'(x) \int_0^x h(\eta) d\eta \right. \\
&\quad \left. + g^2(x) - h^2(x) \right|^2 w(x) dx \\
&\leq 3 \int_{\mathbb{R}_+} |g'(x)|^2 \left| \int_0^x (g(\eta) - h(\eta)) d\eta \right|^2 w(x) dx \\
&\quad + 3 \int_{\mathbb{R}_+} \left| \int_0^x h(\eta) d\eta \right|^2 |g'(x) - h'(x)|^2 w(x) dx \\
&\quad + 3 \int_{\mathbb{R}_+} (g(x) + h(x))^2 w^{\frac{1}{2}}(x) (g(x) - h(x))^2 w^{\frac{1}{2}}(x) dx \\
&\leq 3 \left(\|g\|_w^2 \|g - h\|_{L^1(\mathbb{R}_+)}^2 + \|h\|_{L^1(\mathbb{R}_+)}^2 \|g - h\|_w^2 \right) \\
&\quad + 3 \|(g + h)^4 w\|_{L^1(\mathbb{R}_+)}^{\frac{1}{2}} \|(g - h)^4 w\|_{L^1(\mathbb{R}_+)}^{\frac{1}{2}} \\
&\leq 3(C_3^2 + 2C_4)(\|g\|_w^2 + \|h\|_w^2) \|g - h\|_w^2
\end{aligned}$$

and we have used the universal inequality

$$|x_1 + \dots + x_k|^2 \leq k(|x_1|^2 + \dots + |x_k|^2), \quad k \in \mathbb{N}. \quad (5.14)$$

Setting $C_5 = \sqrt{3(C_3^2 + 2C_4)}$ yields the result. $\square$

5.2 Volatility Specification

Working in the space H_w we can give simple criteria for $\sigma(t, \omega, h)$ to provide a (local) HJM model.

Condition (D2) and (5.13) clearly force $\sigma^j \in H_w^0$, recall (5.12). This is of course a desirable property since we do not want the forward rates $r_t(x)$ being too volatile for large x , see (4.3). In fact, Corollary 5.1.2 yields

Lemma 5.2.1. *Let σ meet (D1) with $H = H_w$ and $L_2^0(H)$ replaced by $L_2^0(H_w^0)$. Then (D2) is satisfied as well.*

From now on we let σ denote a measurable mapping from $(\mathbb{R}_+ \times \Omega \times H_w, \mathcal{P} \otimes \mathcal{B}(H_w))$ into $(L_2^0(H_w^0), \mathcal{B}(L_2^0(H_w^0)))$.

In view of Corollary 2.4.1 it is enough to require local boundedness and local Lipschitz continuity of $\sigma(t, \omega, h)$ and $\alpha_{HJM}(t, \omega, h)$ in h to assert local existence and uniqueness for the HJMM equation (4.23). The choice of H_w gains in interest if we consider the following property.

Lemma 5.2.2. *i) Suppose that $\sigma(t, \omega, h)$ is locally bounded and locally Lipschitz continuous in h . Then the same holds for $\alpha_{HJM}(t, \omega, h)$.*

ii) Suppose $\sigma(t, \omega, h)$ is Lipschitz continuous in h and uniformly bounded,

$$\|\sigma(t, \omega, h)\|_{L_2^0(H_w)} \leq C, \quad \forall (t, \omega, h) \in \mathbb{R}_+ \times \Omega \times H_w, \quad (5.15)$$

for some constant C . Then the same holds for $\alpha_{HJM}(t, \omega, h)$.

Proof. For simplicity of notation we abandon temporarily the (t, ω) -dependence of σ and α_{HJM} .

Let $h_1, h_2 \in B_R(H_w)$ and write $\Delta := \|\alpha_{HJM}(h_1) - \alpha_{HJM}(h_2)\|_w$. By Corollary 5.1.2 and Hölder's inequality

$$\begin{aligned} \Delta &\leq \sum_{j \in \mathbb{N}} \|\mathcal{S}\sigma^j(h_1) - \mathcal{S}\sigma^j(h_2)\|_w \\ &\leq C_5 \sum_{j \in \mathbb{N}} (\|\sigma^j(h_1)\|_w + \|\sigma^j(h_2)\|_w) \|\sigma^j(h_1) - \sigma^j(h_2)\|_w \\ &\leq C_5 \left(\|\sigma(h_1)\|_{L^2_2(H_w)} + \|\sigma(h_2)\|_{L^2_2(H_w)} \right) \|\sigma(h_1) - \sigma(h_2)\|_{L^2_2(H_w)} \\ &\leq C_5 C(R) \left(\|\sigma(h_1)\|_{L^2_2(H_w)} + \|\sigma(h_2)\|_{L^2_2(H_w)} \right) \|h_1 - h_2\|_w. \end{aligned}$$

This proves the first part of i) and ii). The boundedness assertions on α_{HJM} are a consequence of **(H3)**. $\square$

Combining Lemmas 5.2.1–5.2.2 with Theorem 2.4.1 and Corollary 2.4.1 we get our main existence and uniqueness result for the HJMM equation. Denote by $-\gamma$ a market price of risk vector meeting **(D1)** and **(D3)**.

Theorem 5.2.1. *i) Let σ be as in Lemma 5.2.2.i). Then (σ, γ, H_w) is a local HJM model in H_w .
ii) Let σ be as in Lemma 5.2.2.ii). Then (σ, γ, H_w) is an HJM model in H_w and all discounted bond price processes $(Z(t, T))_{t \in [0, T]}$ are true $\mathbb{Q}$ -martingales. Moreover, any mild solution $r^{(t_0, h_0)}$ of the time t_0 -shifted HJMM equation (4.23) is also a weak solution, $(t_0, h_0) \in \mathbb{R}_+ \times H_w$.*

Proof. Only the statement on $(Z(t, T))_{t \in [0, T]}$ remains to be proved. But this is a straightforward consequence of Lemma 4.4.1 and the estimate

$$\|\mathcal{I}_{T-t} \circ \sigma(t, r_t)\|_{\ell^2}^2 \leq (Tk(T))^2 \|\sigma(t, r_t)\|_{L^2_2(H_w)}^2 \leq (Tk(T))^2 C^2,$$

see Lemma 4.3.1. $\square$

We shall examine three types of possible volatility specifications. The first one corresponds to the classical HJM framework where the volatility depends locally on the state variable. It turns out that in this case one merely gets local HJM models. For example the one which is presented in [35, Section 7].

The second type is much more convenient. Here the volatility depends on a functional value of the state variable. It is easy to find sufficient conditions for providing an HJM model in this case.

Lastly, we recapture the BGM [13] model, which we already mentioned in Section 4.4.4. However, since also this type of volatility is of local nature, we merely get a local HJM model.

Subsequently, there follows an intermediary section where we introduce an important financial variable.

5.3 The Yield Curve

We already know by Riesz representation theorem that the functionals $\mathcal{J}_x$ and $\mathcal{I}_x$ can be identified with elements in H_w , see Theorem 5.1.1 and Lemma 4.3.1. Let us just mention how they look like as functions.

Lemma 5.3.1. *We have*

$$\begin{aligned}\mathcal{J}_x(\eta) &= 1 + \int_0^{x \wedge \eta} \frac{1}{w(u)} du, \quad \eta \in \mathbb{R}_+ \\ \mathcal{I}_x(\eta) &= x + \int_0^\eta \frac{x - u \wedge x}{w(u)} du, \quad \eta \in \mathbb{R}_+\end{aligned}$$

as elements in H_w .

Proof. We only proof the statement on $\mathcal{I}_x$. Let $h \in H_w$. Integration by parts yields

$$\begin{aligned}\langle \mathcal{I}_x, h \rangle_w &= xh(0) + \int_{\mathbb{R}_+} (x - \eta \wedge x) h'(\eta) d\eta \\ &= xh(0) + \int_0^x (x - \eta) h'(\eta) d\eta \\ &= xh(0) - xh(0) + \int_0^x h(\eta) d\eta = \int_0^x h(\eta) d\eta.\end{aligned}$$

□

The next result will prove very useful. Recall that A^* denotes the adjoint of A .

Lemma 5.3.2. *We have $\mathcal{I}_x \in D(A^*)$ and $A^* \mathcal{I}_x = \mathcal{J}_x - \mathcal{J}_0$, for all $x \in \mathbb{R}_+$.*

Proof. From Corollary 5.1.1 we know that

$$\langle \mathcal{I}_x, Ah \rangle_w = \int_0^x h'(\eta) d\eta = h(x) - h(0) = \langle \mathcal{J}_x - \mathcal{J}_0, h \rangle_w, \quad \forall h \in D(A).$$

□

Besides the forward curve and the term structure of bond prices there is another measurement of the bond market in between.

Definition 5.3.1. *Given the continuous time t forward curve $x \mapsto r_t(x)$, the time t yield curve $x \mapsto y_t(x)$ is defined by*

$$y_t(x) := \frac{1}{x} \int_0^x r_t(\eta) d\eta, \quad x \in \mathbb{R}_+.$$

Notice that $y_t(0) = r_t(0)$ is well defined.

In contrast to forward rates, yields of zero coupon bonds can be directly observed on the market. We will exploit this fact in Example 5.5.1 below.

For completeness we shall describe the implied dynamics for y . It turns out that the yield curve process enjoys nice regularity properties.

Let σ satisfy the assumptions of Theorem 5.2.1.ii) and let r be a weak solution of the HJMM equation (4.23) in H_w . For a fixed $x > 0$ we have by Lemma 5.3.2 that $(xy_t(x)) = (\mathcal{I}_x(r_t))$ is a real-valued Itô process. Taking into account Lemma 4.3.2 we derive the decomposition

$$\begin{aligned} xy_t(x) &= \mathcal{I}_x(r_0) + \int_0^t \left((\mathcal{J}_x - \mathcal{J}_0)(r_s) + \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_x(\sigma^j(s, r_s)))^2 \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t \mathcal{I}_x(\sigma^j(s, r_s)) d\tilde{\beta}_s^j. \end{aligned}$$

Hence the random field $(Y(t, x)) := (xy_t(x))$ is a classical solution (that is, (t, x) -pointwise) of the stochastic partial differential equation (SPDE)

$$\begin{cases} dY(t, x) = \left(\frac{\partial}{\partial x} Y(t, x) + F_Y(t, x) \right) dt + \sum_{j \in \mathbb{N}} \mathcal{I}_x(\sigma^j(t, r_t)) d\tilde{\beta}_t^j \\ Y(0, x) = \mathcal{I}_x(r_0), \end{cases}$$

where we have written $F_Y(t, x) := \frac{1}{2} \sum_{j \in \mathbb{N}} (\mathcal{I}_x(\sigma^j(t, r_t)))^2 - r_t(0)$.

5.4 Local State Dependent Volatility

HJM present in [35, Section 7] a class of models with local state dependent volatility. In the present setting this reads

$$\sigma^j(t, \omega, h)(x) = \Phi^j(t, \omega, x, h(x)), \quad j \in \mathbb{N} \quad (5.16)$$

for some real valued functions $\Phi^j(t, \omega, x, s)$. We shall formulate sufficient conditions on $(\Phi^j)_{j \in \mathbb{N}}$ such that $\sigma = (\sigma^j)_{j \in \mathbb{N}}$ given by (5.16) satisfies the hypotheses of Theorem 5.2.1.i). Writing $f(t, T) = r_t(T - t)$ for a mild solution $r = r^{(0, h_0)}$ to the HJMM equation (4.23) (if it exists) we get

$$\begin{aligned} f(t, T) &= h_0(T) \\ &\quad + \int_0^t \sum_{j \in \mathbb{N}} \left(\Phi^j(s, T - s, f(s, T)) \int_s^T \Phi^j(s, T - s, f(s, u)) du \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t \Phi^j(s, T - s, f(s, T)) d\tilde{\beta}_s^j, \end{aligned} \quad (5.17)$$

which is [35, Equation (42)].

Proposition 5.4.1. *Let $\Phi^j(t, \omega, x, s) : \mathbb{R}_+ \times \Omega \times \mathbb{R}_+ \times \mathbb{R} =: D \rightarrow \mathbb{R}$ satisfy the following assumptions*

- i) $\Phi^j(t, \omega, x, s)$ is $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+) \otimes \mathcal{B}(\mathbb{R})$ -measurable
- ii) $\Phi^j(t, \omega, x, s)$ is continuously differentiable in (x, s) , for all $(t, \omega, x, s) \in D$
- iii) $\lim_{x \rightarrow \infty} \Phi^j(t, \omega, x, s) = 0$ for all $(t, \omega, s) \in \mathbb{R}_+ \times \Omega \times \mathbb{R}$,

for any $j \in \mathbb{N}$. Suppose there exist $c = (c^j)_{j \in \mathbb{N}} \in C(\mathbb{R}; \ell^2)$ with c^j convex and $c^j(-s) = c^j(s)$, and $\Psi = (\Psi^j)_{j \in \mathbb{N}} \in L^2(\mathbb{R}_+; \ell^2)$ with

$$\int_{\mathbb{R}_+} \|\Psi(x)\|_{\ell^2}^2 w(x) dx < \infty \quad (5.18)$$

such that the growth conditions

$$|\Phi^j(t, \omega, 0, s)| \leq c^j(s) \quad (5.19)$$

$$|\partial_x \Phi^j(t, \omega, x, s)| \leq \Psi^j(x) c^j(s) \quad (5.20)$$

$$|\partial_s \Phi^j(t, \omega, x, s)| \leq c^j(s) \quad (5.21)$$

and the Lipschitz conditions (we suppress the notional dependence on (t, ω))

$$|\Phi^j(0, s_1) - \Phi^j(0, s_2)| \leq (c^j(s_1) + c^j(s_2)) |s_1 - s_2| \quad (5.22)$$

$$|\partial_x \Phi^j(x, s_1) - \partial_x \Phi^j(x, s_2)| \leq \Psi^j(x) (c^j(s_1) + c^j(s_2)) |s_1 - s_2| \quad (5.23)$$

$$|\partial_s \Phi^j(x, s_1) - \partial_s \Phi^j(x, s_2)| \leq (c^j(s_1) + c^j(s_2)) |s_1 - s_2| \quad (5.24)$$

hold for all $(t, \omega, x, s) \in D$, $s_1, s_2 \in \mathbb{R}$ and $j \in \mathbb{N}$. Then $\sigma = (\sigma^j)_{j \in \mathbb{N}}$ defined by (5.16) fulfills the hypotheses of Theorem 5.2.1.i).

Proof. Whenever there is no ambiguity we shall skip the notional dependence on (t, ω) . The proof is divided into four steps.

Step 1: Let us recall the chain rule for the weak derivative. For $h \in H_w$ we have

$$\frac{d}{dx} \Phi^j(x, h(x)) = \partial_x \Phi^j(x, h(x)) + \partial_s \Phi^j(x, h(x)) h'(x).$$

This can be shown, using ii), as in the proof of [15, Corollaire VIII.10].

Step 2: Now we show that $\sigma(h) \in L^0_2(H_w^0)$ for all $h \in H_w$. Let $h \in H_w$. Clearly $\lim_{x \rightarrow \infty} \sigma^j(h)(x) = 0$, by ii) and iii). Conditions (5.19)–(5.21) yield

$$\begin{aligned} \|\sigma^j(h)\|_w^2 &\leq |\Phi^j(0, h(0))|^2 \\ &\quad + 2 \int_{\mathbb{R}_+} (|\partial_x \Phi^j(x, h(x))|^2 + |\partial_s \Phi^j(x, h(x))|^2 |h'(x)|^2) w(x) dx \\ &\leq |c^j(\|h\|_w)|^2 \left(1 + 2 \int_{\mathbb{R}_+} (|\Psi^j(x)|^2 + |h'(x)|^2) w(x) dx \right) \end{aligned}$$

which implies the assertion.

Step 3: Next we check measurability of $(t, \omega, h) \mapsto \sigma^j(t, \omega, h)$. It is enough to show measurability of $(t, \omega, h) \mapsto \langle \sigma^j(t, \omega, h), g \rangle_w$ for all $g \in H_w$, see [19, Proposition 1.3].

Fix $g \in H_w$. By i), ii) and Lemma 4.2.1 the mapping

$$(t, \omega, x, h) \mapsto \partial_x \Phi^j(t, \omega, x, h(x))$$

is $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+) \otimes \mathcal{B}(H_w)$ -measurable. The same applies for $\partial_s \Phi^j(t, \omega, x, h(x))$. Hence for any $f \in H_w$ the function

$$(t, \omega, h) \mapsto I(t, \omega, h)(f) := \int_{\mathbb{R}_+} \partial_s \Phi^j(t, \omega, x, h(x)) f'(x) g'(x) w(x) dx$$

is $\mathcal{P} \otimes \mathcal{B}(H_w)$ -measurable. Here we have used (5.21) and a monotone class argument, see [3, Lemma 23.2]. Since $f \in H_w$ was arbitrary this implies that $(t, \omega, h) \mapsto I(t, \omega, h)$ is a $\mathcal{P} \otimes \mathcal{B}(H_w)$ -measurable application into the dual space of H_w which we identify with H_w as usual (notice that $I(t, \omega, h)$ is a bounded linear functional on H_w , by (5.21)). Similar reasoning, using (5.20), shows that

$$(t, \omega, h) \mapsto J(t, \omega, h) := \int_{\mathbb{R}_+} \partial_x \Phi^j(t, \omega, x, h(x)) g'(x) w(x) dx$$

is $\mathcal{P} \otimes \mathcal{B}(H_w)$ -measurable into $\mathbb{R}$.

But now we can write

$$\langle \sigma^j(t, \omega, h), g \rangle_w = \Phi^j(t, \omega, 0, \mathcal{J}_0(h)) g(0) + J(t, \omega, h) + I(t, \omega, h)(h)$$

and deduce the desired measurability assertion.

Step 4: It remains to check the boundedness and Lipschitz properties. Observe that the local boundedness of σ in h is a direct consequence of (5.19)–(5.20). Let $h_1, h_2 \in H_w$ and write $\Delta := \|\sigma^j(h_1) - \sigma^j(h_2)\|_w^2$. In view of (5.14) we have

$$\begin{aligned} \Delta &\leq |\Phi^j(0, h_1(0)) - \Phi^j(0, h_2(0))|^2 \\ &\quad + 3 \int_{\mathbb{R}_+} |\partial_x \Phi^j(x, h_1(x)) - \partial_x \Phi^j(x, h_2(x))|^2 w(x) dx \\ &\quad + 3 \int_{\mathbb{R}_+} |\partial_s \Phi^j(x, h_1(x)) (h'_1(x) - h'_2(x))|^2 w(x) dx \\ &\quad + 3 \int_{\mathbb{R}_+} |h'_2(x) (\partial_s \Phi^j(x, h_1(x)) - \partial_s \Phi^j(x, h_2(x)))|^2 w(x) dx \\ &\leq C(h_2) (c^j(\|h_1\|_w) + c^j(\|h_2\|_w))^2 \|h_1 - h_2\|_w^2 \end{aligned}$$

where

$$C(h_2) = 1 + 3 \int_{\mathbb{R}_+} \|\Psi(x)\|_{\ell^2}^2 w(x) dx + 3 + 3\|h_2\|_w^2$$

and we have used (5.21)–(5.24). Therefore $\sigma(t, \omega, h)$ is locally Lipschitz continuous in h , and the proposition follows. $\square$

The following two examples are presented in [35, Section 7].

Example 5.4.1. Let $h_0 \in H_w^0$. Define $\Phi^1(t, \omega, x, s) \equiv \Phi^1(x, s) := h_0(x)s$ and $\Phi^j \equiv 0$ for $j > 1$. The assumptions of Proposition 5.4.1 are easily seen to be fulfilled with $c^1(s) = (1 + \|h_0\|_w)(1 + |s|)$ and $\Psi^1(x) = |h'_0(x)|$, and $c^j = \Psi^j \equiv 0$ for $j > 1$. According to Theorem 5.2.1, the so specified volatility mapping σ provides a local HJM model. However, it is shown in Morton's thesis [45] that every mild solution r to the HJMM equation (4.23) explodes with probability one in finite time.

We define

$$\chi(s) := \frac{s}{|s|} (|s| \wedge \lambda), \quad s \in \mathbb{R} \quad (5.25)$$

where λ stands for some positive number.

Example 5.4.2. Like Example 5.4.1, but this time $\Phi^1(x, s) := h_0(x)\chi(s)$. A slight modification of the preceding proof shows that also here σ fulfills the hypotheses of Theorem 5.2.1.i). It is proved in [35, Proposition 4] that (5.17) has a jointly continuous solution $(f(t, T))_{(t, T) \in \Delta^2}$. Yet we still cannot assert global existence for the HJMM equation (4.23) since the norm $\|r_t\|_w$ is not a priori controllable.

In view of Example 5.4.2 one may wish to consider the HJMM equation (4.23) in a larger space, such as $L_\alpha^2(\mathbb{R}_+)$ equipped with norm

$$\|f\|_{L_\alpha^2(\mathbb{R}_+)}^2 := \int_{\mathbb{R}_+} |f(x)|^2 e^{-\alpha x} dx,$$

for some $\alpha > 0$. This approach is used by Goldys and Musiela [32]. Indeed, Example 5.4.2 provides a global mild solution to equation (4.23) if interpreted in $L_\alpha^2(\mathbb{R}_+)$. However, $L_\alpha^2(\mathbb{R}_+)$ does not meet **(H1)** and **(H3)** and it is not clear how the analysis of Chapter 4 could be performed in this space.

5.5 Functional Dependent Volatility

Very convenient for the present framework is a functional dependent volatility

$$\sigma^j(t, \omega, h)(x) = \Phi^j(t, \omega, x, \zeta^j(h)), \quad j \in \mathbb{N} \quad (5.26)$$

where ζ^j are continuous linear functionals on H_w and $\Phi^j(t, \omega, x, s)$ are real-valued functions. The following proposition contains appropriate conditions on Φ^j to make sure that $\sigma = (\sigma^j)_{j \in \mathbb{N}}$ in (5.26) provides an HJM model.

Proposition 5.5.1. *Let $\Phi^j(t, \omega, x, s) : \mathbb{R}_+ \times \Omega \times \mathbb{R}_+ \times \mathbb{R} =: D \rightarrow \mathbb{R}$ satisfy the following assumptions*

- i) $\Phi^j(t, \omega, x, s)$ is $\mathcal{P} \otimes \mathcal{B}(\mathbb{R}_+) \otimes \mathcal{B}(\mathbb{R})$ -measurable
- ii) $\Phi^j(t, \omega, x, s)$ is continuously differentiable in x , for all $(t, \omega, x, s) \in D$
- iii) $\lim_{x \rightarrow \infty} \Phi^j(t, \omega, x, s) = 0$ for all $(t, \omega, s) \in \mathbb{R}_+ \times \Omega \times \mathbb{R}$,

for any $j \in \mathbb{N}$. Suppose there exists $\Psi = (\Psi^j)_{j \in \mathbb{N}} \in L^2(\mathbb{R}_+; \ell^2)$ satisfying (5.18), such that the following growth and Lipschitz conditions

$$|\partial_x \Phi^j(t, \omega, x, s)| \leq \Psi^j(x) \quad (5.27)$$

$$|\partial_x \Phi^j(t, \omega, x, s_1) - \partial_x \Phi^j(t, \omega, x, s_2)| \leq \Psi^j(x) |s_1 - s_2| \quad (5.28)$$

hold for all $(t, \omega, x, s) \in D$, $s_1, s_2 \in \mathbb{R}$ and $j \in \mathbb{N}$. Then $\sigma = (\sigma^j)_{j \in \mathbb{N}}$ defined by (5.26) fulfills the hypotheses of Theorem 5.2.1.ii).

Proof. We proceed as in the proof of Proposition 5.4.1. Again we skip the notional dependence on (t, ω) whenever there is no ambiguity.

Assumptions ii), iii) and (5.27) yield $\Phi^j(\cdot, s) \in H_w^0$ and, using Hölder's inequality,

$$\begin{aligned} \|\Phi^j(\cdot, s)\|_w^2 &\leq |\Phi^j(0, s)|^2 + \int_{\mathbb{R}_+} |\Psi^j(x)|^2 w(x) dx \\ &\leq \left(\int_{\mathbb{R}_+} |\partial_x \Phi^j(x, s)| dx \right)^2 + \int_{\mathbb{R}_+} |\Psi^j(x)|^2 w(x) dx \\ &\leq \left(\int_{\mathbb{R}_+} |\Psi^j(x)|^2 w(x) dx \right) (\|w^{-1}\|_{L^1(\mathbb{R}_+)} + 1) \end{aligned}$$

for all $s \in \mathbb{R}$. From this we deduce $\sigma(h) \in L_2^0(H_w^0)$ for all $h \in H_w$. Moreover, (5.15) holds.

By a similar argument as in Step 3 in the proof of Proposition 5.4.1, using i) and ii), we derive the measurability of

$$\begin{aligned} (t, \omega, h) \mapsto \langle \sigma^j(t, \omega, h), g \rangle_w &= \Phi^j(t, \omega, 0, \zeta^j(h)) g(0) \\ &\quad + \int_{\mathbb{R}_+} \partial_x \Phi^j(t, \omega, x, \zeta^j(h)) g'(x) w(x) dx \end{aligned}$$

for arbitrary $g \in H_w$.

It remains to show that $\sigma(t, \omega, h)$ is Lipschitz continuous in h . But this is a direct consequence of (5.28) and iii), which implies the equality

$$\Phi^j(t, \omega, 0, s) = - \int_{\mathbb{R}_+} \partial_x \Phi^j(t, \omega, x, s) dx.$$

The following example looks promising.

Example 5.5.1. Fix $m \in \mathbb{N}$ and m benchmark yields, see Definition 5.3.1,

$$\frac{1}{\delta_1} \mathcal{I}_{\delta_1}(r_t), \dots, \frac{1}{\delta_m} \mathcal{I}_{\delta_m}(r_t), \quad 0 < \delta_1 < \dots < \delta_m,$$

which are directly *observable* on the market. Let $h_1, \dots, h_m \in H_w^0$ and set

$$\sigma^j(t, \omega, h) \equiv \sigma^j(h) := \begin{cases} h_j \chi \left(\frac{1}{\delta_j} \mathcal{I}_{\delta_j}(h) \right), & j \leq m \\ 0, & j > m \end{cases}$$

see (5.25). Then Proposition 5.5.1 applies with

$$\Phi^j(t, \omega, x, s) \equiv \Phi^j(x, s) = \begin{cases} h_j(x) \chi(s), & j \leq m \\ 0, & j > m \end{cases}$$

and $\Psi^j = \lambda |h'_j|$ for $j \leq m$, resp. $\Psi^j = 0$ for $j > m$.

The functions $h_1, \dots, h_m$ can be inferred by statistical methods.

5.6 The BGM Model

A prominent representative within the HJM framework is the BGM [13] model, which was already motivated in Section 4.4.4. It is desirable to have it included in the present setup. In this section we construct therefore a volatility structure σ satisfying the assumptions of Theorem 5.2.1.i) and which meets (4.20).

Recall the notation from Section 4.4. Assume that we want to price any caplet with settlement date $T + \delta \leq T^*$ according to formula (4.21), for some $T^* \in \mathbb{R}_+$. Without loss of generality $T^* = K^* \delta$ for $K^* \in \mathbb{N}$. Motivated by (4.20) we are looking for a family of deterministic functions $(\theta^j(t, x))_{j \in \mathbb{N}}$ such that for any $(t, h) \in \mathbb{R}_+ \times H_w$

$$\mathcal{I}_x^{x+\delta}(\sigma^j(t, h)) = \left(1 - e^{-\mathcal{I}_x^{x+\delta}(h)}\right) \theta^j(t, x), \quad \forall x \in [0, (K^* - 1)\delta]. \quad (5.29)$$

We suppress temporarily the notional dependence on t and j . Suppose for the moment that $\theta(x)$ is arbitrarily smooth in x . Taking derivative in (5.29) we get

$$\sigma(h)(x + \delta) - \sigma(h)(x) = \phi(h, x), \quad \forall x \in [0, (K^* - 1)\delta], \quad (5.30)$$

where

$$\phi(h, x) := (h(x + \delta) - h(x))e^{-\mathcal{I}_x^{x+\delta}(h)}\theta(x) + \left(1 - e^{-\mathcal{I}_x^{x+\delta}(h)}\right)\partial_x \theta(x).$$

Recurrence relationship (5.30) defines a volatility structure $\sigma(h)(x)$ for $x \in [\delta, K^*\delta)$ provided $\sigma(h)(x)$ is defined on $[0, \delta)$. Notice that (5.30) implies equality (5.29) only up to a integration constant C which is specified by setting $x = 0$

$$\mathcal{I}_\delta(\sigma(h)) = (1 - e^{-\mathcal{I}_\delta(h)})\theta(0) + C.$$

We will later set $\theta(0) = 0$, see (5.31). Thus $C = 0$ if and only if $\mathcal{I}_\delta(\sigma(h)) = 0$ for all $h \in H_w$. Following BGM [13] we set $\sigma(h)(x) = 0$ on $[0, \delta)$. Solving (5.30) we derive

$$\sigma(h)(x) = \sum_{i=1}^k \phi(h, x - i\delta) \quad \text{for } x \in [k\delta, (k+1)\delta) \text{ and } 1 \leq k \leq K^* - 1.$$

This implies $\lim_{x \uparrow (k+1)\delta} \sigma(h)(x) = \sum_{i=1}^k \phi(h, i\delta)$ and we see that $\sigma(h)(x)$ is continuous in $x \in [0, K^*\delta)$ for all $h \in H_w$ if and only if $\phi(h, 0) = 0$ for all $h \in H_w$. That is, if and only if

$$\theta(0) = \partial_x \theta(0) = 0. \quad (5.31)$$

We have to extend the definition of $\sigma(h)(x)$ to the whole $\mathbb{R}_+$. Set

$$\sigma(h)(K^*\delta) := \lim_{x \uparrow K^*\delta} \sigma(h)(x) = \sum_{i=1}^{K^*-1} \phi(h, i\delta)$$

and extrapolate linearly to 0

$$\sigma(h)(x) := \sigma(h)(K^*\delta) \left(1 - \frac{(x - K^*\delta)}{\delta}\right)^+, \quad x \geq K^*\delta.$$

This construction implies that $\sigma(h)'$ exists in $L^1_{\text{loc}}(\mathbb{R}_+)$ and

$$\sigma(h)'(x) = \begin{cases} 0, & x \in [0, \delta) \\ \sum_{i=1}^k \partial_x \phi(h, x - i\delta), & x \in (k\delta, (k+1)\delta), \quad 1 \leq k \leq K^* - 1 \\ -\frac{1}{\delta} \sigma(h)(K^*\delta), & x \in (K^*\delta, (K^* + 1)\delta) \\ 0, & x > (K^* + 1)\delta. \end{cases} \quad (5.32)$$

We have to check whether $\sigma(h) \in H_w^0$ and local Lipschitz continuity. In view of (5.32) we compute

$$\begin{aligned} \partial_x \phi(h, x) &= (h'(x + \delta) - h'(x))e^{-\mathcal{I}_x^{x+\delta}(h)}\theta(x) \\ &\quad - (h(x + \delta) - h(x))^2 e^{-\mathcal{I}_x^{x+\delta}(h)}\theta(x) \\ &\quad + 2(h(x + \delta) - h(x))e^{-\mathcal{I}_x^{x+\delta}(h)}\partial_x \theta(x) \\ &\quad + \left(1 - e^{-\mathcal{I}_x^{x+\delta}(h)}\right) \partial_x^2 \theta(x). \end{aligned}$$

The following estimates are basic. From (5.14) we obtain

$$\begin{aligned} \frac{|\partial_x \phi(h, x)|^2}{4} &\leq \left(1 + e^{2\delta \|h\|_w}\right) \left(2(|h'(x + \delta)|^2 + |h'(x)|^2) |\theta(x)|^2 \right. \\ &\quad + \left(\int_x^{x+\delta} |h'(\eta)| d\eta\right)^4 |\theta(x)|^2 \\ &\quad + 4 \left(\int_x^{x+\delta} |h'(\eta)| d\eta\right)^2 |\partial_x \theta(x)|^2 \\ &\quad \left. + 2|\partial_x^2 \theta(x)|^2\right) \end{aligned}$$

and thus by (5.3)

$$\begin{aligned} \frac{|\partial_x \phi(h, x)|^2}{4} &\leq \left(1 + e^{2\delta \|h\|_w}\right) \left(2(|h'(x + \delta)|^2 + |h'(x)|^2) |\theta(x)|^2 \right. \\ &\quad + C_1^4 \|h\|_w^4 |\theta(x)|^2 \\ &\quad + 4C_1^2 \|h\|_w^2 |\partial_x \theta(x)|^2 \\ &\quad \left. + 2|\partial_x^2 \theta(x)|^2\right). \end{aligned} \tag{5.33}$$

Let $h_1, h_2 \in H_w$. We decompose $\Delta := \partial_x \phi(h_1, x) - \partial_x \phi(h_2, x)$ according to

$$\begin{aligned} \Delta &= (h'_1(x + \delta) - h'_1(x)) \left(e^{-\mathcal{I}_x^{x+\delta}(h_1)} - e^{-\mathcal{I}_x^{x+\delta}(h_2)}\right) \theta(x) \\ &\quad + e^{-\mathcal{I}_x^{x+\delta}(h_2)} (h'_1(x + \delta) - h'_2(x + \delta) + h'_1(x) - h'_2(x)) \theta(x) \\ &\quad - \left(\int_x^{x+\delta} h'_1(\eta) d\eta\right)^2 \left(e^{-\mathcal{I}_x^{x+\delta}(h_1)} - e^{-\mathcal{I}_x^{x+\delta}(h_2)}\right) \theta(x) \\ &\quad - e^{-\mathcal{I}_x^{x+\delta}(h_2)} \left(\int_x^{x+\delta} (h'_1(\eta) + h'_2(\eta)) d\eta\right) \\ &\quad \times \left(\int_x^{x+\delta} (h'_1(\eta) - h'_2(\eta)) d\eta\right) \theta(x) \\ &\quad + 2 \left(\int_x^{x+\delta} h'_1(\eta) d\eta\right) \left(e^{-\mathcal{I}_x^{x+\delta}(h_1)} - e^{-\mathcal{I}_x^{x+\delta}(h_2)}\right) \partial_x \theta(x) \\ &\quad + 2e^{-\mathcal{I}_x^{x+\delta}(h_2)} \left(\int_x^{x+\delta} (h'_1(\eta) - h'_2(\eta)) d\eta\right) \partial_x \theta(x) \\ &\quad - \left(e^{-\mathcal{I}_x^{x+\delta}(h_1)} - e^{-\mathcal{I}_x^{x+\delta}(h_2)}\right) \partial_x^2 \theta(x). \end{aligned}$$

Using

$$\left| e^{-\mathcal{I}_x^{x+\delta}(h_1)} - e^{-\mathcal{I}_x^{x+\delta}(h_2)} \right| \leq \delta \|h_1 - h_2\|_w e^{\delta(\|h_1\|_w + \|h_2\|_w)},$$

(5.3) and (5.14) we get

$$\begin{aligned} \frac{|\Delta|^2}{7} &\leq e^{2\delta(\|h_1\|_w + \|h_2\|_w)} \\ &\times \left(2(|h'_1(x+\delta)|^2 + |h'_1(x)|^2)\delta^2 \|h_1 - h_2\|_w^2 |\theta(x)|^2 \right. \\ &\quad + 2(|h'_1(x+\delta) - h'_2(x+\delta)|^2 + |h'_1(x) - h'_2(x)|^2) |\theta(x)|^2 \\ &\quad + C_1^4 \|h_1\|_w^4 \delta^2 \|h_1 - h_2\|_w^2 |\theta(x)|^2 \\ &\quad + 2C_1^4 (\|h_1\|_w^2 + \|h_2\|_w^2) \|h_1 - h_2\|_w^2 |\theta(x)|^2 \\ &\quad + 4C_1^2 \|h_1\|_w^2 \delta^2 \|h_1 - h_2\|_w^2 |\partial_x \theta(x)|^2 \\ &\quad + 4C_1^2 \|h_1 - h_2\|_w^2 |\partial_x \theta(x)|^2 \\ &\quad \left. + \delta^2 \|h_1 - h_2\|_w^2 |\partial_x^2 \theta(x)|^2 \right). \end{aligned} \quad (5.34)$$

In the sequel, C denotes a real constant which only depends on w , δ and K^* but represents different values from line to line.

In view of (5.32) we have

$$\begin{aligned} \int_{\mathbb{R}_+} |\sigma(h)'(x)|^2 w(x) dx &= \int_0^{K^*\delta} |\sigma(h)'(x)|^2 w(x) dx \\ &\quad + \frac{|\sigma(h)(K^*\delta)|^2}{\delta^2} \int_{K^*\delta}^{(K^*+1)\delta} w(x) dx. \end{aligned}$$

But Hölder's inequality yields

$$|\sigma(h)(K^*\delta)|^2 \leq \left(\int_0^{K^*\delta} |\sigma(h)'(x)| dx \right)^2 \leq C \left(\int_0^{K^*\delta} |\sigma(h)'(x)|^2 w(x) dx \right),$$

and so, by (5.14) and (5.32)

$$\begin{aligned} \int_{\mathbb{R}_+} |\sigma(h)'(x)|^2 w(x) dx &\leq C \int_0^{K^*\delta} |\sigma(h)'(x)|^2 w(x) dx \\ &= C \sum_{k=1}^{K^*-1} \int_{k\delta}^{(k+1)\delta} \left| \sum_{i=1}^k \partial_x \phi(h, x - i\delta) \right|^2 w(x) dx \\ &\leq C \sum_{k=1}^{K^*-1} \sum_{i=1}^k \int_{k\delta}^{(k+1)\delta} |\partial_x \phi(h, x - i\delta)|^2 w(x) dx \\ &\leq C w(K^*\delta) \int_I |\partial_x \phi(h, x)|^2 dx. \end{aligned} \quad (5.35)$$

where

$$I := [0, (K^* - 1)\delta]. \quad (5.36)$$

In the same manner we can see that

$$\begin{aligned} & \int_{\mathbb{R}_+} |\sigma(h_1)'(x) - \sigma(h_2)'(x)|^2 w(x) dx \\ & \leq Cw(K^*\delta) \int_I |\partial_x \phi(h_1, x) - \partial_x \phi(h_2, x)|^2 dx. \end{aligned} \quad (5.37)$$

From (5.33) and (5.35) it follows now easily

$$\begin{aligned} \|\sigma(h)\|_w^2 & \leq C \left(1 + e^{2\delta\|h\|_w} \right) \\ & \quad \times \left(\|\theta\|_{L^\infty(I)}^2 + \|\theta\|_{L^2(I)}^2 + \|\partial_x \theta\|_{L^2(I)}^2 + \|\partial_x^2 \theta\|_{L^2(I)}^2 \right) \\ & \leq C \left(1 + e^{2\delta\|h\|_w} \right) \|\partial_x^2 \theta\|_{L^2(I)}^2 \end{aligned} \quad (5.38)$$

and we have used (5.31). Combining (5.34) and (5.37) gives similarly

$$\|\sigma(h_1) - \sigma(h_2)\|_w^2 \leq Ce^{2\delta(\|h_1\|_w + \|h_2\|_w)} \|h_1 - h_2\|_w^2 \|\partial_x^2 \theta\|_{L^2(I)}^2. \quad (5.39)$$

We are thus led to the following result.

Theorem 5.6.1. *Let $(\theta^j(t, x))_{j \in \mathbb{N}}$ satisfy*

- i) $(t, x) \mapsto \theta^j(t, x) : \mathbb{R}_+ \times I \rightarrow \mathbb{R}$ *is jointly measurable*
- ii) $\theta^j(t, \cdot) \in C^2(I)$, $\forall t \in \mathbb{R}_+$
- iii) $\theta^j(t, 0) = \partial_x \theta^j(t, 0) = 0$, $\forall t \in \mathbb{R}_+$
- iv) $\sup_{t \in \mathbb{R}_+} \sum_{j \in \mathbb{N}} \|\partial_x^2 \theta^j(t, \cdot)\|_{L^2(I)}^2 < \infty$,

where I is given in (5.36). Then the above construction yields a measurable mapping

$$\sigma = (\sigma^j)_{j \in \mathbb{N}} : (\mathbb{R}_+ \times H_w, \mathcal{B}(\mathbb{R}_+) \otimes \mathcal{B}(H_w)) \rightarrow (L_2^0(H_w^0), \mathcal{B}(L_2^0(H_w^0)))$$

which meets (5.29) and the assumptions of Theorem 5.2.1.i).

Proof. The structure and arguments of the proof are similar to those of the proof of Proposition 5.4.1. We shall give a sketch.

Condition iii) is (5.31). Estimate (5.38) and iv) clearly yield $\sigma(t, h) \in L_2^0(H_w^0)$ and local boundedness in h . Let $g \in H_w$. Measurability of $(t, h) \mapsto \langle \sigma^j(t, h), g \rangle_w$ follows, using i) and ii), by similar arguments as in Step 3 in the proof of Proposition 5.4.1.

Finally, iv) and (5.39) imply local Lipschitz continuity of $\sigma(t, h)$ in h . $\square$

The so found local HJM model is a slight modification of the BGM [13] model. The difference is that (4.20) holds for $T \leq T^* - \delta$ with $\pi^j(t, T) = \theta^j(t, T - t)$. Consequently, only caplets with settlement date $T + \delta \leq T^*$ can

be priced according to formula (4.21). From a practical point of view this is no restriction at all, since T^* can be chosen arbitrarily large. From the mathematical point of view this restriction is essential: $T^* = K^*\delta$ has to be finite in estimates (5.35) and (5.37).

Unfortunately, we cannot prove the existence of an HJM model which has property (5.29). In contrast to the preceding result, BGM [13] can assert global existence for equation (4.23) if considered as an SPDE, see [13, Corollary 2.1]. Yet their SPDE methods give no information about $\|r_t\|_w$. Moreover, they do not provide a general framework for (local) HJM models as in the present thesis.

In summary, we have shown that the BGM [13] model is (essentially) recaptured by our setting. In particular, it is available for the consistency results in Chapter 7.

6. Invariant Manifolds for Stochastic Equations

The results of this intermediary chapter are of general form, in the spirit of Chapter 2, and not exclusively related to HJM models. Yet they will enable us to solve our consistency problems.

We study viability and invariance problems related to a stochastic equation in a Hilbert space H . Finite dimensional invariant C^2 submanifolds of H are characterized. We derive Nagumo type conditions and prove a regularity result: any weak solution which is viable in a finite dimensional C^2 submanifold is a strong solution. This chapter is based on the article [29].

6.1 Finite Dimensional Submanifolds in Banach Spaces

For the convenience of the reader we discuss and prove some crucial properties of finite dimensional submanifolds in Banach spaces. The main tool from infinite dimensional analysis is the inverse mapping theorem, see [1, Theorem 2.5.2], which does not rely on a scalar product. Therefore we chose this somewhat oversized framework, although in our setting stochastic equations take place in Hilbert spaces.

Let E denote a Banach space, E' its dual space. We write $\langle e', e \rangle$ for the duality pairing. For a direct sum decomposition $E = E_1 \oplus E_2$ we denote by $\Pi_{(E_2, E_1)}$ the induced projection onto E_1 .

Let $k, m \in \mathbb{N}$. We begin with an important corollary of the inverse mapping theorem, see [1, Theorem 2.5.12].

Proposition 6.1.1. *Let $\phi \in C^k(V; E)$, for some open set $V \subset \mathbb{R}^m$. Suppose $D\phi(y_0)$ is one to one for some $y_0 \in V$. Then $D\phi(y_0)\mathbb{R}^m$ is m -dimensional and complemented in E*

$$E = D\phi(y_0)\mathbb{R}^m \oplus E_2.$$

Moreover, there exist two open neighborhoods V' of $(y_0, 0)$ in $V \times E_2$ and U of $\phi(y_0)$ in E , and a C^k diffeomorphism $\Psi : U \rightarrow V'$ such that

$$\Psi \circ \phi(y) = (y, 0), \quad \forall y \in V' \cap (\mathbb{R}^m \times \{0\}). \quad (6.1)$$

Furthermore, $D\phi(y)$ is one to one and

$$D\phi(y)^{-1} = D\Psi(\phi(y))|_{D\phi(y)\mathbb{R}^m}, \quad \forall y \in V' \cap (\mathbb{R}^m \times \{0\}). \quad (6.2)$$

Proof. Let $\{e_1, \dots, e_m\}$ be a basis of $D\phi(y_0)\mathbb{R}^m$. Every $z \in D\phi(y_0)\mathbb{R}^m$ has a unique representation $z = z_1 e_1 + \dots + z_m e_m$. Set $\alpha_i(z) := z_i$. Each α_i extends to a member e'_i of E' by the Hahn–Banach theorem. Define E_2 as the intersection of the null spaces of all e'_i . Then $E = D\phi(y_0)\mathbb{R}^m \oplus E_2$ and

$$\Pi_{(E_2, D\phi(y_0)\mathbb{R}^m)} = \langle e'_1, \cdot \rangle e_1 + \dots + \langle e'_m, \cdot \rangle e_m.$$

Define $\Phi : V \times E_2 \rightarrow E$ by $\Phi(y, z) := \phi(y) + z$. Then $\Phi \in C^k(V \times E_2; E)$ and $D\Phi(y, z)(v_1, v_2) = D\phi(y)v_1 + v_2$ for $(v_1, v_2) \in \mathbb{R}^m \times E_2$, see [1, Proposition 2.4.12]. Now $D\Phi(y_0, 0) : \mathbb{R}^m \times E_2 \rightarrow E$ is linear, bijective and continuous, hence a linear isomorphism by the open mapping theorem. Accordingly, the inverse mapping theorem yields the existence of U and V' as claimed such that $\Phi : V' \rightarrow U$ is a C^k diffeomorphism with inverse Ψ .

Consequently, $\Psi \circ \phi(y) = \Psi \circ \Phi(y, 0) = (y, 0)$ for $y \in V' \cap (\mathbb{R}^m \times \{0\})$. The chain rule gives $D\Psi(\phi(y))D\phi(y) = Id_{\mathbb{R}^m}$. Hence $D\phi(y)$ is one to one and (6.2) holds. $\square$

Definition 6.1.1. *The mapping ϕ from Proposition 6.1.1 is called a C^k immersion at y_0 . If ϕ is a C^k immersion at each $y_0 \in V$, we just say ϕ is a C^k immersion.*

If $\phi : V \rightarrow E$ is an injective C^k immersion, $V \subset \mathbb{R}^m$ open, we call $\mathcal{M} := \phi(V)$ an m -dimensional immersed C^k submanifold of E .

The next definition straightly extends the concept of a regular surface in $\mathbb{R}^3$.

Definition 6.1.2. *A subset $\mathcal{M} \subset E$ is an m -dimensional (regular) C^k submanifold of E , if for all $h \in \mathcal{M}$ there is a neighborhood U in E , an open set $V \subset \mathbb{R}^m$ and a C^k map $\phi : V \rightarrow E$ such that*

- i) $\phi : V \rightarrow U \cap \mathcal{M}$ is a homeomorphism
- ii) $D\phi(y)$ is one to one for all $y \in V$.

The map ϕ is called a parametrization in h .

$\mathcal{M}$ is a linear submanifold if for all $h \in \mathcal{M}$ there exists a linear parametrization of the form $\phi(y) = h + \sum_{i=1}^m y_i e_i$ in h .

Let $\phi : V \rightarrow E$ be an injective C^k immersion, $V \subset \mathbb{R}^m$ open, whence $\mathcal{M} = \phi(V)$ an immersed submanifold. In the notation of Proposition 6.1.1 we have by (6.1) that $\phi : V' \cap (\mathbb{R}^m \times \{0\}) \rightarrow \phi(V' \cap (\mathbb{R}^m \times \{0\}))$ is a homeomorphism, and hence $\phi(V' \cap (\mathbb{R}^m \times \{0\}))$ is a regular submanifold. Yet in general $\phi : V \rightarrow \mathcal{M}$ is not a homeomorphism. In other words, being an immersed submanifold does not imply being a regular submanifold. This is the crucial difference between an immersion and an *embedding*. For examples and more details see [1, Section 3.5].

In what follows, $\mathcal{M}$ denotes an m -dimensional C^k submanifold of E . As an immediate consequence of Proposition 6.1.1 and Definition 6.1.2 we show that $\mathcal{M}$ shares in fact the characterizing property of a C^k manifold:

Lemma 6.1.1. *Let $\phi_i : V_i \rightarrow U_i \cap \mathcal{M}$, $i = 1, 2$, be two parametrizations such that $W := U_1 \cap U_2 \cap \mathcal{M} \neq \emptyset$. Then the change of parameters*

$$\phi_1^{-1} \circ \phi_2 : \phi_2^{-1}(W) \rightarrow \phi_1^{-1}(W)$$

is a C^k diffeomorphism.

Proof. Of course, $\phi_1^{-1} \circ \phi_2 : \phi_2^{-1}(W) \rightarrow \phi_1^{-1}(W)$ is a homeomorphism.

Now let $y \in \phi_2^{-1}(W)$ and $r = \phi_1^{-1} \circ \phi_2(y)$. Then $E = D\phi_1(r)\mathbb{R}^m \oplus E_2$ by Proposition 6.1.1. Moreover there exist two open neighborhoods $V'_1 \subset V_1 \times E_2$ of $(r, 0)$ and $U' \subset E$ of $\phi_1(r)$, and a C^k diffeomorphism $\Psi : U' \rightarrow V'_1$ such that $\phi_1 = \Psi^{-1}$ on $V'_1 \cap (\mathbb{R}^m \times \{0\})$.

Now again we use the fact that ϕ_1 is a homeomorphism. Hence there exists an open set $U'' \subset E$ with $\phi_1(V'_1 \cap (\mathbb{R}^m \times \{0\})) = U'' \cap \mathcal{M}$. Set $U := U' \cap U''$. Then $\phi_1^{-1} = \Psi$ on $U \cap \mathcal{M}$. By continuity of ϕ_2 there is an open neighborhood $V'_2 \subset \phi_2^{-1}(W)$ of y with $\phi_2(V'_2) \subset U \cap \mathcal{M}$. But then $\phi_1^{-1} \circ \phi_2|_{V'_2} = \Psi \circ \phi_2|_{V'_2}$ is C^k in V'_2 .

Since y was arbitrary, $\phi_1^{-1} \circ \phi_2$ is C^k in $\phi_2^{-1}(W)$. By symmetry the assertion follows. $\square$

Definition 6.1.3. *For $h \in \mathcal{M}$ the tangent space to $\mathcal{M}$ at h is the subspace*

$$T_h \mathcal{M} := D\phi(y)\mathbb{R}^m, \quad y = \phi^{-1}(h),$$

where $\phi : V \subset \mathbb{R}^m \rightarrow \mathcal{M}$ is a parametrization in h .

By Lemma 6.1.1, the definition of $T_h \mathcal{M}$ is independent of the choice of the parametrization.

Definition 6.1.4. *A vector field X on $\mathcal{M}$ is a mapping assigning to each point h of $\mathcal{M}$ an element $X(h)$ of $T_h \mathcal{M}$.*

A vector field X can be represented locally as

$$X(h) = D\phi(y)\alpha(y), \quad y = \phi^{-1}(h), \quad \forall h \in U \cap \mathcal{M}, \quad (6.3)$$

where $\phi : V \rightarrow U \cap \mathcal{M}$ is a parametrization and α is an $\mathbb{R}^m$ -valued vector field on V (uniquely determined by ϕ). This is how smoothness of vector fields on $\mathcal{M}$ can be defined in terms of smoothness of $\mathbb{R}^m$ -valued vector fields.

Definition 6.1.5. *The vector field X is of class C^r , $0 \leq r < k$, if for any parametrization ϕ the corresponding $\mathbb{R}^m$ -valued vector field α in (6.3) is of class C^r .*

Again by Lemma 6.1.1 this is a well defined concept.

It will be useful to extend a parametrization to the whole of $\mathbb{R}^m$. Let $h \in \mathcal{M}$ and $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization in h . Set $y = \phi^{-1}(h)$. Since V is a neighborhood of y there exists $\varepsilon > 0$ such that the open ball

$B_{2\varepsilon}(y) = \{v \in \mathbb{R}^m \mid |y-v| < 2\varepsilon\}$ is contained in V . On $B_{2\varepsilon}(y)$ one can define a function $\psi \in C^\infty(\mathbb{R}^m; [0, 1])$ satisfying $\psi \equiv 1$ on $\overline{B_\varepsilon(y)}$ and $\text{supp}(\psi) \subset B_{2\varepsilon}(y)$, see [12, Theorem (5.1), Chap. II]. Since ϕ is a homeomorphism there exists an open neighborhood U' of h in E with $\phi(B_\varepsilon(y)) = U' \cap \mathcal{M}$. Set $\tilde{\phi} := \psi\phi$. Then $\tilde{\phi} \in C_b^k(\mathbb{R}^m; E)$ and $\tilde{\phi}|_{B_\varepsilon(y)} = \phi|_{B_\varepsilon(y)} : B_\varepsilon(y) \rightarrow U' \cap \mathcal{M}$ is a parametrization in h . We have thus shown

Remark 6.1.1. We may and will assume that any parametrization $\phi : V \rightarrow U \cap \mathcal{M}$ extends to $\phi \in C_b^k(\mathbb{R}^m; E)$.

Let $h \in \mathcal{M}$. As shown in Proposition 6.1.1, the tangent space $T_h\mathcal{M}$ is complemented in E by $E_2 = \cap_{i=1}^m \ker(e'_i)$, where $\{e_1, \dots, e_m\}$ is a basis for $T_h\mathcal{M}$ and $\langle e'_i, e_j \rangle = \delta_{ij}$ (Kronecker delta). The following lemma states that, locally in h , $\Pi_{(E_2, T_h\mathcal{M})}$ is a diffeomorphism on $\mathcal{M}$ into $T_h\mathcal{M}$.

Lemma 6.1.2. *There exists a parametrization $\phi : V \rightarrow U \cap \mathcal{M}$ in h such that*

$$\phi(\langle e'_1, z \rangle, \dots, \langle e'_m, z \rangle) = z, \quad \forall z \in U \cap \mathcal{M}.$$

Proof. Let $\psi : V_1 \rightarrow U_1 \cap \mathcal{M}$ be a parametrization in h and write $y = \psi^{-1}(h)$. Denote by $I_{T_h\mathcal{M}} : T_h\mathcal{M} \rightarrow \mathbb{R}^m$ the canonical isomorphism

$$I_{T_h\mathcal{M}} := (\langle e'_1, \cdot \rangle, \dots, \langle e'_m, \cdot \rangle).$$

Then the map $h := I_{T_h\mathcal{M}} \circ \Pi_{(E_2, T_h\mathcal{M})} \circ \psi : V_1 \rightarrow \mathbb{R}^m$ is C^k and $Dh(y) = I_{T_h\mathcal{M}} \circ D\psi(y)$ is one to one. By the inverse mapping theorem there exist open neighborhoods $W \subset V_1$ of y and $V \subset \mathbb{R}^m$ of $h(y)$ such that $h : W \rightarrow V$ is a C^k diffeomorphism. Since ψ is a homeomorphism there is an open neighborhood $U \subset E$ of h with $\psi(W) = U \cap \mathcal{M}$. The map $\phi := \psi \circ h^{-1} : V \rightarrow U \cap \mathcal{M}$ is then the desired parametrization in h , that is, $\phi^{-1} = I_{T_h\mathcal{M}} \circ \Pi_{(E_2, T_h\mathcal{M})}$ on $U \cap \mathcal{M}$. $\square$

The following result is crucial for our discussion on weak solutions to stochastic equations viable in $\mathcal{M}$.

Proposition 6.1.2. *Let $D \subset E'$ be a dense subset. Then for any $h \in \mathcal{M}$ there exist elements $f'_1, \dots, f'_m$ in D and a parametrization $\phi : V \rightarrow U \cap \mathcal{M}$ in h such that*

$$\phi(\langle f'_1, z \rangle, \dots, \langle f'_m, z \rangle) = z, \quad \forall z \in U \cap \mathcal{M}.$$

If $\mathcal{M}$ is linear, then ϕ is linear: $\phi(v) = e_0 + \sum_{i=1}^m (\sum_{j=1}^m N_{ij} v_j) e_i$, for $v \in V$.

Proof. The idea is to find a decomposition $E = F_1 \oplus F_2$, $\dim F_1 = m$, such that F_1 is “not too far” from $T_h\mathcal{M}$ and such that

$$\Pi_{(F_2, F_1)} = \langle f'_1, \cdot \rangle f_1 + \dots + \langle f'_m, \cdot \rangle f_m$$

with $f'_1, \dots, f'_m \in D$. Thereby the expression “not too far” means that $\Pi_{(F_2, F_1)|T_h\mathcal{M}} : T_h\mathcal{M} \rightarrow F_1$ is an isomorphism.

Let $\{e_1, \dots, e_m\}$ be a basis for $T_h\mathcal{M}$ and $\psi : V_1 \rightarrow U_1 \cap \mathcal{M}$ the parametrization in h given by Lemma 6.1.2. Set $y = \psi^{-1}(h)$. We may assume that $\|e_i\|_E = 1$, for $1 \leq i \leq m$. Since D is dense in E' , there exist elements $f'_1, \dots, f'_m$ in D with $\|f'_i - e'_i\|_{E'} < 2^{-m}$. Notice that by definition $\langle e'_i, e_j \rangle = \delta_{ij}$. Hence

$$|\langle f'_i, e_j \rangle| \leq |\langle e'_i, e_j \rangle| + |\langle f'_i - e'_i, e_j \rangle| < 2^{-m}, \quad \text{if } i \neq j$$

and

$$|\langle f'_i, e_i \rangle| \geq |\langle e'_i, e_i \rangle| - |\langle f'_i - e'_i, e_i \rangle| > 1 - 2^{-m}.$$

The matrix $M := (\langle f'_i, e_j \rangle)_{1 \leq i, j \leq m}$ is therefore invertible, which follows from the theorem of Gerschgorin, see [52]. Consequently, the family $\{f'_1, \dots, f'_m\}$ is linearly independent in E' . Since E is reflexive, by the Hahn–Banach theorem there exists a linearly independent set $\{f_1, \dots, f_m\}$ in E such that $\langle f'_i, f_j \rangle = \delta_{ij}$. Write $F_1 := \text{span}\{f_1, \dots, f_m\}$. The f'_i and f_i induce then a decomposition $E = F_1 \oplus F_2$ with projection $\Pi_{(F_2, F_1)} = \langle f'_1, \cdot \rangle f_1 + \dots + \langle f'_m, \cdot \rangle f_m$. Since

$$\Pi_{(F_2, F_1)} e_j = \sum_{i=1}^m M_{ij} f_i, \quad \text{for } 1 \leq j \leq m,$$

we see that $\Pi_{(F_2, F_1)|T_h\mathcal{M}} : T_h\mathcal{M} \rightarrow F_1$ is an isomorphism. Let $I_{F_1} : F_1 \rightarrow \mathbb{R}^m$ denote the canonical isomorphism $I_{F_1} := (\langle f'_1, \cdot \rangle, \dots, \langle f'_m, \cdot \rangle)$. Then the map $h := I_{F_1} \circ \Pi_{(F_2, F_1)} \circ \psi : V_1 \rightarrow \mathbb{R}^m$ is C^k and $Dh(y) = I_{F_1} \circ \Pi_{(F_2, F_1)} \circ D\psi(y)$ is one to one. Now proceed as in the proof of Lemma 6.1.2 to get the desired parametrization ϕ in h with $\phi^{-1} = I_{F_1} \circ \Pi_{(F_2, F_1)}$ on $U \cap \mathcal{M}$, for some neighborhood U of h .

If $\mathcal{M}$ is linear, we choose $\psi(u) = h + \sum_{i=1}^m u_i e_i$. An easy computation shows that

$$\phi(v) = (h - (\Pi_{(F_2, F_1)|T_h\mathcal{M}})^{-1} \circ \Pi_{(F_2, F_1)} h) + \sum_{i=1}^m \left(\sum_{j=1}^m M_{ij}^{-1} v_j \right) e_i.$$

Setting $N := M^{-1}$ and $e_0 := h - (\Pi_{(F_2, F_1)|T_h\mathcal{M}})^{-1} \circ \Pi_{(F_2, F_1)} h$ completes the proof. $\square$

Lemma 6.1.2 asserts the existence of a particular parametrization $\phi : V \rightarrow U \cap \mathcal{M}$ in $h \in \mathcal{M}$ related to the decomposition $E = T_h\mathcal{M} \oplus E_2$. Actually, E_2 is complemented in E simultaneously by all tangent spaces $T_z\mathcal{M}$, $z \in U \cap \mathcal{M}$.

Lemma 6.1.3. *Let $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization. If there exist elements $e'_1, \dots, e'_m$ in E' with the property that*

$$\phi(\langle e'_1, h \rangle, \dots, \langle e'_m, h \rangle) = h, \quad \forall h \in U \cap \mathcal{M},$$

then $e'_1, \dots, e'_m$ are linearly independent in E' and

$$E = T_h \mathcal{M} \oplus E_2, \quad \forall h \in U \cap \mathcal{M},$$

where $E_2 := \bigcap_{i=1}^m \ker(e'_i)$. Moreover, the induced projections are given by

$$\Pi_{(E_2, T_h \mathcal{M})} = D\phi(y)(\langle e'_1, \cdot \rangle, \dots, \langle e'_m, \cdot \rangle), \quad y = \phi^{-1}(h), \quad \forall h \in U \cap \mathcal{M}. \quad (6.4)$$

Proof. Define $\Lambda := (\langle e'_1, \cdot \rangle, \dots, \langle e'_m, \cdot \rangle) \in L(E; \mathbb{R}^m)$. Fix $h \in U \cap \mathcal{M}$ and let $y = \phi^{-1}(h)$. By assumption $\Lambda \circ \phi(v) = v$ on V , therefore $\Lambda \circ D\phi(y) = Id_{\mathbb{R}^m}$. It follows that $\Lambda|_{T_h \mathcal{M}} : T_h \mathcal{M} \rightarrow \mathbb{R}^m$ is an isomorphism. Consequently, $e'_1, \dots, e'_m$ are linearly independent in E' .

Another consequence is that $\Lambda : E \rightarrow \mathbb{R}^m$ is onto. Now the map $\Pi := D\phi(y) \circ \Lambda$ satisfies (i) $\Pi \in L(E)$, (ii) $\ker(\Pi) = \ker(\Lambda) = E_2$, (iii) $\Pi(E) = T_h \mathcal{M}$ and (iv) $\Pi^2 = \Pi$, hence $E = T_h \mathcal{M} \oplus E_2$ and Π is the corresponding projection. Since $h \in U \cap \mathcal{M}$ was arbitrary, this completes the proof. $\square$

Finally, we derive a result which will prove crucial for Itô calculus on submanifolds. In the remainder of this section we assume $\mathcal{M}$ to be a C^2 submanifold. Let $B \in C^1(E)$ have the property

$$B(h) \in T_h \mathcal{M}, \quad \forall h \in \mathcal{M},$$

i.e. $B|_{\mathcal{M}}$ is a C^1 vector field on $\mathcal{M}$.

Fix $h \in \mathcal{M}$ and let $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization in h . Set $y = \phi^{-1}(h)$. For $\delta > 0$ small enough, the curve

$$c(t) := \phi(y + tD\phi(y)^{-1}B(h)), \quad t \in (-\delta, \delta),$$

is well defined and satisfies

$$c : (-\delta, \delta) \rightarrow U \cap \mathcal{M}, \quad c \in C^1((-\delta, \delta); E), \quad c(0) = h, \quad c'(0) = B(h). \quad (6.5)$$

Hence

$$\frac{d}{dt} B(c(t))|_{t=0} = DB(h)B(h). \quad (6.6)$$

In terms of differential geometry $B(h)$ can be identified with the equivalence class $[c]$ of all curves satisfying (6.5). By the very definition of $DB(h)$ as a mapping from $T_h \mathcal{M}$ into $T_{B(h)} E \equiv E$, the vector $DB(h)B(h)$ is in the same manner identified with $[B \circ c]$, see [1, Section 3.3]. This is just (6.6) and means that the differentiable structure on $\mathcal{M}$ is the one induced by E .

Suppose now that ϕ satisfies the assumptions of Lemma 6.1.3. To shorten notation, we write $\langle e', \cdot \rangle$ instead of $(\langle e'_1, \cdot \rangle, \dots, \langle e'_m, \cdot \rangle)$. By (6.4)

$$B(c(t)) = D\phi(\langle e', c(t) \rangle) \langle e', B(c(t)) \rangle, \quad \forall t \in (-\delta, \delta).$$

Differentiation with respect to t gives

$$\frac{d}{dt} B(c(t))|_{t=0} = D^2\phi(y)(\langle e', B(h) \rangle, \langle e', B(h) \rangle) + D\phi(y) \langle e', DB(h)B(h) \rangle. \quad (6.7)$$

Combining (6.4), (6.6) and (6.7) we get the following proposition.

Proposition 6.1.3. *Let $\mathcal{M}$, B and $\phi : V \rightarrow U \cap \mathcal{M}$ be as above. Then*

$$DB(h)B(h) = D\phi(y)\langle e', DB(h)B(h) \rangle + D^2\phi(y)(\langle e', B(h) \rangle, \langle e', B(h) \rangle),$$

$y = \phi^{-1}(h)$, is the decomposition according to $E = T_h\mathcal{M} \oplus E_2$, for all $h \in U \cap \mathcal{M}$.

6.2 Invariant Manifolds

We adopt the notation from Chapter 2 and let $\mathcal{M}$ denote an m -dimensional C^2 submanifold of H , for some $m \in \mathbb{N}$. The concept of a submanifold is used in more than one sense in the literature, see [12]. We therefore point out that $\mathcal{M}$ represents what is also called a *regular* or *embedded* submanifold, i.e. the topology and differentiable structure on $\mathcal{M}$ is the one induced by H . This allows for deriving global regularity and invariance results, see Theorems 6.2.1 and 6.2.2 below.

Essential for our needs is the following observation. Since H is separable, by Lindelöf's Lemma [1, p. 4] there exists a countable open covering $(U_k)_{k \in \mathbb{N}}$ of $\mathcal{M}$ and for each k a parametrization $\phi_k : V_k \subset \mathbb{R}^m \rightarrow U_k \cap \mathcal{M}$, where $\phi_k \in C_b^2(\mathbb{R}^m; H)$, see Remark 6.1.1. Using the fact that $D(A^*)$ is dense in H , by Proposition 6.1.2 we can assume that for each k there exists a linearly independent set $\{\zeta_{k,1}, \dots, \zeta_{k,m}\}$ in $D(A^*)$ such that

$$\phi_k(\langle \zeta_{k,1}, h \rangle, \dots, \langle \zeta_{k,m}, h \rangle) = h, \quad \forall h \in U_k \cap \mathcal{M}. \quad (6.8)$$

To simplify notation, we write $\langle \zeta_k, h \rangle$ instead of $(\langle \zeta_{k,1}, h \rangle, \dots, \langle \zeta_{k,m}, h \rangle)$ and skip the index k whenever there is no ambiguity.

In the sequel we let X denote a (local) weak solution to (2.20). The main results of this chapter follow under some regularity assumptions, which for clarity are presented here in a collected form.

(A1) There exists a continuous version of X , still denoted by X .

(A2) $B(t, \omega, \cdot) \in C^1(H; L_2^0(H))$ for all $(t, \omega) \in \mathbb{R}_+ \times \Omega$.

(A3) For any $t \in \mathbb{R}_+$ we have

$$\mathbb{E} \left[\int_0^t \left(\|X_s\|_H + \|F(s, X_s)\|_H + \|B(s, X_s)\|_{L_2^0}^2 \right) ds \right] < \infty.$$

(A4) The mappings $F(t, \omega, h)$ and $B(t, \omega, h)$ are locally bounded and locally Lipschitz continuous in h .

(A5) The mappings $F(t, \omega, h)$ and $B(t, \omega, h)$ are right continuous in t , for all $h \in H$ and $\omega \in \Omega$.

Notice that **(A2)** implies $B^j(t, \omega, \cdot) \in C^1(H; H)$ for any $j \in \mathbb{N}$,

$$\sum_{j \in \mathbb{N}} \|DB^j(t, \omega, h)B^j(t, \omega, h)\|_H^2 \leq \|DB(t, \omega, h)\|_{L(H; L_2^0(H))}^2 \|B(t, \omega, h)\|_{L_2^0}^2 < \infty$$

and

$$\sum_{j \in \mathbb{N}} DB^j(t, \omega, \cdot) B^j(t, \omega, \cdot) \in C^0(H; H), \quad \forall (t, \omega) \in \mathbb{R}_+ \times \Omega.$$

For the sake of readability all proofs are postponed to the following section. First we state a (global) regularity result.

Theorem 6.2.1. *Let X be a local weak solution to (2.20) with lifetime τ , assume (A1) and*

$$X_{t \wedge \tau} \in \mathcal{M}, \quad \forall t \in \mathbb{R}_+. \quad (6.9)$$

Then there exists a stopping time $0 < \tau' \leq \tau$, such that X is a local strong solution to (2.20) with lifetime τ' .

If in addition $\tau = \infty$ and (A3) holds, then X is a strong solution to (2.20).

We can give sufficient conditions for a weak solution X to be viable in $\mathcal{M}$. For that purpose we introduce the following Nagumo type *consistency conditions*

$$\mu(t, \omega, h) := Ah + F(t, \omega, h) - \frac{1}{2} \sum_{j \in \mathbb{N}} DB^j(t, \omega, h) B^j(t, \omega, h) \in T_h \mathcal{M} \quad (6.10)$$

$$B^j(t, \omega, h) \in T_h \mathcal{M}, \quad \forall j \in \mathbb{N}. \quad (6.11)$$

Notice that $\mu(t, \omega, h)$ in (6.10) is well defined for any $h \in D(A)$ whenever (A2) holds.

Theorem 6.2.2. *Let X be a weak solution to (2.20) with $X_0 \in \mathcal{M}$. Assume (A1), (A2) and (A4). If $\mathcal{M}$ is closed, lies in $D(A)$ and satisfies the consistency conditions (6.10) and (6.11) for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in \mathbb{R}_+ \times \Omega$, for all $h \in \mathcal{M}$. Then (6.9) holds for $\tau = \infty$.*

Moreover, A is continuous on $\mathcal{M}$. Consequently, (6.10) and (6.11) hold for all $h \in \mathcal{M}$, for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in \mathbb{R}_+ \times \Omega$.

It turns out that the viability condition (6.9) is too weak to imply (6.10) and (6.11). Neither does it imply that $\mathcal{M}$ lies in $D(A)$. As a link between Theorems 6.2.1 and 6.2.2 we shall formulate equivalent conditions for (6.10) and (6.11). Rather than just asking for the assumptions of Theorem 6.2.1 we have to require that (6.9) holds for any space-time starting point (t_0, h_0) .

Definition 6.2.1. *A subset $V \subset H$ is called locally invariant for (2.20), if for all $(t_0, h_0) \in \mathbb{R}_+ \times V$ there exists a continuous local weak solution $X^{(t_0, h_0)}$ to (2.25) with lifetime $\tau = \tau(t_0, h_0)$ such that*

$$X_{t \wedge \tau}^{(t_0, h_0)} \in V, \quad \forall t \in \mathbb{R}_+.$$

Observe that this definition involves local existence.

Theorem 6.2.3. *Assume (A2), (A4) and (A5). Then the following conditions are equivalent:*

- i) $\mathcal{M}$ is locally invariant for (2.20).
- ii) $\mathcal{M} \subset D(A)$ and (6.10)–(6.11) hold for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in \mathbb{R}_+ \times \Omega$, for all $h \in \mathcal{M}$.
- iii) $\mathcal{M} \subset D(A)$ and (6.10)–(6.11) hold for all $(t, h) \in \mathbb{R}_+ \times \mathcal{M}$, for $\mathbb{P}$ -a.e. $\omega \in \Omega$.

We finally mention the important case where $\mathcal{M}$ is a linear submanifold. Then the above results are true under much weaker assumptions.

Theorem 6.2.4. *If $\mathcal{M}$ is an m -dimensional linear submanifold, then Theorems 6.2.2 and 6.2.3 remain valid without assuming (A2) and with $\mu(t, \omega, h)$ in (6.10) replaced by*

$$\nu(t, \omega, h) := Ah + F(t, \omega, h).$$

6.3 Proof of Theorems 6.2.1–6.2.4

Proof (of Theorem 6.2.1). Assume first that X is a weak solution, that is $\tau = \infty$, and that (A3) holds. Fix $T > 0$. Following [24, Lemma (3.5)] there exists an increasing sequence of predictable stopping times $0 = T_0 \leq T_1 \leq T_2 \leq \dots \leq T$ with $\sup_n T_n = T$, such that on each of the intervals

$$[T_n, T_{n+1}] \cap \{T_{n+1} > T_n\}$$

the process X takes values in some $U_{\alpha(n)}$ (here $[0, T] \times \{T_{n+1} > T_n\}$ is identified with $\{T_{n+1} > T_n\}$). This holds due to (A1).

Now let $n \in \mathbb{N}_0$ and $k = \alpha(n)$. We will analyse X locally on U_k . For simplicity we shall skip the index k for ϕ_k , U_k , V_k and ζ_k in what follows. Define the $\mathbb{R}^m$ -valued predictable processes b and ρ^j as

$$\begin{aligned} b_t &:= \langle A^* \zeta, X_t \rangle + \langle \zeta, F(t, X_t) \rangle \\ \rho_t^j &:= \langle \zeta, B^j(t, X_t) \rangle, \end{aligned}$$

where logically $\langle A^* \zeta, X_t \rangle$ stands for $(\langle A^* \zeta_1, X_t \rangle, \dots, \langle A^* \zeta_m, X_t \rangle)$. By the very definition of X we conclude that

$$Y_t := \langle \zeta, X_t \rangle = \langle \zeta, h_0 \rangle + \int_0^t b_s ds + \sum_{j \in \mathbb{N}} \int_0^t \rho_s^j d\beta_s^j$$

is an $\mathbb{R}^m$ -valued Itô process. Hence Itô's formula (Theorem 2.3.1) applies for $\phi \circ Y$. Since moreover $Y_{(T_n+t) \wedge T_{n+1}}$ takes values in V on $\{T_{n+1} > T_n\}$ we have

$$X_{(T_n+t) \wedge T_{n+1}} = (\phi \circ Y)_{(T_n+t) \wedge T_{n+1}}, \quad \text{on } \{T_{n+1} > T_n\}. \quad (6.12)$$

Combining (2.22) and (6.12) we have for $\xi \in D(A^*)$

$$\begin{aligned} 0 = & \int_{T_n}^{(T_n+t) \wedge T_{n+1}} \left(\langle A^* \xi, X_s \rangle + \langle \xi, F(s, X_s) \rangle \right. \\ & \left. - \langle \xi, D\phi(Y_s) b_s + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2 \phi(Y_s)(\rho_s^j, \rho_s^j) \rangle \right) ds \\ & + \sum_{j \in \mathbb{N}} \int_{T_n}^{(T_n+t) \wedge T_{n+1}} \left(\langle \xi, B^j(s, X_s) \rangle - \langle \xi, D\phi(Y_s) \rho_s^j \rangle \right) d\beta_s^j. \end{aligned} \quad (6.13)$$

Notice that the series in the first integral converges point-wise and defines an H -valued predictable process, see Theorem 2.3.1. By (2.26) the last term in (6.13) is a continuous local martingale with respect to the filtration $(\mathcal{F}_{T_n+t})$. Therefore

$$\langle A^* \xi, X_t \rangle = \langle \xi, -F(t, X_t) + D\phi(Y_t) b_t + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2 \phi(Y_t)(\rho_t^j, \rho_t^j) \rangle \quad (6.14)$$

on $[T_n, T_{n+1}] \cap \{T_{n+1} > T_n\}$, $dt \otimes d\mathbb{P}$ -a.s. (see the proof of (3.6)). By separability of H there exists a countable dense subset $\mathcal{D} \subset D(A^*)$ such that (6.14) holds simultaneously for all $\xi \in \mathcal{D}$ and for all $(t, \omega) \in [T_n, T_{n+1}] \cap \{T_{n+1} > T_n\}$ outside an exceptional $dt \otimes d\mathbb{P}$ -nullset. Let (t, ω) be such a point of validity. Then $\langle A^* \xi, X_t(\omega) \rangle$ is a linear functional on $D(A^*)$, which is bounded on $\mathcal{D}$ by (6.14). Hence it is bounded and therefore continuous on $D(A^*)$. Since $A^{**} = A$, see [50, Theorem 13.12], we conclude that $X_t(\omega) \in D(A)$ and – using full notation again –

$$\begin{aligned} AX_t + F(t, X_t) = & D\phi(Y_t) \left(\langle A^* \zeta, X_t \rangle + \langle \zeta, F(t, X_t) \rangle \right) \\ & + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2 \phi(Y_t) (\langle \zeta, B^j(t, X_t) \rangle, \langle \zeta, B^j(t, X_t) \rangle) \end{aligned} \quad (6.15)$$

on $[T_n, T_{n+1}] \cap \{T_{n+1} > T_n\}$, $dt \otimes d\mathbb{P}$ -a.s.

Similarly one shows that (compute the bracket process of the right hand side of (6.13) and β^j)

$$B^j(t, X_t) = D\phi(Y_t) \langle \zeta, B^j(t, X_t) \rangle, \quad \forall j \in \mathbb{N}, \quad (6.16)$$

on $[T_n, T_{n+1}] \cap \{T_{n+1} > T_n\}$, $dt \otimes d\mathbb{P}$ -a.s.

In view of **(A3)** we have

$$\begin{aligned}
& \mathbb{E} \left[\int_0^T \left(\|D\phi(Y_s)b_s\|_H + \frac{1}{2} \sum_{j \in \mathbb{N}} \|D^2\phi(Y_s)(\rho_s^j, \rho_s^j)\|_H + \sum_{j \in \mathbb{N}} \|D\phi(Y_s)\rho_s^j\|_H^2 \right) ds \right] \\
& \leq C_1 \mathbb{E} \left[\int_0^T \left(\|b_s\|_{\mathbb{R}^m} + \sum_{j \in \mathbb{N}} \|\rho_s^j\|_{\mathbb{R}^m}^2 \right) ds \right] \\
& \leq C_2 \mathbb{E} \left[\int_0^T \left(\|X_s\|_H + \|F(s, X_s)\|_H + \|B(s, X_s)\|_{L_2^2}^2 \right) ds \right] < \infty,
\end{aligned} \tag{6.17}$$

where C_1 and C_2 depend only on n .

Summing up over $0 \leq n \leq N-1$ we get from (6.15) and (6.17)

$$\mathbb{E} \left[\int_0^{T_N} \|AX_s\|_H ds \right] < \infty, \quad \forall N \in \mathbb{N}.$$

Since $\lim_N \uparrow T_N = T$

$$\mathbb{P} \left[\int_0^t \|AX_s\|_H ds < \infty \right] = 1, \quad \forall t < T,$$

and by the identities (6.15) and (6.16)

$$X_t = h_0 + \int_0^t (AX_s + F(s, X_s)) ds + \sum_{j \in \mathbb{N}} \int_0^t B^j(s, X_s) d\beta_s^j.$$

Since T was arbitrary the second part of the theorem is proved.

Now suppose X is a local weak solution with lifetime τ . Let $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization in X_0 satisfying (6.8). By (A1) there exists a stopping time $T_1 > 0$, such that $X_{t \wedge \tau}$ takes values in U on $[0, T_1]$. Set $\tau' := \tau \wedge T_1$.

The rest of the proof runs as before, only estimation (6.17) has to be replaced. We give a sketch. Define

$$\begin{aligned}
b_t &:= \left(\langle A^* \zeta, X_{t \wedge \tau'} \rangle + \langle \zeta, F(t, X_{t \wedge \tau'}) \rangle \right) 1_{[0, \tau']}(t) \\
\rho_t^j &:= \langle \zeta, B^j(t, X_{t \wedge \tau'}) \rangle 1_{[0, \tau']}(t).
\end{aligned}$$

Then

$$X_{t \wedge \tau'} = (\phi \circ Y)_t, \quad \forall t \in \mathbb{R}_+ \tag{6.18}$$

where

$$Y_t := \langle \zeta, X_{t \wedge \tau'} \rangle = \langle \zeta, h_0 \rangle + \int_0^t b_s ds + \sum_{j \in \mathbb{N}} \int_0^t \rho_s^j d\beta_s^j. \tag{6.19}$$

Instead of (6.17) we now only have that Y is an $\mathbb{R}^m$ -valued Itô process, by the very definition of X . Combining (6.18) and Theorem 2.3.1 we get the relations (6.15) and (6.16) for X on $[0, \tau']$, $dt \otimes d\mathbb{P}$ -a.s. In particular $X_{t \wedge \tau'} \in D(A)$, $dt \otimes d\mathbb{P}$ -a.s. Hence from (2.15) and (6.15) we derive (2.23) and (2.24). $\square$

A key step in proving Theorems 6.2.2–6.2.4 consists in the following property.

Lemma 6.3.1. *Assume (A2) and (A4). Let $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization satisfying (6.8). Suppose that $U \cap \mathcal{M} \subset D(A)$. Then (6.10) and (6.11) hold for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in \mathbb{R}_+ \times \Omega$, for all $h \in U \cap \mathcal{M}$ if and only if*

$$\begin{aligned} Ah + F(t, \omega, h) &= D\phi(y) \left(\langle A^* \zeta, h \rangle + \langle \zeta, F(t, \omega, h) \rangle \right) \\ &\quad + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\phi(y) (\langle \zeta, B^j(t, \omega, h) \rangle, \langle \zeta, B^j(t, \omega, h) \rangle) \end{aligned} \quad (6.20)$$

$$B^j(t, \omega, h) = D\phi(y) \langle \zeta, B^j(t, \omega, h) \rangle, \quad \forall j \in \mathbb{N}, \quad (6.21)$$

where $y = \langle \zeta, h \rangle$, for all $h \in U \cap \mathcal{M}$, for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in \mathbb{R}_+ \times \Omega$.

Consequently, A is continuous on $U \cap \mathcal{M}$.

Proof. Point-wise equivalence of (6.11) and (6.21), and of (6.10) and the relation

$$\begin{aligned} Ah + F(t, \omega, h) &- \frac{1}{2} \sum_{j \in \mathbb{N}} DB^j(t, \omega, h) B^j(t, \omega, h) \\ &= D\phi(y) \langle \zeta, Ah + F(t, \omega, h) \rangle - \frac{1}{2} \sum_{j \in \mathbb{N}} DB^j(t, \omega, h) B^j(t, \omega, h) \end{aligned} \quad (6.22)$$

follows from Lemma 6.1.3. Assume that (6.10) and (6.11) – and hence (6.21) and (6.22) – hold for $dt \otimes d\mathbb{P}$ -a.e. $(t, \omega) \in \mathbb{R}_+ \times \Omega$, for all $h \in U \cap \mathcal{M}$. But both sides of (6.21) are continuous in h by (A2). Hence there exists a $dt \otimes d\mathbb{P}$ -nullset $\mathcal{N} \subset \mathbb{R}_+ \times \Omega$, such that (6.21) is true simultaneously for all $h \in U \cap \mathcal{M}$ and $(t, \omega) \in \mathcal{N}^c$.

Fix $(t, \omega) \in \mathcal{N}^c$. In view of (A2) Proposition 6.1.3 applies and the last summand in (6.20) equals

$$\frac{1}{2} \sum_{j \in \mathbb{N}} DB^j(t, \omega, h) B^j(t, \omega, h) - \frac{1}{2} D\phi(y) \langle \zeta, \sum_{j \in \mathbb{N}} DB^j(t, \omega, h) B^j(t, \omega, h) \rangle.$$

Hence (6.20) holds for $(t, \omega, h) \in \mathcal{N}^c \times (U \cap \mathcal{M})$ if and only if (6.22) does so.

It remains to show validity of (6.20) for all $h \in U \cap \mathcal{M}$, for $dt \otimes d\mathbb{P}$ -a.e. (t, ω) . We abbreviate the right hand side of (6.20) to $R(t, \omega, h)$. Then it reads

$$Ah = -F(t, \omega, h) + R(t, \omega, h) \quad (6.23)$$

for $dt \otimes d\mathbb{P}$ -a.e. (t, ω) , for all $h \in U \cap \mathcal{M}$. But due to (A4) the right hand side of (6.23) is continuous in h , see (2.6). Hence for any (countable) sequence $h_n \rightarrow h$ in $U \cap \mathcal{M}$ we have $Ah_n \rightarrow Ah$, by the closedness of A . In other words A restricted to $U \cap \mathcal{M}$ is continuous. This establishes the lemma. $\square$

Proof (of Theorem 6.2.2). Define the stopping time

$$\tau_0 := \inf\{t \geq 0 \mid X_t \notin \mathcal{M}\}.$$

By the closedness of $\mathcal{M}$ and (A1) we have $X_{\tau_0} \in \mathcal{M}$ on $\{\tau_0 < \infty\}$. We claim that $\tau_0 = \infty$.

Suppose $\mathbb{P}[\tau_0 < K] > 0$ for a number $K \in \mathbb{N}$. By countability of the covering (U_k) there exists a parametrization $\phi : V \rightarrow U \cap \mathcal{M}$, satisfying (6.8) and $\mathbb{P}[X_{\tau_0} \in U \text{ and } \tau_0 < K] > 0$. Define the bounded stopping time $\tau_1 := \tau_0 \wedge K$ and set $Y_0 := \langle \zeta, X_{\tau_1} \rangle$. Since $\phi \in C_b^2(\mathbb{R}^m; H)$, the $\mathbb{R}^m$ - and $L_2^0(\mathbb{R}^m)$ -valued mappings, respectively,

$$\begin{aligned} b(t, \omega, y) &:= \langle A^* \zeta, \phi(y) \rangle + \langle \zeta, F(\tau_1(\omega) + t, \omega, \phi(y)) \rangle \\ \rho(t, \omega, y) &:= \langle \zeta, B(\tau_1(\omega) + t, \omega, \phi(y))(\cdot) \rangle \\ \rho^j(t, \omega, y) &:= \langle \zeta, B^j(\tau_1(\omega) + t, \omega, \phi(y)) \rangle \end{aligned}$$

are bounded and globally Lipschitz continuous in y on $[0, T] \times \Omega \times \mathbb{R}^m$ for all $T \in \mathbb{R}_+$, which is due to (A4). By Theorem 2.4.1, the stochastic equation in $\mathbb{R}^m$

$$Y_t = Y_0 + \int_0^t b(s, Y_s) ds + \sum_{j \in \mathbb{N}} \int_0^t \rho^j(s, Y_s) d\beta_s^{(\tau), j} \quad (6.24)$$

has a unique continuous strong solution Y .

Now $\mathbb{P}[Y_0 \in V \mid \tau_0 < K] \cdot \mathbb{P}[\tau_0 < K] = \mathbb{P}[Y_0 \in V \text{ and } \tau_0 < K] > 0$ and the distribution of Y_0 under $\mathbb{P}[\cdot \mid \tau_0 < K]$ is regular. Hence by normality of $\mathbb{R}^m$ there exists two open sets V_0, V_1 such that $\bar{V}_0 \subset V_1 \subset \bar{V}_1 \subset V$ and $\mathbb{P}[Y_0 \in \bar{V}_0 \text{ and } \tau_0 < K] > 0$. Define the $(\mathcal{F}_t^{(\tau)})$ -stopping time

$$\tau_2 := \begin{cases} \inf\{t \geq 0 \mid Y_t \notin V_1\}, & \text{if } Y_0 \in \bar{V}_0 \text{ and } \tau_0 < K \\ 0, & \text{otherwise.} \end{cases} \quad (6.25)$$

Then by continuity of Y we have $\mathbb{P}[\tau_2 > 0] = \mathbb{P}[Y_0 \in \bar{V}_0 \text{ and } \tau_0 < K] > 0$ and

$$Y \in V \quad \text{on } [0, \tau_2] \cap \{\tau_2 > 0\}. \quad (6.26)$$

From the boundedness of b and ρ we derive

$$\mathbb{E} \left[\int_0^t \left(\|b(s, Y_s)\|_{\mathbb{R}^m} + \sum_{j \in \mathbb{N}} \|\rho^j(s, Y_s)\|_{\mathbb{R}^m}^2 \right) ds \right] < \infty, \quad \forall t \in \mathbb{R}_+. \quad (6.27)$$

Set $\tilde{X} := \phi \circ Y$. The assumptions of Theorem 2.3.1 are satisfied, hence

$$\begin{aligned} \tilde{X}_t &= X_{\tau_1} + \int_0^t \left(D\phi(Y_s) b(s, Y_s) + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2 \phi(Y_s) (\rho^j(s, Y_s), \rho^j(s, Y_s)) \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t D\phi(Y_s) \rho^j(s, Y_s) d\beta_s^{(\tau), j}, \quad \text{on } [0, \tau_2] \cap \{\tau_2 > 0\}, \end{aligned} \quad (6.28)$$

is a well-defined H -valued Itô process. Moreover, by (6.26) we have

$$\tilde{X} \in (U \cap \mathcal{M}) \subset D(A) \quad \text{on } [0, \tau_2] \cap \{\tau_2 > 0\}. \quad (6.29)$$

Hence due to (A2) and (A4) Lemma 6.3.1 applies and

$$\begin{aligned} A\tilde{X}_t + F(\tau_1 + t, \tilde{X}_t) &= D\phi(Y_t)b(t, Y_t) + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\phi(Y_t)(\rho^j(t, Y_t), \rho^j(t, Y_t)) \\ B^j(\tau_1 + t, \tilde{X}_t) &= D\phi(Y_t)\rho^j(t, Y_t), \quad \forall j, \end{aligned}$$

on $[0, \tau_2] \cap \{\tau_2 > 0\}$, $dt \otimes d\mathbb{P}$ -a.s. This together with (6.27) and (6.28) implies

$$\mathbb{P}\left[\int_0^{t \wedge \tau_2} \|A\tilde{X}_s\|_H ds < \infty\right] = 1, \quad \forall t \in \mathbb{R}_+$$

and

$$\begin{aligned} \tilde{X}_{t \wedge \tau_2} &= X_{\tau_1} + \int_0^{t \wedge \tau_2} \left(A\tilde{X}_s + F(\tau_1 + s, \tilde{X}_s) \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^{t \wedge \tau_2} B^j(\tau_1 + s, \tilde{X}_s) d\beta_s^{(\tau), j}, \quad \text{on } \{\tau_2 > 0\}. \end{aligned}$$

On the other hand we know by Corollary 2.4.1 that (X_{τ_1+t}) is the unique continuous weak solution to the time τ_1 -shifted version of equation (2.20). Whence $\tilde{X}_t = X_{\tau_1+t}$ on $[0, \tau_2] \cap \{\tau_2 > 0\}$. Because of (6.29) in particular $X_{\tau_1+\tau_2} \in \mathcal{M}$. But by construction $\{\tau_0 < K\} = \{\tau_0 < K\} \cap \{\tau_0 = \tau_1\}$ and

$$0 < \mathbb{P}[\tau_2 > 0] \leq \mathbb{P}[\{\tau_1 + \tau_2 > \tau_1\} \cap \{\tau_0 < K\}] \leq \mathbb{P}[\tau_1 + \tau_2 > \tau_0],$$

which is absurd. Hence $\tau_0 = \infty$.

The last statement of the theorem follows from Lemma 6.3.1 and (A2). $\square$

Proof (of Theorem 6.2.3). i) $\Rightarrow$ ii): Fix $(t_0, h_0) \in \mathbb{R}_+ \times \mathcal{M}$ and denote by $X = X^{(t_0, h_0)}$ a continuous local weak solution to (2.25) with lifetime τ . We proceed as in the proof of Theorem 6.2.1 and adapt the notation. Define the $(\mathcal{F}_t^{(t_0)})$ -stopping time $T_1 > 0$ with the property that $X_{t \wedge \tau'}$ takes values in $U_{\alpha(0)}$ for $\tau' := \tau \wedge T_1$. Analogously to (6.14) we derive the equality

$$\begin{aligned} \langle A^*\xi, X_t \rangle &= \left\langle \xi, -F(t_0 + t, X_t) + D\phi(Y_t) \left(\langle A^*\zeta, X_t \rangle + \langle \zeta, F(t_0 + t, X_t) \rangle \right) \right. \\ &\quad \left. + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\phi(Y_t) (\langle \zeta, B^j(t_0 + t, X_t) \rangle, \langle \zeta, B^j(t_0 + t, X_t) \rangle) \right\rangle, \end{aligned} \quad (6.30)$$

for all $\xi \in D(A^*)$ on $[0, \tau']$, $dt \otimes d\mathbb{P}$ -a.s. Due to (A4) and (A5) both sides of (6.30) are right continuous in t and by (2.6) we can interchange the limiting

$t \downarrow 0$ with the summation over j . It is not hard to see that therefore both sides of (6.30) are equal at $t = 0$, $\mathbb{P}$ -a.s. Arguing as for (6.15) we conclude that $h_0 \in D(A)$ and

$$\begin{aligned} Ah_0 + F(t_0, \omega, h_0) &= D\phi(y_0) \left(\langle A^* \zeta, h_0 \rangle + \langle \zeta, F(t_0, \omega, h_0) \rangle \right) \\ &\quad + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2 \phi(y_0) (\langle \zeta, B^j(t_0, \omega, h_0) \rangle, \langle \zeta, B^j(t_0, \omega, h_0) \rangle), \end{aligned}$$

$\mathbb{P}$ -a.s. Similarly

$$B^j(t_0, \omega, h_0) = D\phi(y_0) \langle \zeta, B^j(t_0, \omega, h_0) \rangle, \quad \forall j \in \mathbb{N}, \quad \mathbb{P}\text{-a.s.}$$

Since (t_0, h_0) was arbitrary, Lemma 6.3.1 yields condition ii).

ii) $\Rightarrow$ i): Let $(t_0, h_0) \in \mathbb{R}_+ \times \mathcal{M}$ and let $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization in h_0 satisfying (6.8). As in the proof of Theorem 6.2.2 (setting $\tau_0 = \tau_1 = 0$) we get a unique continuous strong solution Y to

$$\begin{aligned} Y_t &= \langle \zeta, h_0 \rangle + \int_0^t \left(\langle A^* \zeta, \phi(Y_s) \rangle + \langle \zeta, F(t_0 + s, \phi(Y_s)) \rangle \right) ds \\ &\quad + \sum_{j \in \mathbb{N}} \int_0^t \langle \zeta, B^j(t_0 + s, \phi(Y_s)) \rangle d\beta_s^{(t_0),j}. \end{aligned}$$

The stopping time τ_2 given by (6.25) is strictly positive and $Y_{t \wedge \tau_2} \in V$. Analysis similar to that in the proof of Theorem 6.2.2 shows that $X_t^{(t_0, h_0)} := (\phi \circ Y)_{t \wedge \tau_2}$ is a continuous local strong solution to (2.25) with lifetime τ_2 and $X_t^{(t_0, h_0)} \in U \cap \mathcal{M}$, for all $t \in \mathbb{R}_+$.

ii) $\Leftrightarrow$ iii): This is a direct consequence of Lemma 6.3.1 and (A5). $\square$

Proof (of Theorem 6.2.4). In view of Proposition 6.1.2 we can choose any parametrization ϕ_k in (6.8) such that $D^2 \phi_k \equiv 0$ on V_k . By straightforward inspection of the proofs we see that Lemma 6.3.1 and Theorems 6.2.2 and 6.2.3 remain valid under the conditions stated in the theorem. $\square$

6.4 Consistency Conditions in Local Coordinates

In this section we express the consistency conditions (6.10) and (6.11) in local coordinates and we identify the underlying coordinate process Y . From now on the assumptions of Theorem 6.2.3 are in force and $\mathcal{M}$ is locally invariant for (2.20).

Let $\phi : V \rightarrow U \cap \mathcal{M}$ be a parametrization. We assume that (6.1) and (6.2) hold for all $y \in V$ (otherwise replace V by $V \cap V'$). Theorem 6.2.3 says that

$$\mu(t, \omega, \phi(y)) = D\phi(y) \tilde{b}(t, \omega, y) \tag{6.31}$$

$$B^j(t, \omega, \phi(y)) = D\phi(y) \rho^j(t, \omega, y), \quad \forall j \in \mathbb{N}, \tag{6.32}$$

for all $(t, \omega, y) \in \mathbb{R}_+ \times \Omega \times V$ (after removing a $\mathbb{P}$ -nullset from Ω), where $\tilde{b}$ and ρ^j are uniquely specified measurable mappings from $(\mathbb{R}_+ \times \Omega \times V, \mathcal{P} \otimes \mathcal{B}(V))$ into $(\mathbb{R}^m, \mathcal{B}(\mathbb{R}^m))$, see (6.3). We write $\rho(t, \omega, y) = (\rho^j(t, \omega, y))_{j \in \mathbb{N}}$. To simplify notation, we suppress the argument ω whenever there is no ambiguity. In view of (A2) and (6.2) we then have

$$\rho(t, \cdot) = D\Psi(\phi(\cdot)) \circ B(t, \phi(\cdot)) \in C^1(V; L_2^0(\mathbb{R}^m)). \quad (6.33)$$

Fix $(t, y) \in V$. It is shown at the end of Section 6.1 that

$$DB^j(t, \phi(y))B^j(t, \phi(y)) = \frac{d}{ds}B^j(t, c(s))|_{s=0}$$

where $c(s) := \phi(y + s\rho^j(t, y))$. On the other hand, by (6.32),

$$\begin{aligned} \frac{d}{ds}B^j(t, c(s))|_{s=0} &= \frac{d}{ds}(D\phi(y + s\rho^j(t, y))\rho^j(t, y + s\rho^j(t, y)))|_{s=0} \\ &= D^2\phi(y)(\rho^j(t, y), \rho^j(t, y)) + D\phi(y)(D\rho^j(t, y)\rho^j(t, y)), \end{aligned}$$

compare with (6.7). Inserting this in (6.31) yields

$$A\phi(y) + F(t, \phi(y)) - \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\phi(y)(\rho^j(t, y), \rho^j(t, y)) = D\phi(y)b(t, y), \quad (6.34)$$

where we have set

$$b(t, \omega, y) := \tilde{b}(t, \omega, y) + \frac{1}{2} \sum_{j \in \mathbb{N}} D\rho^j(t, \omega, y)\rho^j(t, \omega, y). \quad (6.35)$$

Starting from (6.33) we now compute

$$\begin{aligned} D\rho^j(t, y)\rho^j(t, y) &= D^2\Psi(\phi(y))(B^j(t, \phi(y)), B^j(t, \phi(y))) \\ &\quad + D\Psi(\phi(y))DB^j(t, \phi(y))B^j(t, \phi(y)), \end{aligned} \quad (6.36)$$

where we have used (6.32). Combining (6.31), (6.35) and (6.36), we obtain

$$\begin{aligned} b(t, y) &= D\Psi(\phi(y)) \left(A\phi(y) + F(t, \phi(y)) - \frac{1}{2} \sum_{j \in \mathbb{N}} DB^j(t, \phi(y))B^j(t, \phi(y)) \right) \\ &\quad + \frac{1}{2} \sum_{j \in \mathbb{N}} D\rho^j(t, y)\rho^j(t, y) \\ &= D\Psi(\phi(y))(A\phi(y) + F(t, \phi(y))) \\ &\quad + \frac{1}{2} \sum_{j \in \mathbb{N}} D^2\Psi(\phi(y))(B^j(t, \phi(y)), B^j(t, \phi(y))). \end{aligned} \quad (6.37)$$

Note that, by (2.6), (6.33) and (A2), all series converge absolutely and are continuous in y .

We now show how equation (2.25) can be written in local coordinates. Observe that, in general, $b(t, \omega, y)$ is continuous but *not* locally Lipschitz continuous in y , on $\mathbb{R}_+ \times \Omega \times V$. The latter holds if, for instance, ϕ is a C^3 map. This can be easily deduced from Proposition 6.1.1, (A4), (6.20) and (6.37).

Nevertheless, equation (6.38) below is uniquely solvable. Indeed, let $X^{(t_0, h_0)}$ denote a $(U \cap \mathcal{M})$ -valued continuous local strong solution to the time t_0 -shifted stochastic equation (2.25), for some $(t_0, h_0) \in \mathbb{R}_+ \times (U \cap \mathcal{M})$. Itô's formula implies, together with (6.33) and (6.37), that $\Psi \circ X^{(t_0, h_0)}$ is a V -valued continuous local strong solution to the time t_0 -shifted stochastic equation

$$\begin{cases} dY_t = b(t_0 + t, Y_t) dt + \sum_{j \in \mathbb{N}} \rho^j(t_0 + t, Y_t) d\beta_t^{(t_0), j} \\ Y_0 = y_0, \end{cases} \quad (6.38)$$

where $y_0 = \phi^{-1}(h_0)$. Conversely, any V -valued continuous local (and perforce strong) solution $Y^{(t_0, y_0)}$ of (6.38) is, by Itô's formula, (6.32) and (6.34), a continuous local strong solution to the time t_0 -shifted stochastic equation (2.25) with $h_0 = \phi(y_0)$, and hence unique.

Summarizing, we have proved

Theorem 6.4.1. *Under the above assumptions, the consistency conditions (6.10) and (6.11) hold for all $(t, \omega, h) \in \mathbb{R}_+ \times \Omega \times (U \cap \mathcal{M})$ if and only if (6.34) and (6.32) hold for all $(t, \omega, y) \in \mathbb{R}_+ \times \Omega \times V$.*

Equation (2.25) transforms locally into equation (6.38), which is uniquely solvable. That is, $X^{(t_0, h_0)}$ is a continuous local strong solution to (2.25) in $U \cap \mathcal{M}$ if and only if so is $\phi^{-1} \circ X^{(t_0, h_0)}$ to (6.38) in V , where $y_0 = \phi^{-1}(h_0)$.

The coefficients $b(t, \omega, y)$ and $\rho(t, \omega, y)$ are locally bounded and continuous in $y \in V$. They are both Lipschitz continuous in $y \in V$, if ϕ is a C^3 map.

If ϕ is linear ($D^2\phi \equiv 0$), then all conclusions can be drawn without assuming (A2).

7. Consistent HJM Models

With the tools developed in the preceding sections we are now ready to solve the consistency problems (P2)–(P4) which were introduced in Section 1.3. In particular, we shall invert equation (1.6). Consistent HJM models are defined and the exponential-polynomial families revisited. We finally explore the class of affine HJM models.

7.1 Consistency Problems

We take as given a parametrized family $\mathcal{G} = \{G(\cdot, z) \mid z \in \mathcal{Z}\}$ of functions in H_w . We shall assume throughout that the parameter set $\mathcal{Z} \subset \mathbb{R}^m$ is open, for some $m \in \mathbb{N}$. By slight abuse of notation, we use the same letter G for the mapping $z \mapsto G(\cdot, z) : \mathcal{Z} \rightarrow H_w$. The parameter $z \in \mathcal{Z}$ is daily chosen in such a way that the forward curve $x \mapsto G(x, z)$ optimally fits the data available on the bond market. This means that one observes the time t evolution $(r_t)_{t \in \mathbb{R}_+}$ of the forward curve by its $\mathcal{Z}$ -valued coordinate process $(Z_t)_{t \in \mathbb{R}_+}$ given by

$$G(Z_t) = r_t. \quad (7.1)$$

Relation (7.1) implies $r_t \in \mathcal{G}$. If we assume that r is given by an HJM model (σ, γ, I) in H_w then this requires a certain consistency between (σ, γ, I) and the family $\mathcal{G}$. The terminology from Chapter 6 is tailor-made for the appropriate reformulation of the problems (P2) and (P4):

(P5) What are the conditions on the HJM model (σ, γ, I) such that the set $\mathcal{G} \cap I$ is locally invariant for the HJMM equation (4.23)?

All these questions are related to the invertibility of (7.1).

Definition 7.1.1. *A local HJM model (σ, γ, I) in H_w is called consistent with $\mathcal{G}$ if*

- i) $\mathcal{G} \subset I$
- ii) $\mathcal{G}$ is locally invariant for the HJMM equation (4.23) equipped with σ .

This definition replies to (P5).

Remark 7.1.1. Recall our convention that $\tilde{W}$, appearing in the HJMM equation (4.23), is an infinite dimensional Brownian motion with respect to the risk neutral measure $\mathbb{Q}$, see Section 4.5. Whereas the observations of Z in (7.1) are of course made under the physical measure $\mathbb{P}$, given by (4.25). However, this does not cause any difficulty for the present discussion, since consistency evidently is a property of σ and I only. This fact is also reflected by the invariance of the Nagumo type consistency conditions (6.10)–(6.11) with respect to the Girsanov transformation (4.25).

Here is a stronger version of Definition 7.1.1, which corresponds to problem **(P3)**. The following terminology is in accordance with Definition 4.1 in [8].

Definition 7.1.2. *A local HJM model (σ, γ, I) in H_w is called r -consistent with $\mathcal{G}$ if*

i) $\mathcal{G} \subset I$

ii) *For any $(t_0, h_0) \in \mathbb{R}_+ \times \mathcal{G}$ there exists a Z -valued $(\mathcal{F}_t^{(t_0)})$ -Itô process*

$$Z_t = Z_t^{(t_0, h_0)} = Z_0 + \int_0^t b_s ds + \int_0^t \rho_s d\tilde{W}_s^{(t_0)} \quad (7.2)$$

such that $G(Z)$ is a local mild solution to the time t_0 -shifted HJMM equation (4.23) equipped with σ and initial condition $G(Z_0) = h_0$.

Remark 7.1.2. Notice that the state space process Z in Definition 7.1.2 is consistent with $\mathcal{G}$ if $G(Z)$ is a global mild solution (see Definition 3.1.2).

It is clear that r -consistency implies consistency. Indeed, suppose that the local HJM model (σ, γ, I) is r -consistent with $\mathcal{G}$. By the uniqueness of the local mild solution $r = r^{(t_0, h_0)}$ we have (locally) $r_t = G(Z_t) \in \mathcal{G}$ where $Z = Z^{(t_0, h_0)}$ is given by (7.2). Hence (σ, γ, I) is consistent with $\mathcal{G}$.

The converse is not true in general, but we shall give sufficient conditions for the converse to hold too.

As in Section 5.2, we let from now on σ denote a measurable mapping from $(\mathbb{R}_+ \times \Omega \times H_w, \mathcal{P} \otimes \mathcal{B}(H_w))$ into $(L_2^0(H_w^0), \mathcal{B}(L_2^0(H_w^0)))$. A restatement of Corollary 5.1.2 and Lemma 5.2.2 yields the applicability of the results from Chapter 6 (where we have to replace systematically F and B by α_{HJM} and σ , respectively).

Lemma 7.1.1. *Suppose that $\sigma(t, \omega, h)$ is locally bounded and locally Lipschitz continuous in h . Then **(A4)** is satisfied. If in addition $\sigma(t, \omega, h)$ is right continuous in $t \in \mathbb{R}_+$ for all $h \in H_w$ and $\omega \in \Omega$, then **(A5)** holds too.*

We can now rephrase Theorem 6.4.1 as follows.

Theorem 7.1.1. *Let σ satisfy **(A2)** and the hypotheses of Lemma 7.1.1. Suppose that $\mathcal{G}$ is an m -dimensional C^2 submanifold of H_w . Then consistency of (σ, γ, I) with $\mathcal{G}$ is equivalent to r -consistency.*

*If $\mathcal{G}$ is linear, the same conclusion can be drawn without assumption **(A2)**.*

There is a subtle but important difference between $\mathcal{G}$ being a *regular* or merely an *immersed* submanifold. Assume that G is a C^2 immersion at some $z_0 \in \mathcal{Z}$. From Proposition 6.1.1 we deduce that there exists an open neighborhood V of z_0 in $\mathcal{Z}$ such that $G(V)$ is an m -dimensional C^2 submanifold in H_w and $G : V \rightarrow G(V)$ is a parametrization. But $\mathcal{G}$ can have self-intersections or can accumulate on $G(V)$ as shown in Figure 7.1.

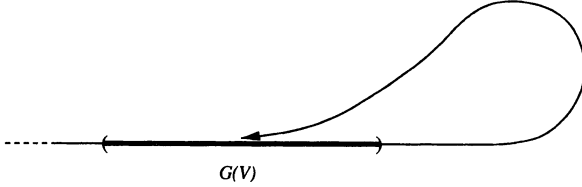


Fig. 7.1. An immersed submanifold

To get an illustration, let us imagine for the moment that the object, call it $\mathcal{O}$, in Figure 7.1 is the real line immersed in $\mathbb{R}^2$. Then it is not hard to construct an $\mathcal{O}$ -valued continuous stochastic process whose trajectories “branch out” at the accumulation point with probability 1/2, respectively.

Thus, in principle, (σ, γ, I) may very well fail to be consistent with $G(V)$ while still being consistent with $\mathcal{G}$. The same reasoning applies for seeing that consistency does not imply r -consistency: even if $\mathcal{G}$ is an immersed submanifold, the coordinate process Z may have a jump at z_0 if $G(z_0)$ is an accumulation point of $\mathcal{G}$.

On the other hand it is true that r -consistency with $\mathcal{G}$ implies r -consistency with $G(V)$. If furthermore σ satisfies the hypotheses of Theorem 7.1.1, we can express the consistency conditions (6.10)–(6.11) in local coordinates (see Theorem 7.2.1 below). But first we have to check differentiability of G .

7.2 A Simple Regularity Criterion for G

Suppose that $G \in C^2(\mathbb{R}_+ \times \mathcal{Z})$. We can give a simple criterion for differentiability of G in H_w .

Let $\xi_1, \dots, \xi_m$ denote the standard basis in $\mathbb{R}^m$ and let $z_0 \in \mathcal{Z}$. The *partial derivative* $D_i G(z_0)$ is the derivative of the mapping $s \mapsto G(z_0 + s\xi_i) : \mathbb{R} \rightarrow H_w$ at $s = 0$. For the differential calculus in Banach spaces we refer to [1, Chapter 2]. By (H1) we have

$$\mathcal{J}_x \left(\frac{d}{ds} G(z_0 + s\xi_i) \Big|_{s=0} \right) = \frac{d}{ds} \mathcal{J}_x (G(z_0 + s\xi_i)) \Big|_{s=0} = \partial_{z_i} G(x, z_0), \quad x \in \mathbb{R}_+.$$

Therefore $D_i G(z_0) = \partial_{z_i} G(\cdot, z_0)$ if it exists.

Lemma 7.2.1. *Let $\tilde{w} : \mathbb{R}_+ \rightarrow [1, \infty)$ be a non-decreasing function such that $w\tilde{w}^{-1} \in L^1(\mathbb{R}_+)$ and let V be an open subset of $\mathcal{Z}$. Suppose*

$$\sup_{(x,z) \in \mathbb{R}_+ \times V} |\partial_{z_i} \partial_x G(x, z)|^2 \tilde{w}(x) < \infty.$$

Then the partial derivatives $D_i G(z)$ exists for all $z \in V$ and $z \mapsto D_i G(z)$ is continuous from V into H_w , for all $1 \leq i \leq m$. Consequently, we have $G \in C^1(V; H_w)$ and

$$DG(z) = (\partial_{z_1} G(\cdot, z), \dots, \partial_{z_m} G(\cdot, z)) \in L(\mathbb{R}^m; H_w). \quad (7.3)$$

Proof. Let $z \in V$. Write $\varepsilon_i = \varepsilon \xi_i$ where $\varepsilon > 0$. We have to show that

$$T(\varepsilon) := \left\| \frac{G(z + \varepsilon_i) - G(z)}{\varepsilon} - \partial_{z_i} G(\cdot, z) \right\|_w^2 \rightarrow 0 \quad \text{for } \varepsilon \rightarrow 0.$$

We decompose $T(\varepsilon) = T_1(\varepsilon) + T_2(\varepsilon)$, where

$$T_1(\varepsilon) := \left| \frac{G(0, z + \varepsilon_i) - G(0, z)}{\varepsilon} - \partial_{z_i} G(0, z) \right|^2 \rightarrow 0 \quad \text{for } \varepsilon \rightarrow 0$$

and

$$\begin{aligned} T_2(\varepsilon) &:= \int_{\mathbb{R}_+} \left| \frac{\partial_x G(x, z + \varepsilon_i) - \partial_x G(x, z)}{\varepsilon} - \partial_{z_i} \partial_x G(x, z) \right|^2 w(x) dx \\ &\leq \int_{\mathbb{R}_+} \int_0^1 \left| \partial_{z_i} \partial_x G(x, z + s\varepsilon_i) - \partial_{z_i} \partial_x G(x, z) \right|^2 w(x) ds dx. \end{aligned}$$

For ε small enough we have $z + s\varepsilon_i \in V$ for all $s \in [0, 1]$ and by assumption

$$\left| \partial_{z_i} \partial_x G(x, z + s\varepsilon_i) - \partial_{z_i} \partial_x G(x, z) \right|^2 w(x) \leq 2Cw(x)\tilde{w}^{-1}(x) \in L^1([0, 1] \times \mathbb{R}_+)$$

where C is independent of x and ε . Hence $T_2(\varepsilon) \rightarrow 0$ for $\varepsilon \rightarrow 0$, by dominated convergence. We conclude that $D_i G(z)$ exists. Continuity of $z \mapsto D_i G(z) : V \rightarrow H_w$ follows similarly. This proves the first part of the lemma.

For the second part we refer to [1, Proposition 2.4.12]. □

By [1, Proposition 2.4.12], $G \in C^2(\mathcal{Z}; H_w)$ if and only if $D_i G \in C^1(\mathcal{Z}; H_w)$ for $1 \leq i \leq m$, which again can be checked with Lemma 7.2.1. Moreover

$$D^2 G(z) = (\partial_{z_i} \partial_{z_j} G(\cdot, z))_{1 \leq i, j \leq m} \quad (7.4)$$

which is a bilinear mapping from $\mathbb{R}^m \times \mathbb{R}^m$ into H_w .

Assume now that $G \in C^2(\mathcal{Z}; H_w)$ and that $\partial_{z_1} G(\cdot, z_0), \dots, \partial_{z_m} G(\cdot, z_0)$ are linearly independent functions, for some $z_0 \in \mathcal{Z}$. Then G is a C^2 immersion at z_0 and we can go on with the reasoning at the end of the preceding section. Denote by V an open neighborhood of z_0 in $\mathcal{Z}$ such that $G : V \rightarrow G(V)$ is a parametrization.

Theorem 7.2.1. *Let G and V be as above. Suppose that (σ, γ, I) is r -consistent with $\mathcal{G}$ and that σ satisfies the hypotheses of Theorem 7.1.1. Then the coordinate process Z in (7.2) is a diffusion with coefficients $b(t, \omega, z)$ and $\rho(t, \omega, z)$, which are locally bounded and continuous in $z \in V$. The coefficients are locally Lipschitz continuous in $z \in V$ if $G \in C^3(\mathcal{Z}; H_w)$.*

Moreover, after removing a nullset from Ω , the following consistency condition holds

$$\begin{aligned} \partial_x G(x, z) = & \sum_{i=1}^m b^i(t, \omega, z) \partial_{z_i} G(x, z) + \frac{1}{2} \sum_{k,l=1}^m a^{k,l}(t, \omega, z) \partial_{z_k} \partial_{z_l} G(x, z) \\ & - \sum_{k,l=1}^m a^{k,l}(t, \omega, z) \partial_{z_k} G(x, z) \int_0^x \partial_{z_l} G(\eta, z) d\eta \end{aligned} \quad (7.5)$$

for all $(t, \omega, z) \in \mathbb{R}_+ \times \Omega \times V$. Here $a^{k,l} := \sum_{j \in \mathbb{N}} \rho^{j,k} \rho^{j,l}$ denotes the diffusion matrix.

Proof. By Theorem 6.2.3 necessarily $G(V) \in D(A)$. Combining (7.3)–(7.4) with Theorem 6.4.1 yields the assertion. $\square$

Remark 7.2.1. Notice that we have essentially got back (3.4). Yet (7.5) is a stronger (since pointwise) condition. This is because a consistent state space Itô process Z induces a singular process $r = G(Z)$ which is viable in $\mathcal{G}$ (see (6.9)). Whereas r -consistency has to do with local invariance of $\mathcal{G}$, a much stronger concept (see Definition 6.2.1).

If $\mathcal{G} = G(\mathcal{Z})$ is a regular submanifold then condition (7.5) can be used to check whether *any* local HJM model is consistent with $\mathcal{G}$. In some cases this leads to the complete characterization of all consistent local HJM models. So for regular exponential-polynomial families.

7.3 Regular Exponential-Polynomial Families

We restate the analysis of Chapter 3 from the preceding geometric point of view. However, this needs a slight modification.

Throughout this section we choose as weighting function $w(x) = (1+x)^\alpha$ for some $\alpha > 3$, see Example 5.1.2. We adapt the notion of Section 3.2 and write

$$G(x, z) = \sum_{i=1}^K p_i(x, z) e^{-z_i \cdot n_i + 1 \cdot x} \quad (7.6)$$

where $p_i(x, z) = \sum_{j=0}^{n_i} z_{i,j} x^j$, for some vector $n = (n_1, \dots, n_K) \in \mathbb{N}_0^K$ and $K \in \mathbb{N}$. Since H_w contains only bounded functions, we restrict our considerations to the set of bounded exponential-polynomial families $BEP(K, n)$,

see Definition 3.2.1. But $BEP(K, n)$ is not a submanifold in H_w . We have to extract the singularities. An easy computation shows

$$\partial_{z_{i,j}} G(x, z) = \begin{cases} x^j e^{-z_{i,n_i+1}x}, & 0 \leq j \leq n_i \\ -xp_i(x, z)e^{-z_{i,n_i+1}x}, & j = n_i + 1. \end{cases}$$

Hence $\{\partial_{z_{i,j}} G(\cdot, z) \mid 0 \leq j \leq n_i + 1, 1 \leq i \leq K\}$ are linearly independent functions if and only if

$$z_{i,n_i+1} \neq z_{j,n_j+1} \quad \text{and} \quad z_{i,n_i} \neq 0, \quad \forall 1 \leq i \neq j \leq K. \quad (7.7)$$

Under these constraints, $G(\cdot, z) \in H_w$ if and only if $z_{i,n_i+1} > 0$ or both $z_{i,n_i+1} = 0$ and $n_i = 0$, compare with (3.12). But $\mathcal{Z}$ is an open set. To include the latter case, we will therefore assume that

$$\begin{aligned} n_1 &= 0 \\ z_{1,1} &\equiv 0 \quad \text{and} \quad z_{i,n_i+1} > 0 \quad \text{for } 2 \leq i \leq K. \end{aligned} \quad (7.8)$$

Consequently $G(\infty, z) = z_{1,0}$. For regularity reasons, see Remark 7.3.1 below, we have to sharpen condition (7.7) and let the distance between different exponents be bounded away from zero by some fixed $\varepsilon > 0$. This leads to the following parameter set

$$\mathcal{Z}_\varepsilon := \mathbb{R} \times \left\{ (z_{2,0}, \dots, z_{K,n_K+1}) \left| \begin{array}{l} z_{i,n_i} \neq 0, \\ z_{i,n_i+1} > 0, \\ |z_{i,n_i+1} - z_{j,n_j+1}| > \varepsilon, \end{array} \quad \forall 2 \leq i \neq j \leq K \right. \right\}$$

in $\mathbb{R}^m$ where $m := 1 + (n_2 + 2) + \dots + (n_K + 2)$. To avoid self-intersections of G we further require that

$$n_i \neq n_j, \quad 2 \leq i \neq j \leq K. \quad (7.9)$$

Now the representation (7.6) of $G(\cdot, z)$ with $z \in \mathcal{Z}_\varepsilon$ is unique: $G(\cdot, z) = G(\cdot, \tilde{z})$ with $z, \tilde{z} \in \mathcal{Z}_\varepsilon$ implies $z = \tilde{z}$.

Definition 7.3.1. For $K \in \mathbb{N}$, $n \in \mathbb{N}_0^K$ satisfying (7.8)–(7.9), $\varepsilon > 0$ and $\mathcal{Z}_\varepsilon$ as above we define

$$REP(K, n, \varepsilon) := G(\mathcal{Z}_\varepsilon)$$

the regular exponential-polynomial family.

Clearly, we have $REP(K, n, \varepsilon) \subset BEP(K, n)$.

Setting $\bar{w}(x) = (1 + x)^{\alpha+2}$, we conclude easily by Lemma 7.2.1 that $G \in C^2(\mathcal{Z}_\varepsilon, H_w)$. Up to now, $G : \mathcal{Z}_\varepsilon \rightarrow H_w$ is an injective immersion and $REP(K, n, \varepsilon)$ is an immersed submanifold in H_w .

Proposition 7.3.1. The mapping $G : \mathcal{Z}_\varepsilon \rightarrow H_w$ is an embedding. Its image $REP(K, n, \varepsilon)$ is consequently an m -dimensional C^2 submanifold in H_w , with m given above.

Proof. We have to show that $G : \mathcal{Z}_\varepsilon \rightarrow REP(K, n, \varepsilon)$ is a homeomorphism.

Let (z^k) be a sequence in $\mathcal{Z}_\varepsilon$ and $z \in \mathcal{Z}_\varepsilon$ such that $G(z^k) \rightarrow G(z)$ in H_w . Inequality (5.4) yields $G(z^k) \rightarrow G(z)$ in $L^\infty(\mathbb{R}_+)$. The proof is completed by showing that $z^k \rightarrow z$.

In view of (5.3) we conclude $z_{1,0}^k = G(\infty, z^k) \rightarrow G(\infty, z) = z_{1,0}$. Thus $(G(\infty, z^k) - z_{1,0}^k) \rightarrow (G(\infty, z) - z_{1,0})$ in H_w and we may assume in the sequel that $z_{1,0}^k \equiv z_{1,0} = 0$.

Write $\bar{\theta}_i := \limsup_{k \rightarrow \infty} z_{i,n_i+1}^k \leq \infty$, $i = 2, \dots, K$. Then there exists a subsequence (z^{k_μ}) of (z^k) such that $\lim_{\mu \rightarrow \infty} z_{i,n_i+1}^{k_\mu} = \bar{\theta}_i$ for all i .

Suppose first $\bar{\theta}_i < \infty$ for all i . By definition of $\mathcal{Z}_\varepsilon$ it follows that

$$|\bar{\theta}_i - \bar{\theta}_j| \geq \varepsilon, \quad \forall 2 \leq i \neq j \leq K. \quad (7.10)$$

We write $N := m - K$ and define the $\mathbb{R}^N$ -vectors

$$\begin{aligned} v^\mu(x) &:= \left(e^{-z_{2,n_2+1}^{k_\mu} x}, x e^{-z_{2,n_2+1}^{k_\mu} x}, \dots, x^{n_2} e^{-z_{2,n_2+1}^{k_\mu} x}, \dots, x^{n_K} e^{-z_{K,n_K+1}^{k_\mu} x} \right) \\ v(x) &:= \left(e^{-\bar{\theta}_2 x}, x e^{-\bar{\theta}_2 x}, \dots, x^{n_2} e^{-\bar{\theta}_2 x}, \dots, x^{n_K} e^{-\bar{\theta}_K x} \right) \\ y^\mu &:= \left(z_{2,0}^{k_\mu}, z_{2,1}^{k_\mu}, \dots, z_{2,n_2}^{k_\mu}, \dots, z_{K,n_K}^{k_\mu} \right). \end{aligned}$$

Then

$$\langle v^\mu(x), y^\mu \rangle_{\mathbb{R}^N} = G(x, z^{k_\mu}) \rightarrow G(x, z), \quad \forall x \in \mathbb{R}_+. \quad (7.11)$$

Fix a sequence $x_1 < \dots < x_N$ in $\mathbb{R}_+$. The vectors $v^\mu(x_1), \dots, v^\mu(x_N)$ are linearly independent for any μ and so are $v(x_1), \dots, v(x_N)$, by (7.10). Hence the $N \times N$ -matrices

$$M^\mu = M^\mu(x_1, \dots, x_N) := \begin{pmatrix} v^\mu(x_1) \\ \vdots \\ v^\mu(x_N) \end{pmatrix}, \quad M = M(x_1, \dots, x_N) := \begin{pmatrix} v(x_1) \\ \vdots \\ v(x_N) \end{pmatrix}$$

are regular and $M^\mu \rightarrow M$ for $\mu \rightarrow \infty$. Consequently, $(M^\mu)^{-1} \rightarrow M^{-1}$ and

$$y^\mu \rightarrow M^{-1} \begin{pmatrix} G(x_1, z) \\ \vdots \\ G(x_N, z) \end{pmatrix}.$$

Hence $y_{i,j} := \lim_{\mu \rightarrow \infty} z_{i,j}^{k_\mu}$ exists in $\mathbb{R}$ for $0 \leq j \leq n_i$, $2 \leq i \leq K$. We conclude

$$\lim_{\mu \rightarrow \infty} G(x, z^{k_\mu}) = \sum_{i=2}^K \left(\sum_{j=1}^{n_i} y_{i,j} x^j \right) e^{-\bar{\theta}_i x} = G(x, z), \quad \forall x \in \mathbb{R}_+. \quad (7.12)$$

Combining (7.9) with (7.12) we get $\bar{\theta}_i = z_{i,n_i+1}$ and $y_{i,j} = z_{i,j}$. That way one concludes that any subsequence of (z^k) contains a convergent subsequence

with limit z . Hence $\lim_{k \rightarrow \infty} z^k = z$, and the proposition is proved in case $\bar{\theta}_i < \infty$ for all $i = 2, \dots, K$.

Now assume that $I := \{2 \leq i \leq K \mid \bar{\theta}_i = \infty\}$ is not empty. Let $i \in I$. Then there exists $x_i^* \in \mathbb{R}_+$ such that

$$|p_i(x, z^{k_\mu})| e^{-z_{i,n_i+1}^{k_\mu} x} \leq \sum_{j=0}^{n_i} |z_{i,j}^{k_\mu}| x^j \left(e^{-z_{i,n_i+1}^{k_\mu}} \right)^x \rightarrow 0 \quad \text{for } \mu \rightarrow \infty, \quad \forall x \geq x_i^*.$$

We write $\tilde{N} := N - \sum_{i \in I} (n_i + 1)$ and define the $\mathbb{R}^{\tilde{N}}$ -vectors $\tilde{v}^\mu(x)$, $\tilde{v}(x)$ and $\tilde{y}^\mu$ by skipping all components of $v^\mu(x)$, $v(x)$, resp. y^μ which include $z_{i,j}^{k_\mu}$ or $\bar{\theta}_i$ for $i \in I$. Set $x^* := \max_{i \in I} x_i^*$. Instead of (7.11) we now have

$$\langle \tilde{v}^\mu(x), \tilde{y}^\mu \rangle_{\mathbb{R}^{\tilde{N}}} = G(x, z^{k_\mu}) - \sum_{i \in I} p_i(x, z^{k_\mu}) e^{-z_{i,n_i+1}^{k_\mu} x} \rightarrow G(x, z) \quad \forall x \geq x^*.$$

Choose a sequence $x^* \leq x_1 < \dots < x_{\tilde{N}}$. Inequality (7.10) still holds for $i, j \notin I$. Accordingly, the vectors $\tilde{v}(x_1), \dots, \tilde{v}(x_{\tilde{N}})$ are linearly independent and so are $\tilde{v}^\mu(x_1), \dots, \tilde{v}^\mu(x_{\tilde{N}})$ for all μ . A passage to the limit similar to the above implies

$$\lim_{\mu \rightarrow \infty} G(x, z^{k_\mu}) = \sum_{i \notin I} \left(\sum_{j=1}^{n_i} y_{i,j} x^j \right) e^{-\bar{\theta}_i x} = G(x, z), \quad \forall x \geq x^*,$$

which is impossible since $[x^*, \infty)$ is a set of uniqueness for analytic functions. Hence I is empty, and the proof is complete. $\square$

Remark 7.3.1. Estimate (7.10) is essential for the preceding proof. Although Proposition 7.3.1 seems to be true also for $\varepsilon = 0$, we do not see how to prove it for that case.

We now can summarize the main results from Section 3.2.

Theorem 7.3.1. *Let σ satisfy the assumptions of Lemma 7.1.1, and suppose the local HJM model (σ, γ, I) is consistent with $REP(K, n, \varepsilon)$. Then*

$$\sigma(t, \omega, \cdot) \equiv 0 \quad \text{on} \quad \overline{REP(K, n, \varepsilon)}, \quad \forall (t, \omega) \in \mathbb{R}_+ \times \Omega.$$

Notice that we do not need **(A2)** here.

Proof. We write

$$\mathcal{B} := \{ \beta = (0, \beta_2, \dots, \beta_K) \in \mathbb{R}^K \mid \beta_i > 0, |\beta_i - \beta_j| > \varepsilon, \forall 2 \leq i \neq j \leq K \}.$$

For $\beta \in \mathcal{B}$ we define

$$E_\beta := REP(K, n, \varepsilon) \cap \text{span} \{ x^j e^{-\beta_i x} \mid 0 \leq j \leq n_i, 1 \leq i \leq K \}. \quad (7.13)$$

We claim that (σ, γ, I) is consistent with E_β . Indeed, let $(t_0, h_0) \in \mathbb{R}_+ \times E_\beta$. That is, $h_0 = G(z^*)$ with $z^* \in \mathcal{Z}_\varepsilon$ and $z_{i, n_i+1}^* = \beta_i$ for $1 \leq i \leq K$. By Theorem 6.2.1 there exists a $\mathcal{Z}_\varepsilon$ -valued $(\mathcal{F}_t^{(t_0)})$ -Itô process

$$Z_t^{i,j} = z_{i,j}^* + \int_0^t b_s^{i,j} ds + \sum_{k \in \mathbb{N}} \int_0^t \rho_s^{i,j;k} d\tilde{\beta}_s^{(t_0),k}$$

and a stopping time $\tau > 0$ such that $r_{t \wedge \tau}^{(t_0, h_0)} = G(Z_{t \wedge \tau})$ for all $t \in \mathbb{R}_+$ (in fact, this holds for $Z = (G^{-1} \circ \phi)(Y)$ where ϕ is as in the proof of Theorem 6.2.1 and Y is given by (6.19)). Accordingly, $r^{(t_0, h_0)}$ is a local strong solution to the time t_0 -shifted HJMM equation (4.23). This in turn implies relation (3.4) for $G(x, Z_{\cdot \wedge \tau})$. By Theorem 3.2.1 we conclude that there exists a stopping time $0 < \tau' \leq \tau$ such that

$$b^{i, n_i+1} = \sum_{k \in \mathbb{N}} (\rho^{i, n_i+1; k})^2 = 0, \quad dt \otimes dQ\text{-a.s. on } [0, \tau'].$$

Therefore $Z_{t \wedge \tau'}^{i, n_i+1} \equiv \beta_i$ and the claim follows.

Now write

$$\tilde{\mathcal{B}} := \{\beta \in \mathcal{B} \mid 2\beta_i = \beta_j \text{ for some } i \neq j\}. \quad (7.14)$$

The set $\tilde{\mathcal{B}}$ is contained in a finite union of hyperplanes in $\mathbb{R}^K$. Therefore $(\mathcal{B} \setminus \tilde{\mathcal{B}}) \cap \mathcal{B} = \mathcal{B}$.

Since E_β is a linear submanifold, Theorem 7.1.1 yields that (σ, γ, I) is r -consistent with E_β . Theorem 7.2.1 implies (3.4). Hence Theorem 3.6.2 applies and $\sigma(t, \omega, h) \equiv 0$ for $h \in E_\beta$, for all $\beta \in \mathcal{B} \setminus \tilde{\mathcal{B}}$. By continuity of $h \mapsto \sigma(t, \omega, h)$ the assertion follows. $\square$

Remark 7.3.2. Notice that $\{E_\beta\}_{\beta \in \mathcal{B}}$, defined by (7.13), forms a foliation of $REP(K, n)$. Theorem 7.3.1 states that there does not exist any non-trivial HJM model which is consistent with the *whole* submanifold $REP(K, n)$. However, for a *single* leave E_β with $\beta \in \tilde{\mathcal{B}}$ (see (7.14) and Corollary 3.2.2) there might exist a particular consistent HJM model.

Let us recall the two leading examples from Chapter 3.

7.3.1 The Nelson–Siegel Family

$$G_{NS}(x, z) = z_{1,0} + (z_{2,0} + z_{2,1}x)e^{-z_{2,2}x}$$

In the preceding context we have $G_{NS}(\mathcal{Z}_\varepsilon) = REP(2, (0, 1), \varepsilon)$, where actually ε is redundant.

7.3.2 The Regular Svensson Family

$$G_S(x, z) = z_{1,0} + (z_{2,0} + z_{2,1}x)e^{-z_{2,2}x} + z_{3,1}xe^{-z_{3,2}x}$$

Obviously, condition (7.9) is not satisfied. So the preceding discussion does not apply immediately. Yet we can “regularize” also the Svensson family. Let $\varepsilon > 0$ and define

$$\mathcal{Z}_{S,\varepsilon} := \mathbb{R} \times \left\{ (z_{2,0}, z_{2,1}, z_{2,2}, z_{3,1}, z_{3,2}) \left| \begin{array}{l} |z_{2,0}| > \varepsilon, \\ z_{2,2}, z_{3,2} > 0, \\ |z_{2,2} - z_{3,2}| > \varepsilon \end{array} \right. \right\}.$$

Similar as in the proof of Proposition 7.3.1 one shows that $G_S : \mathcal{Z}_{S,\varepsilon} \rightarrow H_w$ is an embedding. As before we conclude $\sigma(t, \omega, h) \equiv 0$ for $h \in \overline{G_S(\mathcal{Z}_{S,\varepsilon})}$, for every consistent local HJM model (σ, γ, I) .

In Section 3.7.2 we have identified the only non-trivial state space Itô process Z which is consistent with the full (bounded) Svensson family. We have shown that it induces a generalized Vasicek model. It is an amazing fact that the support of the non-deterministic component $Z_t^{2,0}$ contains 0 for arbitrary small t (it is Gaussian distributed). Thus $\mathbb{Q}[Z_t \notin \mathcal{Z}_{S,\varepsilon}] > 0$ for every $t > 0$.

Remark 7.3.3. Consistency of HJM models with exponential-polynomial families has been checked first by Björk and Christensen [8] for deterministic σ . We have extended their results in Chapter 3, where we discussed r -consistency for $EP(K, n)$ and $BEP(K, n)$. Theorem 7.3.1 now asserts the generality of our approach.

7.4 Affine Term Structure

In this section we apply Theorem 7.2.1 to identify the particular class of *affine local HJM models*. That is, those local HJM models which are consistent with finite dimensional linear submanifolds.

It is required throughout that σ meets the assumptions of Lemma 7.1.1. Suppose that $\mathcal{M}$ is an m -dimensional linear submanifold of H_w which is locally invariant for the HJMM equation (4.23). Without loss of generality, $\mathcal{M}$ is connected. According to Theorem 6.2.4 there exists $V \subset \mathbb{R}^m$ open and connected and a global parametrization $\phi : V \rightarrow \mathcal{M}$ with $\phi(y) = g_0 + \sum_{i=1}^m y_i g_i$ where $g_0, \dots, g_m \in D(A)$ are linearly independent. Clearly $D\phi(y) \equiv (g_1, \dots, g_m)$.

Applying Theorem 7.2.1 yields

$$g'_0 + \sum_{i=1}^m y_i g'_i = \sum_{i=1}^m b^i(t, \omega, y) g_i - \frac{1}{2} \left(\sum_{i,j=1}^m a^{i,j}(t, \omega, y) G_i G_j \right)' \quad (7.15)$$

for all $(t, \omega, y) \in \mathbb{R}_+ \times \Omega \times V$ (after removing a $\mathbb{Q}$ -nullset from Ω), where $G_i(x) := \int_0^x g_i(\eta) d\eta$. To shorten notation, we let $'$ stand for ∂_x .

Write $\gamma_i = g_i(0)$. Integrating (7.15) yields

$$g_0 - \gamma_0 + \sum_{i=1}^m y_i (g_i - \gamma_i) = \sum_{i=1}^m b^i(t, \omega, y) G_i - \frac{1}{2} \sum_{i,j=1}^m a^{i,j}(t, \omega, y) G_i G_j. \quad (7.16)$$

Now if G_i and $G_i G_j$, $1 \leq i, j \leq m$, are linearly independent functions, we can invert the linear equation (7.16) for $b^i(t, \omega, y)$ and $a^{i,j}(t, \omega, y)$. Since the left hand side of (7.16) is affine in y , we obtain that also they are affine functions in y :

$$\begin{aligned} b^i(t, \omega, y) &= b^i(t, \omega) + \sum_{j=1}^m \beta^{i,j}(t, \omega) y_j, \quad 1 \leq i \leq m \\ a^{i,j}(t, \omega, y) &= a^{i,j}(t, \omega) + \sum_{k=1}^m \alpha^{i,j,k}(t, \omega) y_k, \quad 1 \leq i, j \leq m \end{aligned}$$

for some predictable functions $b^i(t, \omega)$, $\beta^{i,j}(t, \omega)$, $a^{i,j}(t, \omega)$ and $\alpha^{i,j,k}(t, \omega)$ being right continuous in t .

Since affine functions are analytic and V is open, equation (7.16) splits into the following system of Riccati equations for the G_i 's

$$\begin{aligned} G'_0 &= \gamma_0 + \sum_{k=1}^m b^k(t, \omega) G_k - \frac{1}{2} \sum_{k,l=1}^m a^{k,l}(t, \omega) G_k G_l \\ G'_i &= \gamma_i + \sum_{k=1}^m \beta^{k,i}(t, \omega) G_k - \frac{1}{2} \sum_{k,l=1}^m a^{k,l,i}(t, \omega) G_k G_l, \quad 1 \leq i \leq m \end{aligned} \quad (7.17)$$

with initial conditions $G_0(0) = \dots = G_m(0) = 0$.

Hence we derived some of the results in [23] as a special case of our general point of view. Clearly, Theorem 7.2.1 imposes constraints on the choice of V in order that $b(t, \omega, y)$ and $\rho(t, \omega, y)$ are locally bounded and locally Lipschitz continuous in $y \in V$ (since trivially the parametrization ϕ is a C^3 map). Moreover, there is a tradeoff between the functions g_i and the coefficients $a^{i,j}$, $\alpha^{i,j,k}$, b^i , $\beta^{i,j}$ and γ_i by the Riccati equations (7.17). We shall discuss this problem for the case $m = 1$. For a fuller treatment we refer to [23].

For simplicity we assume for the remainder of the chapter that $\bar{W}$ is a real-valued Brownian motion and that all coefficients are autonomous, that is

$$\begin{aligned} b(t, \omega, y) &\equiv b^1(y) = b + \beta y \\ a(t, \omega, y) &\equiv a^{1,1}(y) = a + \alpha y. \end{aligned}$$

Accordingly

$$\rho(t, \omega, y) \equiv \rho^{1,1}(y) = \sqrt{a + \alpha y}.$$

We rewrite the Riccati equations (7.17) for G_0 and G_1 :

$$\begin{aligned} G'_0 &= \gamma_0 + bG_1 - \frac{1}{2}aG_1^2, & G_0(0) &= 0 \\ G'_1 &= \gamma_1 + \beta G_1 - \frac{1}{2}\alpha G_1^2, & G_1(0) &= 0. \end{aligned} \quad (7.18)$$

Equation (7.18) admits a non-exploding solution $G_1 : \mathbb{R}_+ \rightarrow \mathbb{R}$ if and only if $\alpha\gamma_1 \geq 0$ or both $\alpha\gamma_1 < 0$ and $\beta \leq -\sqrt{2|\alpha\gamma_1|}$.

Suppose $\alpha > 0$. Theorem 7.2.1 tells us that $y \mapsto \sqrt{a + \alpha y}$ has to be locally Lipschitz continuous on V . Consequently, $a + \alpha y$ has to be bounded away from zero. Thus $V \subset [\frac{\varepsilon - a}{\alpha}, \infty)$ for some $\varepsilon > 0$. The dynamics of the coordinate process, see (6.38), is given by the stochastic differential equation

$$dY_t = (b + \beta Y_t) dt + \sqrt{a + \alpha Y_t} d\tilde{W}_t, \quad Y_0 \in V. \quad (7.19)$$

It can be shown (see [36, p. 221] and substitute $X := Y + \frac{a}{\alpha}$) that (7.19) admits a (global) unique continuous strong solution on $[\frac{-a}{\alpha}, \infty)$ if $b \geq \frac{a}{\alpha}\beta$.

If $\alpha < 0$, then $V \subset (-\infty, \frac{\varepsilon - a}{\alpha}]$ for some $\varepsilon > 0$ and (7.19) remains valid.

7.4.1 The Cox–Ingersoll–Ross (CIR) Model

For $\alpha > 0$, $\gamma_1 = 1$ and $a = \gamma_0 = 0$ (and hence $b \geq 0$) we get the CIR [18] short rate model. There exists a closed form solution to (7.18), see [18, Formula (23)],

$$G_1(x) = \frac{2 \left(e^{(\sqrt{\beta^2 + 2\alpha})x} - 1 \right)}{\left(\sqrt{\beta^2 + 2\alpha} - \beta \right) \left(e^{(\sqrt{\beta^2 + 2\alpha})x} - 1 \right) + 2\sqrt{\beta^2 + 2\alpha}}$$

and

$$G_0(x) = b \int_0^x G_1(\eta) d\eta.$$

It holds by inspection¹ that $g_0 = G'_0 = bG_1 \in H_w$ and $g_1 = G'_1 \in H_w^0$ for any polynomial weighting function w as in Example 5.1.2. Hence the CIR model is contained in our setting.

From (6.32) and the identity $y = g_0(0) + yg_1(0) = \mathcal{J}_0(\phi(y))$ we obtain

$$\sigma(t, \omega, h) \equiv \sigma^1(h) = \sqrt{\alpha \mathcal{J}_0(h)} g_1, \quad h \in \phi((0, \infty)).$$

¹ More general it can be shown that the critical point $y_0 = \frac{\beta + \sqrt{\beta^2 + 2\alpha}}{\alpha} \in \mathbb{R}_+$ for (7.18) is globally exponentially stable, see [2, Chapter IV]. A fact which also applies for the multi-dimensional system (7.17). We will investigate this further in connection with general affine models.

Denote by $\tilde{\sigma} : H_w \rightarrow H_w^0$ a measurable continuation of σ . Let Y be the continuous solution of (7.19) with $Y_0 > \varepsilon$ and write $\tau_\varepsilon = \inf\{t \mid Y_t \leq \varepsilon\}$. It is straightforward that $(r_{t \wedge \tau_\varepsilon}) = (\phi(Y_{t \wedge \tau_\varepsilon}))$ with $r_0 = \phi(Y_0)$ is the unique continuous local strong solution to the HJMM equation (4.23) equipped with $\tilde{\sigma}$, see Theorem 6.4.1

However, $\sigma(h)$ – and hence $\tilde{\sigma}(h)$ – cannot be extended Lipschitz continuously to $h = \phi(0)$ (naturally $\sigma(\phi(0)) = 0$). Thus, although $\phi(Y)$ is a strong solution to the HJMM equation (4.23) with $\tilde{\sigma}(\phi(0)) = 0$, we cannot assert its global uniqueness. Uniqueness, however, holds for the $\mathbb{R}_+$ -valued solution Y , by the preceding discussion.

7.4.2 The Vasicek Model

The case $\alpha = 0$ leads to a Gaussian coordinate process Y . There are no constraints on V , but a has to be non-negative.

For $\gamma_0 = 0$, $\gamma_1 = 1$, $b \geq 0$ and $\beta < 0$ this is the Vasicek [56] short rate model. The solution for (7.18) is given by

$$G_1(x) = \frac{1}{\beta} (e^{\beta x} - 1)$$

and

$$G_0(x) = b \int_0^x G_1(\eta) d\eta - \frac{1}{2}a \int_0^x G_1^2(\eta) d\eta.$$

Obviously, $g_0 = bG_1 - \frac{1}{2}aG_1^2 \in H_w$ and $g_1(x) = e^{\beta x} \in H_w^0$. Hence also this classical model is contained in our setting.

From (6.32) we deduce that $\sigma(t, \omega, h) \equiv \sigma^1(h) \equiv \sqrt{a}g_1$ for $h \in \phi(\mathbb{R})$.

8. Appendix: A Summary of Conditions

For the reader's convenience we summarize in this appendix all the conditions made in the text. Recall the definition of $\mathcal{S}$,

$$(\mathcal{S}f)(x) := f(x) \int_0^x f(\eta) d\eta, \quad x \in \mathbb{R}_+$$

for $f \in C(\mathbb{R}_+)$, see (4.10).

8.1 Axioms for the Forward Curve Space

The following axioms are introduced in Sections 4.2–4.3.

- (H1) The functions $h \in H$ are continuous, and the pointwise evaluation $\mathcal{J}_x(h) := h(x)$ is a continuous linear functional on H , for all $x \in \mathbb{R}_+$.
- (H2) The semigroup $\{S(t) \mid t \in \mathbb{R}_+\}$ is strongly continuous in H with infinitesimal generator denoted by A .
- (H3) There exists a constant K such that

$$\|Sh\|_H \leq K\|h\|_H^2$$

for all $h \in H$ with $Sh \in H$.

8.2 Conditions on the Forward Curve Movements

The following conditions are introduced in Sections 4.2–4.3.

- (C1) The initial forward curve $r_0 = f(0, \cdot)$ lies in H .
- (C2) The processes α and σ are H -, resp. $L^0_2(H)$ -valued predictable and

$$\mathbb{P} \left[\int_0^t \left(\|\alpha_s\|_H + \|\sigma_s\|_{L^0_2(H)}^2 \right) ds < \infty \right] = 1, \quad \forall t \in \mathbb{R}_+.$$

- (C3) There exists a continuous version of r , still denoted by r .
- (C4) The processes $\mathcal{S}\sigma^j$ are H -valued predictable.

(C5) There exists an ℓ^2 -valued predictable process $\gamma = (\gamma^j)_{j \in \mathbb{N}}$ satisfying Novikov's condition (2.18) and

$$\sum_{j \in \mathbb{N}} \gamma^j \sigma^j = \sum_{j \in \mathbb{N}} S \sigma^j - \alpha, \quad dt \otimes d\mathbb{P}\text{-a.s.}$$

8.3 Conditions for HJM Models

The following conditions are introduced in Section 4.5.

- (D1) The mappings σ and γ are measurable from $(\mathbb{R}_+ \times \Omega \times H, \mathcal{P} \otimes \mathcal{B}(H))$ into $(L_2^0(H), \mathcal{B}(L_2^0(H)))$, resp. $(\ell^2, \mathcal{B}(\ell^2))$.
- (D2) The mappings $S\sigma^j$ are measurable from $(\mathbb{R}_+ \times \Omega \times H, \mathcal{P} \otimes \mathcal{B}(H))$ into $(H, \mathcal{B}(H))$.
- (D3) There exists a function $\Gamma \in L^2(\mathbb{R}_+)$ such that

$$\|\gamma(t, \omega, h)\|_{\ell^2} \leq \Gamma(t), \quad \forall (t, \omega, h) \in \mathbb{R}_+ \times \Omega \times H.$$

8.4 Assumptions for Characterizing Invariant Manifolds

The following assumptions are introduced in Section 6.2.

- (A1) There exists a continuous version of X , still denoted by X .
- (A2) $B(t, \omega, \cdot) \in C^1(H; L_2^0(H))$ for all $(t, \omega) \in \mathbb{R}_+ \times \Omega$.
- (A3) For any $t \in \mathbb{R}_+$ we have

$$\mathbb{E} \left[\int_0^t \left(\|X_s\|_H + \|F(s, X_s)\|_H + \|B(s, X_s)\|_{L_2^0}^2 \right) ds \right] < \infty.$$

- (A4) The mappings $F(t, \omega, h)$ and $B(t, \omega, h)$ are locally bounded and locally Lipschitz continuous in h .
- (A5) The mappings $F(t, \omega, h)$ and $B(t, \omega, h)$ are right continuous in t , for all $h \in H$ and $\omega \in \Omega$.

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